# MATH 140 Fall 2026

notes by

Marcel Goh

**A note on these notes.** This document will periodically be updated with the material that we cover. The datestamps in the left margin indicate when the notes from each day start. Subsections labelled with a $*$ are optional. These notes should *not* be used as a substitute for the textbook or for attending class. (For instance, on the blackboard I will often draw graphs or diagrams, and these for the most part will be omitted from the notes.) The best way to do well in this class is to do as many exercises from Stewart's textbook as you can stomach. This document may contain errors, typographical or mathematical. Please email me if you find any.

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# I. FOUNDATIONS

*Que si cependant la Geometrie a toûjours*
*quelque obscurité essentiel, qu'on ne puisse dissiper,*
*& ce fera uniquement, à ce que je crois, du côté de l'Infini,*
*c'est que de ce côté-là la Geometrie tient à la Phisique,*
*à la nature intime des Corps que nous connoissons peu,*
*& peut-être aussi à une Metaphisique trop élevée,*
*dont il ne nous est permis que d'appercevoir quelques rayons.*

— BERNARD LE BOVIER DE FONTENELLE, *Élements de la Geometrie de l'Infini* (1727)

## 1. Rates of change

31.VIII This set of notes assumes that the reader is familiar with all the pre-calculus concepts that one learns in the later years of secondary school: functions, trigonometry, exponential functions, logarithms, etc. However, given that the very definition of what differential calculus is requires us to talk about functions and their inputs, it is perhaps prudent to begin with a refresher on real-valued functions with real-valued outputs.

A *function* is a collection of ordered pairs $(x, y)$ with the rule that if $(x, y)$ and $(x, z)$ are both in the collection, then $y = z$. We can view $x$ as the *input* to the function, and $y$ as the *output* of the function; under this view, the rule above is simply that the function never outputs two different values when given the same input. People like to use the letter $f$ (or $g$ or $h$, but really we can use any letter) to denote a function, and $f(x)$ to denote its value evaluated at a given input $x$. However, many particular functions are difficult to write without involving the input variable $x$ in the expression, and in these cases we'll write $f(x)$ to mean the function as well.

Here are some examples of functions:

a) The function $f(x) = 3x$, defined on all real numbers, that outputs triple its input value.

b) The function that assigns to each real number $x$ its square $x^2$.

c) The function $y = f(x)$ that equals 0 whenever $x$ is a negative real number, and equals $x$ whenever $x \geq 0$.

d) The function that takes as input any nonzero real number $x$ and outputs its reciprocal $1/x$.

e) The function $f(x)$, defined on all real numbers, given by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational;} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Note that in each case above, we were careful to specify what the possible inputs to the function were. Going back to the definition of a function as a collection of ordered pairs $(x, y)$, the set of all $x$ such that $(x, y)$ is in the function is called the *domain* of $f$; this is precisely the set of all valid inputs to the function $f$. (The set of all $y$ such that $(x, y)$ is in $f$ is called the *range* of $f$, but this definition won't be as important to us.)

The title of this section is "rates of change," but determining the rate at which a given function changes is actually what this *entire class* will essentially be about. We want to study how much the output of a function changes when you change its input.

For example, consider a car moving from Montreal to Quebec City. As time goes on, it gets further away from Montreal and closer to Quebec City. So we can model the trip as a function $d(t)$, where for each positive real number $t$ we let $d(t)$ equal the distance of the car from Montreal (in kilometres), exactly $t$

minutes since it set off. The rate of change of this function at any given point in time $t$ is the car's *velocity* at that point.

How do we calculate velocity? Well, we know that it is measured in units such as "kilometres per hour" or "metres per second," so it is not too hard to see that velocity is calculated by dividing distance over time.

Let's go back to the car example. Suppose your car is equipped with a very good GPS that can measure your distance from Montreal; however, it does not have a working speedometer. Your friend (who is driving) asks you, "How fast am I driving?" You have taken high school physics, so you know how to solve this problem. You get out a stopwatch and start it exactly when the GPS measures your distance at 135.2 kilometres from Montreal. After one minute on the stopwatch, the GPS reads 137.4, so you report to your friend that she is driving $137.4 - 135.2 = 2.2$ kilometres per minute, or 132 kilometres per hour.

But your friend is not satisfied. "You told me my average speed over the course of a minute, not my speed at the moment I asked you the question. During that minute, I had to speed up to overtake a U-Haul and then slow down because someone cut me off in the fast lane."

So you try and do better. You get out the stopwatch again, and when you start it the GPS reads 137.7. You wait ten seconds, then observe that the GPS reads 138.1. So you report your friend's new speed at $138.1 - 137.7 = 0.4$ kilometres every ten seconds, or 2.4 kilometres per minute, or 144 kilometres per hour.

Your friend is a little more satisfied, but wonders if you can do better. So you try one last time. You start the stopwatch when the GPS reads 138.5, wait exactly one second, and then check the GPS again. To everyone's surprise and disappointment, it still reads 138.5, because it only measures distance to the tenth part of a kilometre. So the calculation would report that the car has moved 0 kilometres in that one second, that is, it's going 0 kilometres per hour.

Obviously in this example you were hamstrung by the precision of your measuring instruments, but it is still worth pondering the following point. Your friend was asking for her speed at some particular instant. However, during any particular instant she does not move, nor does time move. So any naïve calculation as to her speed would go along the lines of $0/0$, which is an undefined value.

Yet it is undeniable that something's "speed at a particular instant" is a concept that holds meaning in everyday life. If you are in a plane and your friend is in a car below you, then it's certainly not nonsensical to say that you are moving faster than your friend at that moment, even though technically it could also be said that both of you have moved "0 kilometres over 0 hours."

The way we make this precise is suggested by your technique in the first scenario. To avoid dividing by zero, we let time elapse just a tiny bit, then measure how much the object's position has changed in that minute interval, and perform the division. Assuming we have arbitrarily sophisticated measuring instruments, we can keep taking smaller and smaller intervals, and arrive at something closer and closer to the "instantaneous velocity." But it is worth

stating that "instantaneous velocity" is a theoretical concept, and not one that can be measured exactly. So when you are driving in a real car, and your speedometer reads some value, say, 127 kilometres per hour, what it's really doing is measuring your car's change in distance over a very small period of time, and reporting to you the quotient (not unlike what you were doing in that first scenario!).

**Secants and tangents.** The average rate of change of a function $f(x)$ between the input $x$ and the input $x+h$ (for some small $h>0$) is the total amount it changes, namely $f(x+h)-f(x)$, divided by the amount the input changed, namely $(x+h)-x=h$. If you graph the function $f(x)$ on the $(x,y)$ plane, then geometrically what you are doing is drawing a line between the points $\big(x, f(x)\big)$ and $\big(x+h, f(x+h)\big)$ and calculating its *slope* (the "rise over run"). Such a line cutting through the function's graph is called a *secant line*, because it *cuts* through the function, at the very least at those two aforementioned points.

By taking $h$ smaller and smaller (but never zero!) we continue to obtain secant lines, but the distance between the two points $\big(x, f(x)\big)$ and $\big(x+h, f(x+h)\big)$ will get smaller and smaller. You can imagine these points getting so close that it's as if the *line touches the function exactly once*, at the point $\big(x, f(x)\big)$ and nowhere else. This line, if it exists, will be called the *tangent line*, but we're not ready to precisely define it just yet.

All we note at this point is that if we admit the concept of the "instantaneous rate of change," and allow that it can be approximated by taking the the average rate of change over smaller and smaller intervals, then the slope of this tangent line (if it exists!) is exactly the instantaneous rate of change that we seek. So it seems that the slope of the tangent line should be the value of the quotient

$$\frac{f(x+h)-f(x)}{(x+h)-x}=\frac{f(x+h)-f(x)}{h}$$

when $h=0$, if only we could plug in $h=0$. But we cannot, since division by zero is undefined.

What we have learned from this preliminary section is that to understand rates of change, we shall have to develop a precise concept of what it means to take a function close to some input at which it is not necessarily defined. This is the notion of a *limit*.

## 2. Limits

02.IX In this section we establish the notion of a limit, which will be a rigorous way of formalising the idea of a function approaching a value at some point where it may or may not be defined. To understand the definition of a limit, it might be helpful to think about it as a generalisation of the concept of a continuous function $f$ equalling some value $y$ at the input $f(x)$.

Consider the very simple function $f(x)=x^2$. We know $f(3)=3^2=9$. But since the function is continuous, it is also the case that if you want the output

to be close to 9, then all you have to do is get the input close to 3. This is true in a *quantitative* sense!

If you want to get $f(x)$ to be within 0.1 of 9, then you just need to get $x$ within 0.01 of the input value 3. How do we know this? Well $2.99^2 = 8.9401$ and $3.01^2 = 9.0601$, so (since the function $x^2$ is nondecreasing for positive inputs) for any $x$ in the range $(2.99, 3.01)$, the value of $f(x)$ is in the range $(8.9, 9.1)$.

To sum up the previous paragraph in one sentence, you could say that in order for $f(x)$ to get 0.1-close to 9, it suffices to pick any input $x$ that is 0.01-close to 3. But since $f(3) = 9$ and $f$ is continuous, we can do much better. To guarantee that $f(x)$ be 0.01-close to 9, we just need to ensure that $x$ is within 0.0001 of 3. And in fact, for any small number $\epsilon > 0$, there exists some $\delta$ such that whenever $x \in (3 - \delta, 3 + \delta)$, we must have $f(x) \in (9 - \epsilon, 9 + \epsilon)$. (In math, we often use the Greek letters $\epsilon$ and $\delta$ to denote very small numbers.)

**The quintessential non-example.** Here is an example to illustrate what we *don't* mean when we are talking about a function getting arbitrarily close to a number when its input gets close to some number. Let

$$f(x) = \sin\Big(\frac{\pi}{x}\Big);$$

this function is defined everywhere except at $x = 0$. The question is, does it "approach" anything when $x = 0$? This function oscillates between $-1$ and 1, but as $x$ approaches 0 it oscillates very wildly. You might be fooled into thinking that as $x$ approaches 0, the function $f$ approaches 0. This is because

$$0 = \sin(\pi) = \sin(2\pi) = \sin(3\pi) = \cdots,$$

so $f(1) = 0$, $f(1/2) = 0$, $f(1/3) = 0$, and so on. But we also have

$$1 = \sin\Big(\frac{\pi}{2}\Big) = \sin\Big(\frac{5\pi}{2}\Big) = \sin\Big(\frac{9\pi}{2}\Big) = \cdots,$$

giving us $f(2) = 1$, $f(2/5) = 1$, $f(2/9) = 1$, etc., so this is a case of $x$ approaching 0 but $f(x)$ *not* getting closer and closer to 0.

So the notion we end up defining for $f$ getting close to something as $x$ approaches something should not accidentally apply to this instance of $f$ above. We are just about ready to state this notion formally.

**The formal definition of a limit.** First we define what a *neighbourhood* of a number is; it is simply some (open) interval containing that number. For example, the interval $(2.5, 3.5)$ is a neighbourhood of the number 3. So is $(2.9, 3.2)$, but $(10, 100)$ is not, and neither is $(1.5, 3)$, since we are considering open intervals.

Suppose that $f$ is defined on all inputs in some neighbourhood of the number $a$, except possibly at $a$ itself. (In other words, suppose that there is some neighbourhood of $a$ contained in the domain of $f$, except possibly the point $a$.)

We say that the the *limit* of the function $f$ at the input $a$ is equal to $L$, and write

$$\lim_{x \to a} f(x) = L,$$

if for every $\epsilon > 0$, there exists some $\delta > 0$ such that whenever $x \neq a$ and $x \in (a - \delta, a + \delta)$, we have $f(x) \in (L - \epsilon, L + \epsilon)$.

(If the use of the "element of" symbol $\in$ disturbs you, you can write the latter two expressions as $a - \delta < x < a + \delta$ and $L - \epsilon < f(x) < L + \epsilon$, respectively. Another way to write the definition of a limit is: For every $\epsilon > 0$ there is some $\delta > 0$ such that whenever $0 < |x - a| < \delta$, we have $\big|f(x) - L\big| < \epsilon$. Convince yourself that this means the exact same thing mathematically as the definition in terms of intervals that we had above—where is the condition $x \neq a$ encoded in the second definition?)

Because the formal definition of a limit is rather cumbersome, involving parameters $\epsilon$ and $\delta$, in many cases we can make do with the following "simplified" or "intuitive" definition of a limit. *The limit of a function $f$ at the input $a$ is $L$ if we can make $f(x)$ as close as we like to $L$ by requiring that $x$ be sufficiently close (but not necessarily equal to) $a$.*

In the example at the beginning of this section, we were convincing ourselves that

$$\lim_{x \to 3} x^2 = 9.$$

This should not be a mindblowing fact, since we of course have $3^2 = 9$. But we were careful in defining the concept of a limit in such a way that it now applies to many more situations as well, not just when $f(a)$ equals $L$.

**A slightly harder example.** Let us now show an example of a function $f$ and an input $a$ such that $\lim_{x \to a} f(x)$ exists and equals some number $L$, even when $f$ is not defined at the input $a$ itself. Suppose we let

$$f(x) = \frac{3x^2 - 2x - 1}{x - 1}.$$

Then we see that $f(x)$ is not defined at the input $x = 1$, since $1 - 1 = 0$ and dividing by 0 is illegal. Nevertheless, if you graph this function $f$ using a graphing calculator or something like Desmos, you will see that the function $f$ appears very simple. It is defined everywhere except at $x = 1$, and besides that little gap, its graph is simply a line; it is the unique line passing through the points $(0, 1)$ and $(2, 7)$. So even though $f$ is not defined when $x = 1$, we might suspect from this picture that $\lim_{x \to 1} f(x) = 4$. Let us state this as a proposition and prove it rigorously (via the formal definition of a limit).

**Proposition 1.** *We have*

$$\lim_{x \to 1} \frac{3x^2 - 2x - 1}{x - 1} = 4.$$

*Proof.* Let $f(x) = (3x^2 - 2x - 1)/(x - 1)$ again for short. Suppose that $\epsilon > 0$ is given to us; we need to prescribe some $\delta > 0$ such that whenever $0 < |x - 1| < \delta$, we have $0 < \big|f(x) - 4\big| < \epsilon$. We shall show that picking $\delta = \epsilon/3$ works.

Here's why. Assume that $0 < |x - 1| < \delta = \epsilon/3$. Multiplying the second inequality by 3 gives us

$$|3x - 3| < \epsilon.$$

We want a term of $-4$ inside the absolute-value bars, so let's just add and subtract 1 in that expression to make that happen:

$$|3x + 1 - 4| < \epsilon \tag{1}$$

Now we're almost done—we would be done if we had $f(x)$ instead of $3x+1$. But note that

$$3x^2 - 2x - 1 = (3x + 1)(x - 1),$$

so it's time to use the other assumption we had, which was $|x - 1| > 0$, to see that $(x - 1)/(x - 1) = 1$. In other words, under the assumption $|x - 1| > 0$, we have

$$3x + 1 = (3x + 1) \cdot 1 = \frac{(3x + 1)(x - 1)}{x - 1} = \frac{3x^2 - 2x - 1}{x - 1} = f(x).$$

Substituting this into (1) gives us $0 < \big|f(x) - 4\big| < \epsilon$, which is precisely what we wanted to show. ∎

Picking $\delta = \epsilon/3$ seemed to require a great deal of algebraic ingenuity here; let's try to convince ourselves that it's not so mysterious a choice. If you draw the graph of $f$, you'll note that the slope near $x = 1$ is 3 (well it's actually 3 everywhere!). So to guarantee no more than $\epsilon$ error in the output, we need to be three times as precise in the input: we can have no more than $\epsilon/3$ error in the input. This leads to the choice $\delta = \epsilon/3$. Note that the proof would also work if we picked $\delta$ even smaller, e.g., $\delta = \epsilon/10$. But $\delta = \epsilon/3$ is the largest one can take $\delta$ for the proof to work, and it is a common stylistic choice for a mathematician to pick a maximal value such as this.

Before moving on, we note here that you may at some point encounter authors who write "$f(x) \to L$ as $x \to a$." This is just an alternative way of writing $\lim_{x\to a} f(x) = L$. We will try to stick with the $\lim_{x\to a}$ version in these notes, as that is the convention the textbook uses, but there is a strong argument for the $f(x) \to L$ notation as well, as it makes it clear that $f(x)$ is not required to actually equal $L$ at any stage, it just needs to get arbitrarily close to it.

04.IX **Formal negation of limit definition.** To fully understand the meaning of $\lim_{x\to a} f(x)$, it may be enlightening for us to consider the negation of this statement. Concretely, what does it mean for a number $L$ *not* to be the limit of a function $f$ at some input $a$? Negating long mathematical statements (such as the definition of limit) properly is not an easy task, and the untrained mathematician is liable to make errors especially concerning quantifiers such as "for all" and "there exists."

The correct formal negation is the following: *The real number $L$ is not a limit of $f$ at the input $a$ if there exists some $\epsilon > 0$ such that for all $\delta > 0$, there is some $x$ in the domain of $f$ satisfying $0 < |x - a| < \delta$ with the property that $\big|f(x) - L\big| \geq \epsilon$.*

This is a fairly convoluted statement, but if you ponder it for a while you will convince yourself that it does mean the opposite of what the limit definition says. When we write $\lim_{x\to a} f(x) = L$, we mean that you can get the values of $f(x)$ arbitrarily close to $L$ by taking $x$ sufficiently close to $a$, and the opposite of this statement is there is some fixed positive distance $\epsilon$ such that even if you take $x$ arbitrarily close to $a$, there will always be some badly-behaved choices of $x$ that result in $f(x)$ being too far (at a distance at least $\epsilon$) from $f(x)$.

(Note that the negation of the mathematical statement $\lim_{x\to a} f(x) = L$ should not be recorded as $\lim_{x\to a} f(x) \neq L$, since even writing down the left-hand side of this inequality suggests that it exists (is a real number), and as we shall see later in this section, the limit of a function $f$ need not always exist at some given $a$.)

Here's an example to illustrate how you disprove the claim that the limit of a function $f$ at some $a$ is $L$.

**Proposition 2.** *Let*

$$f(x) = \begin{cases} -1, & \text{if } x < 0; \\ 1, & \text{if } x > 0 \end{cases}.$$

*Then the limit of $f$ at 0 is not 0.*

*Proof.* We just need to find some $\epsilon > 0$ such that for any choice of $\delta > 0$, there will always be some real number $x$ with $0 < |x - 0| < \delta$ but $\big|f(x) - 0\big| \geq \epsilon$. It turns out that choosing $\epsilon = 1/2$ works.

To see this, let $\delta > 0$ be given. We need to pick some $x$ with $|x - 0| < \delta$ but with $\big|f(x) - 0\big| \geq 1/2$. This is not so hard—we have lots of choices! Choosing $x = \delta/2$ works, since $f(x) = f(\delta/2) = 1$, and $|1| \geq 1/2$.

Hence the limit of $f$ at 0 is not 0. ∎

***A game with the Adversary.** Let $f$ be a function, let $L$ be a number, and let $a$ be a number such that $f$ is defined in some neighbourhood of $a$. The statement "$\lim_{x\to a} f(x) = L$" is either true or false (the "law of the excluded middle" tells us that every mathematical statement is either true or false). We can interpret its truthhood or falsity as the result of a game played between ourselves and a supernatural Adversary. Here's the how the game goes.

**Game L** (*The limit game*). Let $f$ be a function, $L$ a number, and $a$ a number with a neighbourhood contained in the domain of $f$.

**L1.** [Adversary's turn.] The Adversary is allowed to pick any $\epsilon > 0$.

**L2.** [Our turn.] We can then pick any $\delta > 0$.

**L3.** [Adversary's turn.] The Adversary is then allowed to pick any $x$ in the domain of $f$ that satisfies

$$0 < |x - a| < \delta.$$

(This means that $x$ is at distance less than $\delta$ from $a$, but doesn't equal $a$.)

**L4.** [Verdict?] If

$$\big|f(x) - L\big| < \epsilon,$$

that is, if the distance between $f(x)$ and $L$ is less than $\epsilon$, then we win; otherwise the Adversary wins. ∎

If $\lim_{x\to a} f(x) = L$ is true, then we can always win; the very definition of limit tells us that no matter what $\epsilon$ is chosen in step L1 by the Adversary, in step L2 there is always a choice of $\delta$ that will force the Adversary into letting us win in step L3. On the other hand, if $\lim_{x\to a} f(x) = L$ is false (either because this limit does not exist or because it equals something else), then the Adversary can always win. There is a choice of $\epsilon$ in step L1 such that no matter what we pick in step L2, the Adversary can beat us by making a further nefarious choice for $x$ in step L3.

**Nonexistent limits.** We alluded earlier to the possibility of a function $f$ not having any limits at some point $a$. We say that the limit of $f$ at $a$ *does not exist* if for all real numbers $L$, the limit of $f$ at $a$ is not $L$. In fact, the function $f$ from the previous proposition has this property at $a = 0$. The proof is largely unchanged from the easier case above when we had just set $L$ to 0, requiring only a little more care in the second paragraph.

**Proposition 3.** *For $x \neq 0$, let*

$$f(x) = \frac{|x|}{x}.$$

*Then the limit of $f$ as $x$ approaches 0 does not exist.*

*Proof.* We can re-express $f$ by writing

$$f(x) = \begin{cases} -1, & \text{if } x < 0; \\ 1, & \text{if } x > 0 \end{cases},$$

which may be clearer to some readers (and also shows this is the same function from Proposition 2). Let $L$ be arbitrary. We need to show that the limit of $f$ at 0 is not $L$. To do this, we shall pick $\epsilon > 0$ such that for any choice of $\delta > 0$, there is some real number $x$ with $0 < |x - 0| < \delta$ but $\big|f(x) - L\big| \geq \epsilon$. Just as in the previous proof for the special case $L = 0$, the choice $\epsilon = 1/2$ works.

Let $\delta > 0$ be given. We need to pick some $x$ with $|x - 0| < \delta$ but with $\big|f(x) - L\big| \geq 1/2$. Here it matters whether $L$ is positive or negative (if $L = 0$ then just appeal to the proof of Proposition 2). If $L$ is positive, then choose $x = -\delta/2$. We have $0 < |-\delta/2 - 0| < \delta$ as required, and also

$$\big|f(-\delta/2) - L\big| = |-1 - L| > |-1 - 0| \geq 1/2 = \epsilon.$$

On the other hand, if $L$ is negative, we instead choose $x = \delta/2$, so we still have $0 < |\delta/2 - 0| < \delta$ as needed, and this time

$$\big|f(\delta/2) - L\big| = |1 - L| > |1 - 0| \geq 1/2 = \epsilon.$$

Either way, we have satisfied the criterion of the limit at $f$ at $a$ not being $L$.

But the previous paragraph works for all numbers $L$, so the limit of $f$ at $a$ cannot equal any number! Hence we have proved that it does not exist. ∎

In the example just given, we now know that the limit of $f$ at 0 does not exist. However, we often think of a limit intuitively as the function "approaching" some value, and in this sense it's wrong to say that $f$ doesn't approach anything at all when the input gets close to 0. When you approach from the right, it is equal to 1. And when you approach from the left, the function equals $-1$. This leads us to the concept of a one-sided limit.

**One-sided limits.** Let $f$ be a function, let $L$ be a number, and let $a$ be a number such that $f$ is defined to the right of $a$. (More precisely, we assume that $f$ is defined on the interval $(a, a+\eta)$ for some $\eta > 0$.) We say that *limit of $f(x)$ as $x$ approaches $a$ from the right is $L$*, and write

$$\lim_{x \to a^+} f(x) = L,$$

if for all $\epsilon > 0$ there is $\delta > 0$ such that any number $x$ in the domain of $f$ with

$$a < x < a + \delta$$

must satisfy

$$\big|f(x) - L\big| < \epsilon.$$

We might say in this case that $L$ is the *right-hand limit* of $f$.

Likewise, let $f$ be a function, $L$ be a number, and suppose that $a$ is such that $f$ is defined to the *left* of $a$. We say that *limit of $f(x)$ as $x$ approaches $a$ from the left is $L$*, and write

$$\lim_{x \to a^-} f(x) = L,$$

if for all $\epsilon > 0$ there is $\delta > 0$ such that any number $x$ in the domain of $f$ with

$$a - \delta < x < a$$

must satisfy

$$\big|f(x) - L\big| < \epsilon.$$

If this is true then we say $L$ is the *left-hand limit* of $f$.

More loosely, $L$ is the right-hand limit of $f$ at $a$ if you can get $f(x)$ to be arbitrarily close to $L$ by requiring that $x$ be sufficiently close to $a$ *and greater than* $a$, and it's the left-hand limit of $f$ at $a$ if you can get $f(x)$ as close as you like to $L$ by requiring that $x$ be sufficiently close to $a$ *and smaller than* $a$.

Both of these definitions are relaxations of the definition of limit; it is "easier" to be a left-hand limit than it is to be a *bona fide* limit). For example, we saw in Proposition 3 that 1 is not a limit of $f$ from that proposition, because

$f$ has no limits. However, once you have digested the definitions of one-sided limits, you should be able to check that 1 is the right-hand limit of $f$ at 0, and $-1$ is the left-hand limit at 0.

If the actual limit of $f$ at a point $a$ is $L$, then it is a consequence of the definitions that the right-hand limit and the left hand limit are also $L$. Conversely, if the one-sided limits both equal the same $L$, then the limit is actually $L$. This can be summed up symbolically in the following statement, which we'll call a theorem since it comes in handy now and again.

**Theorem 4.** *For any function $f$ and numbers $a$ and $L$, we have*

$$\lim_{x\to a} f(x) = L$$

*if and only if*

$$\lim_{x\to a^+} f(x) = L \qquad \text{and} \qquad \lim_{x\to a^-} f(x) = L. \quad \blacksquare$$

A simple application of Theorem 4 is to prove that the function $f(x) = |x|$ goes to 0 as $x$ approaches 0. We'll see this as an example in the next section, but it's really not a difficult proof, so you could try it now. Show that both one-sided limits approach 0, then simply invoke the theorem.

Here's a related theorem (no less useful) that tells us what happens if the one-sided limits of a function do not agree at some given point. It shows that the example exhibited by Proposition 3 is an instance of a very general phenomenon.

**Theorem 5.** *If*

$$\lim_{x\to a^+} f(x) = L, \qquad \lim_{x\to a^-} f(x) = M,$$

*and $L \neq M$, then $\lim_{x\to a} f(x)$ does not exist.*

*Proof.* We skip this proof as it's beyond the scope of this course (you would see it in, e.g., MATH 242 or MATH 254), but it is not so different in spirit to the proof of Proposition 3. The motivated student is encouraged to try it as an advanced exercise. (If you try for a while and get stuck, email me or come to office hours with what you have and I could help you fill in the details.) $\blacksquare$

If we were told Theorem 5 from the start, we could have used it to prove Proposition 3 without much ado.

*Proof of Proposition 3 using Theorem 5.* Recall that we are trying to prove that the limit of the function $f(x) = |x|/x$ does not exist as $x$ approaches 0. Since $|x| = x$ for all $x > 0$, we see that

$$\lim_{x\to 0^+} \frac{|x|}{x} = \lim_{x\to 0^+} \frac{x}{x} = \lim_{x\to 0^+} 1 = 1.$$

(The last equality is easy to see if you just graph the constant function 1. If you are a serious student and insist on a rigorous proof, then please do Exercise 24 of

Section 2.4 in the textbook, which invites you to furnish a proof that $\lim_{x\to a} c = c$ for all numbers $a$ and $c$.)

On the other hand, since $|x| = -x$ for all $x < 0$, we have

$$\lim_{x\to 0^-} \frac{|x|}{x} = \lim_{x\to 0^-} \frac{-x}{x} = \lim_{x\to 0^+} -1 = -1.$$

So the left- and right-hand limits do not agree, and Theorem 5 applies in this situation to tell us that the limit does not exist. ▮

One-sided limits not agreeing is one way that the limit of a function at a given point can fail to exist. Another way is for the limit to get arbitrarily large, which is the situation we tackle next.

09.IX **Infinite limits and vertical asymptotes.** We say that the *limit of $f(x)$ is infinity as $x$ approaches $a$ from the right*, and write

$$\lim_{x\to a^+} f(x) = \infty,$$

if for every large (positive) number $M$, there exists $\delta > 0$ such that every $x$ satisfying

$$a < x < a + \delta$$

has

$$f(x) > M.$$

If, instead, for every large positive number $M$ there is $\delta > 0$ such that every $x$ with

$$a < x < a + \delta$$

satisfies

$$f(x) < -M,$$

then we say that the limit of $f(x)$ is *minus infinity* as $x$ approaches $a$ from the right, and write

$$\lim_{x\to a-} f(x) = -\infty.$$

The same definition can be made for left-hand limits by replacing the assumption $a < x < a + \delta$ with the assumption $a - \delta < x < a$ above. This makes for four situations in total, and if any one of these situations occurs, then $f$ is said to have a *vertical asymptote* at the line $x = a$.

Just as in the case of finite limits, when the left- and right-hand limits agree (and are both the same type of infinity), then we simply say that the *limit* is infinity (or minus infinity, accordingly). In other words, we have

$$\lim_{x\to a} f(x) = \infty$$

if and only if

$$\lim_{x\to a^-} f(x) = \infty \qquad \text{and} \qquad \lim_{x\to a^+} f(x) = \infty,$$

and we have

$$\lim_{x\to a} f(x) = -\infty$$

if and only if

$$\lim_{x\to a^-} f(x) = -\infty \qquad \text{and} \qquad \lim_{x\to a^+} f(x) = -\infty.$$

(Let $p$ and $q$ be mathematical statements. The phrase "$p$ if $q$" means that $q$ implies $p$, and the phrase "$p$ only if $q$" means that $p$ implies $q$—think about it for a second, it makes sense! So the statement "$p$ if and only if $q$" means that $p$ is true exactly when $q$ is true, and vice versa.)

One should not interpret the symbol $\infty$ above as a number, since there is no such number. When we write $\lim_{x\to a} f(x) = \infty$, we simply mean that this expression can be made arbitrarily large as by requiring that $x$ be sufficiently close to $a$. It should also be stressed that the equality symbol in this expression should *not* be interpreted as meaning that the limit of $f$ at $a$ exists, because the terminology of a limit "existing" is reserved for when the limit equals a number (and, once again, $\infty$ is not a number). As with finite limits, sometimes authors write, e.g., $f(x) \to -\infty$ as $x \to a$ instead of $\lim_{x\to a} f(x) = -\infty$. This notation is also valid, and in some sense preferable as it does not suggest that $f(x)$ ever "equals" some "number" $\infty$.

Let's look at a couple of simple examples of functions with vertical asymptotes.

**Proposition 6.** *We have*

$$\lim_{x\to 0^+} \frac{1}{x} = \infty$$

*and*

$$\lim_{x\to 0^-} \frac{1}{x} = -\infty.$$

*Proof.* We prove both statements separately. The first asserts that $1/x$ can be made arbitrarily large by taking $x \to 0$ from the right. To prove it, let $M > 0$ be an arbitrarily large number (given to us). Pick $\delta = 1/M$, and assume that $0 < x < \delta = 1/M$. Because $x > 0$, we are allowed to write $1/x$, and the fact that $x < 1/M$ implies that $1/x > M$, which is what we needed to prove in the definition of infinite limit.

The second statement is proved extremely similarly, but we include the full proof for completeness. Let $M$ be given, and pick $\delta = 1/M$. Assume that $-\delta < x < 0$; the right inequality here assures us that the expression $1/x$ is well-defined, and the left inequality here can be rewritten as $x > -\delta = -1/M$. Rearranging the equation now yields $1/x < -M$, which is what we needed to show in order to prove that the limit is $-\infty$. ∎

To sum up this proof, we can make $1/x$ arbitrarily large to the right of 0 by requiring that $x > 0$ be sufficiently close to 0, and we can make $1/x$ an arbitrarily

large negative number by requiring that $x < 0$ be sufficiently close to (the left of) 0.

Here's another example where the left- and right-hand limits agree on the type of infinity.

**Proposition 7.** *The limit of*

$$f(x) = \frac{1}{(x-3)^2}$$

*is infinity as $x$ approaches* 3.

*Proof.* We can do the left- and right-hand limits separately, but it is not hard to prove this whole thing all at once (i.e., to consider both sides of 3 at the same time).

Let $M > 0$ be given. We want to prescribe $\delta > 0$ such that all $x \neq 3$ with

$$3 - \delta < x < 3 + \delta$$

satisfy

$$\frac{1}{(x-3)^2} > M.$$

(The assumption $3 - \delta < x < 3 + \delta$ is what we mean by considering both sides of 3 at the same time; note that this assumption together with the assumption that $x \neq 3$ can be summed up in the single inequality $0 < |x - 3| < \delta$.) Setting $\delta = 1/\sqrt{M}$ works this time, so our assumption on $x$ will be

$$3 - \frac{1}{\sqrt{M}} < x < 3 + \frac{1}{\sqrt{M}},$$

and that $x \neq 3$ of course, so that $f(x)$ is even defined. The difference $x - 3$ may be positive or negative; we don't know which. But what we do know, from our assumption, is that the distance $|x - 3|$ between $x$ and 3 satisfies

$$|x - 3| < \frac{1}{\sqrt{M}}.$$

Squaring both sides (and using the fact that $|x-3|^2 = (x-3)^2$, since the square of a negative number is positive anyway) gives us

$$(x-3)^2 < \frac{1}{M},$$

and rearranging yields

$$f(x) = \frac{1}{(x-3)^2} > M,$$

which is what we needed to show. (The number $M$ was arbitrary in this proof, so we can keep repeating with larger and larger $M$ to show that $f(x)$ can be made arbitrarily large in the way shown above.) ∎

Once again, picking $\delta = 1/\sqrt{M}$ made the algebra work out nicely, but you could have written a valid proof by picking, say, $\delta = 1/M$ instead, but then there'd be some issues with having to assume that $M > 1$. (Morally this is O.K., since we're only worried about what happens when $M$ is very large, but *technically* our definition had $M > 0$.) Don't stress too much about picking the "best" value for $\delta$. Generally, you find $\delta$ by first trying out various values of $M$ and seeing what $\delta$ is necessary to deal with that $M$; after a few tries you should be able to come up with a formula that works generally. The broader thing to take away from reading this proof or trying to do it yourself is that no matter what $M$ is given, you can pick *some* $\delta$ that makes the proof work out.

11.IX **Limits at infinity and horizontal asymptotes.** There is one last type of limit that remains to be defined. This is the case in which the values of $f(x)$ can be taken arbitrarily close to a value by making $x$ sufficiently large, either in the positive direction or the negative direction. Formally, we write

$$\lim_{x\to\infty} f(x) = L$$

and say that the *limit of $f(x)$ is $L$ as $x$ approaches infinity* if for all $\epsilon > 0$ there exists some large $M > 0$ such that every $x$ with $x > M$ has $\big|f(x) - L\big| < \epsilon$. Likewise, we say the *limit of $f(x)$ is $L$ as $x$ approaches negative infinity* and write

$$\lim_{x\to-\infty} f(x) = L$$

if the same condition applies, except the inequality $x > M$ is replaced by $x < -M$. In either of these situations, we say that $y = f(x)$ has a *horizontal asymptote* at the line $y = L$.

Note that the same example from Proposition 6 also exhibits a horizontal asymptote.

**Proposition 8.** *We have*

$$\lim_{x\to\infty} 1/x = 0 \qquad \textit{and} \qquad \lim_{x\to-\infty} 1/x = 0.$$

*Proof.* Let $\epsilon > 0$ and pick $M = 1/\epsilon$. Then any number $x$ with $x > M$ has

$$\left|\frac{1}{x} - 0\right| = \frac{1}{x} < \frac{1}{M} = \epsilon,$$

so $\lim_{x\to\infty} 1/x = 0$.

For the other inequality, the proof is very similar. Let $\epsilon > 0$ and again we choose $M = 1/\epsilon$. Now any (negative) number $x$ with $x < -M$ has

$$\left|\frac{1}{x} - 0\right| = \left|\frac{1}{x}\right| = \frac{1}{-x} < \frac{1}{M} = \epsilon,$$

so $\lim_{x\to-\infty} 1/x = 0$. ∎

Since $\infty$ is not a number, it does not make sense to write an expression such as, say, $1/\infty$. On the other hand, you may have heard people say, informally, something like "anything divided by infinity is zero." This statement can be made into a proper mathematical one by expressing it in terms of limits. It is true that for any fixed number $K$, the limit $\lim_{x\to\infty} K/x$ equals 0. (As an exercise, prove this directly by very slightly modifying the proof of Proposition 8 above. Hint: The proposition already gives a proof when $K = 1$. So for the general case, instead of setting $M = 1/\epsilon$, you might try choosing something very similar but involving $K$.)

**Infinite limits at infinity.** As with limits as $x$ approaches some finite $a$, the infinite limit is said not to exist if it does not equal any finite number $L$. There is one special case of this worth mentioning, which is when the function $f(x)$ can be made arbitrarily large (positive or negative) by requiring that $x$ be sufficiently large (positive or negative). Hence there are four cases of infinite limits at infinity, namely

$$\lim_{x\to\infty} f(x) = \infty, \qquad \lim_{x\to-\infty} f(x) = \infty,$$

$$\lim_{x\to\infty} f(x) = -\infty, \qquad \text{and} \qquad \lim_{x\to-\infty} f(x) = -\infty.$$

Tedious as it is, let us formally define all four of these statements, for the sake of completeness.

a) We say that $f(x)$ *goes to infinity as $x$ goes to infinity* and write

$$\lim_{x\to\infty} f(x) = \infty$$

if for all $M > 0$, there exists $N > 0$ such that for all $x$ with $x > N$, we have $f(x) > M$.

b) We say that $f(x)$ *goes to infinity as $x$ goes to negative infinity* and write

$$\lim_{x\to-\infty} f(x) = \infty$$

if for all $M > 0$, there exists $N > 0$ such that for all $x$ with $x < -N$, we have $f(x) > M$.

c) We say that $f(x)$ *goes to negative infinity as $x$ goes to infinity* and write

$$\lim_{x\to\infty} f(x) = -\infty$$

if for all $M > 0$, there exists $N > 0$ such that for all $x$ with $x > N$, we have $f(x) < -M$.

d) We say that $f(x)$ *goes to negative infinity as $x$ goes to negative infinity* and write

$$\lim_{x\to-\infty} f(x) = -\infty$$

if for all $M > 0$, there exists $N > 0$ such that for all $x$ with $x < -N$, we have $f(x) < -M$.

That was a mouthful. But by this point it is hoped that the reader would be comfortable enough with these formal definitions to be able to formulate, say (a), (c), and (d) after being shown (b).

We finish off this very long section by presenting an example of a function that seems to grow fairly slowly $x \to \infty$, but nevertheless still goes to infinity as $x$ is taken to be large.

**Proposition 9.** *We have*

$$\lim_{x\to\infty} \sqrt[3]{x} = \infty$$

*and*

$$\lim_{x\to-\infty} \sqrt[3]{x} = -\infty.$$

*Proof.* Let $M > 0$ be a large positive number given to us. We want to pick $N$ such that for all $x > N$, we have $\sqrt[3]{x} = x^{1/3} > M$. After a moment's pause we see that choosing $N = M^3$ works, since if we assume that $x > N$, then

$$x^{1/3} > N^{1/3} = (M^3)^{1/3} = M.$$

The negative infinite limit is proved similarly; we choose $N = M^3$ this time as well, so that whenever $x < -N$, we have

$$x^{1/3} < (-N)^{1/3} = (-M^3)^{1/3} = -M.$$

This completes the proof. ∎

## 3. Rules for evaluating limits

In the previous section, we already encountered two rules (namely, Theorems 4 and 5) that can help us find limits of functions or determine that they do not exist. This section will be devoted to establishing a few more useful laws. First we record a list of algebraic laws, which show that the limit behaves well under a number of algebraic operations such as addition, multiplication, etc.

**Theorem 10** (*Algebraic limit laws*). *Let $c$ be any fixed number, and suppose that $f$ and $g$ are functions such that*

$$\lim_{x\to a} f(x) \qquad \textit{and} \qquad \lim_{x\to a} g(x)$$

*both exist. Then*

*i)* $\lim_{x\to a}\big(f(x) + g(x)\big) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$;

*ii)* $\lim_{x\to a}\big(f(x) - g(x)\big) = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$;

*iii)* $\lim_{x\to a} cf(x) = c \cdot \lim_{x\to a} f(x)$;

*iv)* $\lim_{x\to a}\big(f(x)g(x)\big) = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$*;*

*v)* $\lim_{x\to a} f(x)/g(x) = \lim_{x\to a} f(x)\big/\lim_{x\to a} g(x)$*, provided that* $\lim_{x\to a} g(x) \neq 0$*;*

*vi) for all positive integers* $n$*,* $\lim_{x\to a} f(x)^n = \big(\lim_{x\to a} f(x)\big)^n$*; and*

*vii) for all positive integers* $n$*,* $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$*, (provided that* $\lim_{x\to a} f(x) > 0$ *in the case that* $n$ *is even ,so that the* $n$*th root is well-defined).*

*These statements are all true even when* $x \to a$ *is replaced with* $x \to \infty$*,* $x \to -\infty$*,* $x \to a^+$*, or* $x \to a^-$*.*

The identities above are known as (i) the sum law, (ii) the difference law, (iii) the constant multiple law, (iv) the product law, (v) the quotient law, (vi) the power law, and (vii) the root law, respectively. They can be expressed in words as well: the limit of a sum is the sum of the limits, the limit of a difference is the difference of the limits, and so on.

14.IX *Proof.* We only prove (i) in these notes. Laws (ii) through (v) are proved in Appendix F of the textbook. Law (vi) is a consequence of repeating law (iv) $n$ times (formally, you would need to use the *principle of mathematical induction*, which is outside the scope of this course), and the proof of law (vii) is assigned to you in the textbook as Exercise 69 of Section 2.5. All of these proofs only deal with the case that $x \to a$ for some finite number $a$, as opposed to $x \to \infty$; it is an instructive exercise to try and adapt the proof below for the $x \to \infty$ case, to test that you have fully digested this proof and the definition of limits at infinity.

Enough talk. On to the proof of (i). We shall make use of the *triangle inequality*, which says that for any numbers $a$, $b$, and $c$, we have

$$|a - c| \leq |a - b| + |b - c|.$$

In other words, the length of any side in a triangle is always smaller than the sum of the other two sides.

Suppose $\lim_{x\to a} f(x) = L_1$ and $\lim_{x\to a} g(x) = L_2$. We want to show that the limit of $f(x) + g(x)$ as $x \to a$ is $L_1 + L_2$. Let $\epsilon > 0$. We wish to pick a $\delta > 0$ such that for all $x$ with $0 < |x - a| < \delta$, we have

$$\big|f(x) + g(x) - (L_1 + L_2)\big| < \epsilon.$$

Well, since $\lim_{x\to a} f(x) = L_1$, we can make $f(x)$ arbitrarily close to $L_1$ by prescribing that $x$ be sufficiently close to $a$. So there is some $\delta_1 > 0$ such that if we assume $0 < |x - a| < \delta_1$, then

$$\big|f(x) - L_1\big| < \frac{\epsilon}{2}.$$

(We choose $\epsilon/2$ so that there is some additional room for error when we deal with $g$.) Likewise, there is some $\delta_2 > 0$ such that if we assume $0 < |x - a| < \delta_2$, then

$$\big|g(x) - L_2\big| < \frac{\epsilon}{2}.$$

Now, set $\delta = \min\{\delta_1, \delta_2\}$. This guarantees both $\delta \leq \delta_1$ and $\delta \leq \delta_2$, so that if we assume $0 < |x - a| < \delta$, then the inequalities

$$0 < |x - a| < \delta_1 \qquad \text{and} \qquad 0 < |x - a| < \delta_2$$

*both* hold. As a result, by our observations in the previous paragraph, both of the inequalities

$$\big|f(x) - L_1\big| < \frac{\epsilon}{2} \qquad \text{and} \qquad \big|g(x) - L_2\big| < \frac{\epsilon}{2}$$

are true as well. Finally we see that

$$\begin{aligned} \big|f(x) + g(x) - (L_1 + L_2)\big| &= \big|f(x) - L_1 + g(x) - L_2\big| \\ &\leq \big|f(x) - L_1\big| + \big|g(x) - L_2\big| \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon, \end{aligned}$$

where the second line follows from the aforementioned triangle inequality. This completes the proof. ∎

These laws are extremely powerful in general, allowing us to express the limit of a complex expression in terms of the limits of simpler expressions. For example, using laws (i), (iii), and (iv), we can deduce that

$$\lim_{x \to 3} 4x^2 - 7 = 4 \cdot \Big(\lim_{x \to 3} x\Big)\Big(\lim_{x \to 3} x\Big) - \lim_{x \to 3} 7. \tag{2}$$

We already encountered an expression of the form $\lim_{x \to a} c$, where $c$ is a constant number, in an earlier proof. It should be obvious that this limit is $c$, and that $\lim_{x \to a} x = a$. Let us state these two facts, along with two more as a theorem, so that we can easily invoke them later on. (We continue the numbering from Theorem 10 in order to remain consistent with the textbook.)

**Theorem 11** (*Special limits*). *For any positive integer $n$ and any numbers $a$ and $c$,*

*viii)* $\lim_{x \to a} c = c$;

*ix)* $\lim_{x \to a} x = a$;

*x)* $\lim_{x \to a} x^n = a^n$; *and*

*xi)* $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$.

*Laws (viii) and (ix) remain true when $x \to a$ is replaced by $x \to \infty$ or $x \to -\infty$.*

*Proof.* As mentioned before, the proof of law (viii) is presented to you as Exercise 24 in Section 2.4; the proof of law (ix) is Exercise 23 in the same section. This is in the case where $x$ approaches some finite $a$; you might like to prove the corresponding limits at infinity to test your understanding of those definitions.

Laws (x) and (xi) are easy to prove once we have all the other laws established. To prove law (x), apply the power law with $f(x) = x$ and then law (ix) to obtain

$$\lim_{x\to a} x^n = \left(\lim_{x\to a} x\right)^n = a^n.$$

To prove law (xi), apply the root law with $f(x) = x$ and then law (ix) to obtain

$$\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{\lim_{x\to a} x} = \sqrt[n]{a}. \quad \blacksquare$$

With these new laws in hand, we can revisit the polynomial $4x^2 - 7$ from earlier and conclude that

$$\lim_{x\to 3}(4x^2 - 7) = 4 \cdot 3^2 - 7 = 29.$$

In fact, we don't even need to break up $x^2$ into $x \cdot x$ like we did in (2), as law (x) takes care of monomials such as these.

When applying the limit laws to a determine what happens when $x \to 3$ to a simple polynomial such as $4x^2 - 7$, it seemed like we could just substitute in $x = 3$ in the expression and get the right answer. This turns out to be true in the general case $x \to a$ as long as $a$ is in the domain of $f$.

**Theorem 12** (*Direct substitution*). *Let $f$ be a polynomial or a rational function (a quotient of two polynomials). If $a$ is in the domain of $f$, then*

$$\lim_{x\to a} f(x) = f(a).$$

*Proof.* Every rational function $f$ is of the form

$$f(x) = \frac{a_m x^m + a_{m-1}x^{m-1} + \cdots + a_1 x + a_0}{b_n x^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0},$$

where $m$ and $n$ are nonnegative integers, and $a_0, a_1, \ldots, a_m, b_0, b_1, \ldots, b_n$ are real (possibly negative or zero) coefficients. Taking the limit of both sides, the quotient law deals with the fraction, the sum law splits the limit of the sum into the sum of limits, and from there we use law (x) and the constant multiple law to break up these terms further. We then apply laws (viii) and (ix) as to finish off the proof. $\blacksquare$

This theorem applies to functions involving roots as well, thanks to the root law. (It more generally applies to all *continuous* functions—we will discuss the

notion of continuity very soon.) Using the direct substitution rule is fairly easy whenever its hypotheses are satisfied, so we will not dwell on this. There is, however, a more advanced version of this rule that is useful in certain cases, such as when $f(x)$ is a rational function and $a$ is *not* in the domain of $f$ (e.g., because plugging in $x = a$ makes the denominator 0). Then we may have to invoke the following theorem in order to use the direct substitution rule.

**Theorem 13.** *Let $a$ be a number, and suppose that $f$ and $g$ are two functions that are equal for all $x$ in some interval containing $a$, except possibly at $a$ itself. Then if $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ both exist, we must have*

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x).$$

**Proof.* This proof is a bit more advanced than you should fairly be expected to follow in this course, though if you take a second-year mathematics class such as MATH 242 one day, you'd find it a routine exercise by the end of that course. We include it in these notes for completeness.

We know that $f$ and $g$ agree in some interval containing $a$, except possibly at $a$ itself. To be concrete, this means there exists some number $\eta > 0$ be such that $f(x) = g(x)$ for every $x$ with $0 < |x-a| < \eta$. Suppose $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} = M$. We want to show that $L = M$; we shall show that $|L - M| < \epsilon$ for all $\epsilon > 0$, which is an equivalent statement (a nonnegative quantity that is less than any positive quantity must be zero).

Let $\epsilon > 0$. We will use an "$\epsilon/2$ trick" again, just as in the proof of the sum law above. Since $\lim_{x\to a} f(x) = L$, there is $\delta_1$ such that for all $x$ with $0 < |x - a| < \delta_1$, we have

$$\big|f(x) - L\big| < \epsilon/2. \tag{3}$$

Likewise, since $\lim_{x\to a} g(x) = M$, there is $\delta_2$ such that for all $x$ with $0 < |x-a| < \delta_2$, we have

$$\big|g(x) - M\big| < \epsilon/2. \tag{4}$$

Now pick $\delta = \min\{\eta, \delta_1, \delta_2\}$, so that every $x$ with $0 < |x-a| < \delta$ has $f(x) = g(x)$ *and* satisfies both (3) and (4). We now combine all three of these facts and invoke the triangle inequality to obtain

$$\begin{aligned} |L - M| &= \big|L - g(x) + g(x) - M\big| \\ &\le \big|L - g(x)\big| + \big|g(x) - M\big| \\ &= \big|f(x) - L\big| + \big|g(x) - M\big| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

As we said above, since this proof shows that $|L - M| < \epsilon$ for every $\epsilon > 0$, we must have $|L - M| = 0$; that is, $L = M$. ∎

If we'd had these theorems from the start, we wouldn't have had to use so much elbow grease in proving Proposition 1. Here's how we could have proceeded.

*Proof of Proposition 1 using Theorems 12 and 13.* Let

$$f(x) = \frac{3x^2 - 2x - 1}{x - 1};$$

recall that we are trying to show that $\lim_{x\to 1} f(x) = 4$. We factor $3x^2 - 2x - 1 = (3x + 1)(x - 1)$ and observe that for any $x \neq 1$,

$$f(x) = \frac{3x^2 - 2x - 1}{x - 1} = 3x + 1.$$

Letting $g(x) = 3x+1$, on any open interval containing 1 we have $f(x) = g(x)$ everywhere except at $x = 1$, so Theorem 13 applies to inform us that $\lim_{x\to 1} f(x) = \lim_{x\to 1} g(x)$. Moreover, $g(x)$ is a polynomial with $x = 1$ in its domain, so

$$\lim_{x\to 1} g(x) = 3 \cdot 1 + 1 = 4$$

by Theorem 12. ∎

Here's another example in which we use the root law.

**Proposition 14.** *We have*

$$\lim_{x\to 0} \frac{\sqrt{x^2 + 4} - 2}{x^2} = \frac{1}{4}.$$

*Proof.* We cannot use the quotient law, since $x^2 \to 0$ as $x \to 0$. If we can massage the expression to obtain a factor of $x^2$ in the numerator, then (in an open interval containing 0 but not at 0 itself), we can cancel the $x^2$ in the denominator and perhaps make some headway. This strategy is facilitated by the observation that

$$\Big(\sqrt{x^2 + 4} - 2\Big)\Big(\sqrt{x^2 - 4} + 2\Big) = \Big(\sqrt{x^2 - 4}\Big)^2 - 2^2 = x^2.$$

Hence we have

$$\begin{aligned}
\lim_{x\to 0} \frac{\sqrt{x^2 + 4} - 2}{x^2} &= \lim_{x\to 0} \frac{\sqrt{x^2 + 4} - 2}{x^2} \cdot \frac{\sqrt{x^2 + 4} + 2}{\sqrt{x^2 + 4} + 2} \\
&= \lim_{x\to 0} \frac{x^2}{x^2\big(\sqrt{x^2 + 4} + 2\big)} \\
&= \lim_{x\to 0} \frac{1}{\sqrt{x^2 + 4} + 2}.
\end{aligned}$$

We no longer have the issue of the limit of the denominator being 0, and can happily apply the algebraic limit laws as well as the direct substitution property:

$$\begin{aligned}\lim_{x\to 0}\frac{1}{\sqrt{x^2+4}+2} &= \frac{1}{\lim_{x\to 0}\left(\sqrt{x^2+4}+2\right)}\\ &= \frac{1}{\lim_{x\to 0}\sqrt{x^2+4}+\lim_{x\to 0}2}\\ &= \frac{1}{\sqrt{\lim_{x\to 0}(x^2+4)}+2}\\ &= \frac{1}{\sqrt{4}+2}\\ &= \frac{1}{4} \quad \blacksquare\end{aligned}$$

**Using one-sided limits.** Recall Theorem 4, which states that a limit exists and equals some $L$ if and only if both one-sided limits exist and equal $L$. This can be used to compute $\lim_{x\to 0}|x|$, even though the algebraic limit laws do not apply to the function $|x|$ directly.

**Proposition 15.** *We have*

$$\lim_{x\to 0}|x| = 0.$$

*Proof.* Towards the invocation of Theorem 4, we calculate the one-sided limits of $|x|$ using the direct substitution rule. When $x > 0$, we have $|x| = x$, so

$$\lim_{x\to 0^+}|x| = \lim_{x\to 0^+} x = 0,$$

since For all $x < 0$ we instead have $|x| = -x$, so

$$\lim_{x\to 0^-}|x| = \lim_{x\to 0^-} -x = 0.$$

Both one-sided limits agree, so we conclude that the limit is 0. $\blacksquare$

**Evaluating limits at infinity.** Recall that most of the algebraic limit laws established earlier in this section apply to limits at infinity. Hence we can combine the root and power laws with Proposition 8 to prove the following useful theorem.

**Theorem 16.** *Let $r > 0$ be a rational number. Then*

$$\lim_{x\to\infty}\frac{1}{x^r} = 0.$$

*If $r > 0$ is such that $x^r$ is defined for negative values of $x$ as well, then*

$$\lim_{x\to-\infty}\frac{1}{x^r} = 0.$$

*Proof.* Let $r > 0$ be a rational number. This means there are positive integers $m$ and $n$ such that $r = m/n$, so we can write

$$\lim_{x\to\infty} \frac{1}{x^r} = \lim_{x\to\infty} \sqrt[n]{\left(\frac{1}{x}\right)^m} = \sqrt[n]{\left(\lim_{x\to\infty} \frac{1}{x}\right)^m} = 0,$$

where the third equality follows from the root and power laws, and the last follows from Proposition 8.

This works for $x \to -\infty$ as well, as long as $x^r$ is defined for negative values of $x$. (If $n$ is even and $m$ is odd, $x^m$ is negative and the $n$th root of a negative is not defined, so it does not makes sense to write $x^r = \sqrt[n]{x^m}$ in this case.) ∎

When studying the behaviour of a function $f(x)$ as $x \to \infty$, we can always multiply the whole expression by $x^r/x^r$ for any $r$, since we are interested in the case where $x$ is large and don't have to worry about the denominator possibly being 0. This is very useful in practice when applying Theorem 16.

Here's a simple example.

16.IX **Proposition 17.** *We have*

$$\lim_{x\to\infty} \frac{5x^2 - 3x + 4}{3x^2 + 10x - 3} = \frac{5}{3}$$

*Proof.* Since we're only interested in the behaviour when $x$ is very large, we can assume $x > 0$, so that $x^2/x^2$ is defined. We divide by this function, obtaining

$$\lim_{x\to\infty} \frac{5x^2 - 3x + 4}{3x^2 + 10x - 3} = \lim_{x\to\infty} \frac{(5x^2 - 3x + 4)/x^2}{(3x^2 + 10x - 3)/x^2} = \lim_{x\to\infty} \frac{5 - 3/x + 4/x^2}{3 + 10/x - 3/x^2}.$$

Invoking the quotient and sum laws now, we take the limits of each term in the numerator and denominator individually, and Theorem 16 applies to give us

$$\lim_{x\to\infty} \frac{5 - 3/x + 4/x^2}{3 + 10/x - 3/x^2} = \lim_{x\to\infty} \frac{5 - 0 + 0}{3 + 0 - 0} = \frac{5}{3}.$$

(We technically used the constant multiple law here too, e.g., when performing the operation $\lim_{x\to\infty} 4/x^2 = 4 \cdot \lim_{x\to\infty} 1/x^2$.) ∎

The limit as $x \to -\infty$ of the function in the above proposition is also $5/3$ (work it out!), but it is not always the case that the limits as $x \to \infty$ and $x \to -\infty$ agree.

**Proposition 18.** *Let*

$$f(x) = \frac{\sqrt{2x^2 + 1}}{5x - 1}.$$

*We have*

*i)* $\lim_{x\to\infty} f(x) = \sqrt{2}/5$;

*ii)* $\lim_{x\to-\infty} f(x) = -\sqrt{2}/5$;
*iii)* $\lim_{x\to(1/5)^+} f(x) = \infty$; *and*
*iv)* $\lim_{x\to(1/5)^-} f(x) = -\infty$.

*Proof.* To prove part (i), we perform the same strategy. The largest power of $x$ in the denominator is just $x$, so we divide top and bottom by $x$, noting that for $x > 0$ one has $x = \sqrt{x^2}$:

$$\begin{aligned}\lim_{x\to\infty} \frac{\left(\sqrt{2x^2+1}\right)/x}{(5x-1)/x} &= \lim_{x\to\infty} \frac{\left(\sqrt{2x^2+1}\right)/\sqrt{x^2}}{5-1/x} \\ &= \lim_{x\to\infty} \frac{\left(\sqrt{2+1/x^2}\right)}{5-1/x} \\ &= \frac{\sqrt{2+0}}{5-0} \\ &= \frac{\sqrt{2}}{5}\end{aligned}$$

It looks like we can do the same thing when $x < 0$, and morally we can, except we have to remember that for negative $x$ one instead has $x = -\sqrt{x^2}$. So

$$\begin{aligned}\lim_{x\to-\infty} \frac{\left(\sqrt{2x^2+1}\right)/x}{(5x-1)/x} &= \lim_{x\to-\infty} \frac{\left(\sqrt{2x^2+1}\right)/\left(-\sqrt{x^2}\right)}{5-1/x} \\ &= \lim_{x\to-\infty} \frac{-\left(\sqrt{2+1/x^2}\right)}{5-1/x} \\ &= \frac{-\sqrt{2+0}}{5-0} \\ &= \frac{-\sqrt{2}}{5},\end{aligned}$$

settling claim (ii). We might as well take a moment here to practise finding infinite limits. The input $x = 1/5$ is outside the domain of $f$, since it makes the denominator 0, so we should investigate the behaviour of this function around this point. Note that when $x = 1/5$ exactly, the numerator is a positive number (since squaring a positive number preserves its positivity, and all the other operations in that expression do as well—if you were extra insecure, you'd calculate the numerator fully and see that it equals $\sqrt{27}/5$, a positive quantity). Hence slightly to the left of $x = 1/5$ and slightly to the right, $f$ is also positive. When $x$ just slightly to the right of $1/5$, the denominator is a very small positive number, and we can make this denominator arbitrarily small by taking $x$ sufficiently close to $x$ from the right. So

$$\lim_{x\to(1/5)^+} = \infty,$$

since the numerator is positive as well. On the other hand, when $x$ is very close to $1/5$ from the left, the denominator is a very small negative number, and again

this can be taken arbitrarily close to 0 by taking $x$ sufficiently close to $x$ from the left. So

$$\lim_{x\to(1/5)^-} = -\infty. \quad \blacksquare$$

The examples we've seen of limits at infinity have been finite, so before moving on, let's see an example where the limit at infinity is infinite.

**Proposition 19.** *We have*

$$\lim_{x\to\infty} x - x^2 = -\infty.$$

*Proof.* We might be tempted to try the difference law here, but since the limits are both infinite we get $\infty - \infty$, which is nonsense. A better thing to do is to factor $x - x^2 = x(1-x)$. When $x$ gets large, $x$ is a large positive number and $(1-x)$ is a large negative number. Hence the product is a (very) large negative number. We conclude that the limit must be $-\infty$. $\blacksquare$

**The squeeze theorem.** We end this section by highlighting the following intuitive theorem, which also goes by the names *sandwich theorem* or *two policemen and a drunk theorem*.

**Theorem 20** (*Squeeze theorem*). *Let $a$ be a number and suppose that for all $x \neq a$ in an open interval containing $a$, we have*

$$f(x) \leq g(x) \leq h(x).$$

*If*

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L,$$

*then $\lim_{x\to a} g(x)$ exists and equals $L$.*

*Proof.* This proof is given in Appendix F of the textbook, but we include it in the notes as well for completeness.

Let $\epsilon > 0$. Since $\lim_{x\to a} f(x) = L$, there is $\delta_1 > 0$ such that every $x$ with $0 < |x-a| < \delta_1$ satisfies

$$\big|f(x) - L\big| < \epsilon. \tag{5}$$

Likewise, there is $\delta_2 > 0$ such that every $x$ with $0 < |x-a| < \delta_2$ satisfies

$$\big|h(x) - L\big| < \epsilon. \tag{6}$$

Let $\eta > 0$ be a number such that for all $x$ with $0 < |x-a| < \eta$, we have $f(x) \leq g(x) \leq h(x)$. Subtracting $L$ from this inequality gives us

$$f(x) - L \leq g(x) - L \leq h(x) - L \tag{7}$$

for all such $x$. Hence, letting $\delta = \min\{\delta_1, \delta_2, \eta\}$, every $x$ with $0 < |x-a| < \delta$ satisfies all three of (5), (6), and (7). Combining these three inequalities yields

$$-\epsilon < f(x) - L \leq g(x) - L \leq h(x) - L < \epsilon,$$

which implies that

$$\big|g(x) - L\big| < \epsilon,$$

as desired. ∎

Near the beginning of the course we encountered the pathological function $\sin(\pi/x)$ and convinced ourselves that it does not possess a limit as $x \to 0$. However, by combining the squeeze theorem with Proposition 15, we can show that a modification of $\sin(\pi/x)$ does in fact approach a limit.

**Proposition 21.** *We have*

$$\lim_{x\to 0} x \sin\Big(\frac{\pi}{x}\Big) = 0.$$

*Proof.* It looks like this follows from the product law, but we cannot actually apply that law because $\lim_{x\to 0} \sin(\pi/x)$ does not exist.

The thing to note is that the function sin is bounded from above by the constant function 1 and bounded from below by the constant function $-1$. In other words, for all real numbers $t$ we have

$$-1 \leq \sin(t) \leq 1.$$

So for all $x > 0$ we have

$$-x \leq x \sin\Big(\frac{\pi}{x}\Big) \leq x$$

and for all $x < 0$ we have

$$x \leq x \sin\Big(\frac{\pi}{x}\Big) \leq -x.$$

In other words, the inequality

$$-|x| \leq x \sin\Big(\frac{\pi}{x}\Big) \leq |x|$$

holds for all $x \neq 0$. By Proposition 15, $\lim_{x\to 0} |x| = 0$, so we also have $\lim_{x\to 0} -|x| = 0$ by the constant multiple law. Hence by the squeeze theorem (with $f(x) = -|x|$, $g(x) = x\sin(\pi/x)$, and $h(x) = |x|$), we have

$$\lim_{x\to 0} x \sin\Big(\frac{\pi}{x}\Big) = 0.$$

which is what we wanted to show. ∎

We will see more applications of the squeeze theorem in the future, for example, when we work out the derivative (whatever that is) of the sine function.

## 4. Continuity

18.IX We have spent the entire course so far discussing the limit of functions $f(x)$ at numbers $a$. We now understand that this is the value (if any) that the outputs $f(x)$ must approach as $x$ is taken closer and closer to $a$. This definition of a limit does not concern itself with $f(a)$ equalling this value that the function approaches, or in fact equalling any value at all. But in many cases, $f(a)$ does equal the limit of $f$ at $a$, and this situation turns out to be pleasant and important enough to warrant significant attention.

We say that a function $f$ is *continuous* at a number $a$ if

i) $f(a)$ is defined (i.e., $a$ is in the domain of $f$);

ii) $\lim_{x\to a} f(x)$ exists; and

iii) $\lim_{x\to a} f(x) = f(a)$.

If $f$ is defined near $a$ (on some interval containing $a$ but not necessarily at $a$ itself) and yet fails to be continuous by violating one or more of these three conditions, then we say $f$ is *discontinuous* at $a$.

Clearly by this definition there are various ways for a function to be discontinuous, and some of these cases have special names.

a) If $\lim_{x\to a} f(x)$ exists but $f(a)$ is either undefined or does not equal this limit, we say that $f$ has a *removable discontinuity* at $a$, because we could eliminate this discontinuity by (re)defining the value of $f$ at the single point $a$. For instance, the function $f(x) = 3x^2 - 2x - 1/(x-1)$ we saw before has a removable discontinuity at $x = 1$.

b) If (ii) fails to hold because $\lim_{x\to a} f(x) = \pm\infty$, then we say $f$ has an *infinite discontinuity* at $a$ (regardless of whether $f(a)$ is defined or not). The function $f(x) = 1/(x-3)^2$ has an infinite discontinuity at $x = 3$.

c) Lastly, if the one-sided limits of $f$ at $a$ both exist, but (ii) fails because they do not agree, then we say $f$ has a *jump discontinuity* at $a$—the function "jumps" from having some value directly to the left of $a$ to having some other value directly to the right of $a$. The function

$$f(x) = \begin{cases} 0, & \text{if } x \le 0; \\ 1, & \text{if } x > 0 \end{cases} \tag{8}$$

exhibits this behaviour at $x = 0$.

The appearance of one-sided limits in the definition of a jump discontinuity suggests that we should have a special name for the general case when $f(a)$ equals a one-sided limit of $f$ at $a$ but perhaps not the other. We say that a function $f$ is *continuous from the right* at a number $a$ if

$$\lim_{x\to a^+} f(x),$$

and *continuous from the left* at $a$ if

$$\lim_{x \to a^-} f(x).$$

For example, the function $f(x)$ appearing in (8) is continuous from the left but not from the right. Of course, a function is continuous at $a$ if and only if it is continuous from the right and from the left at $a$, by Theorem 4.

Since the definition of continuous is given directly in terms of the definition of limit, we can use the algebraic limit laws in order to prove corresponding algebraic laws for continuity.

**Theorem 22.** *Suppose that $f$ and $g$ are continuous at $a$. Then*

*i)* $f + g$;
*ii)* $f - g$;
*iii)* $cf$, *where $c$ is any real number; and*
*iv)* $fg$

*are all continuous at $a$. Furthermore, if $g(a) \neq 0$, then*

*v)* $f/g$

*is also continuous at $a$.*

*Proof.* We just prove (i), leaving the rest to the reader as they are similarly easy now that we have access to the algebraic limit laws.

Let $f$ and $g$ be continuous at $a$. This means that

$$\lim_{x \to a} f(x) = f(a) \qquad \text{and} \qquad \lim_{x \to a} g(x) = g(a).$$

By the sum law for limits, we have

$$\begin{aligned} \lim_{x \to a} (f + g)(x) &= \lim_{x \to a} \big(f(x) + g(x)\big) \\ &= \lim_{x \to a} f(x) + \lim_{x \to a} g(x) \\ &= f(a) + g(a) \\ &= (f + g)(a), \end{aligned}$$

which is what we needed to prove. ∎

So if we know certain functions are continuous, then combining them algebraically in the five ways listed above preserves the property of being continuous. This begs the question of which basic functions are continuous. We already know about one class of examples, because the direct substitution property we learned in the previous section directly asserts that any rational function is continuous at all points in its domain. (Go back and read this theorem and you'll see for yourself!) In particular, since every polynomial is a rational function and the domain of a polynomial is the entire real line, polynomials are continuous everywhere.

The following theorem presents a more exhaustive catalogue of continuous functions.

**Theorem 23.** *If a function $f$ is a*

*i) polynomial (e.g., $3x^5 - 4x + 1$);*
*ii) rational function (e.g., $(4x^2 - 3)/(5x - 1)$);*
*iii) root function (e.g., $\sqrt{x}$, $\sqrt[3]{x}$);*
*iv) trigonometric function (e.g., $\sin x$, $\tan x$);*
*v) inverse trigonometric function (e.g., $\arcsin x$, $\arccos x$);*
*vi) exponential function (e.g., $e^x, 2^x$); or*
*vii) logarithmic function (e.g., $\log_{10} x, \ln x$);*

*then $f$ is continuous at all points in its domain.*

*Proof.* We shall skip the proofs of (iv) through (vii) since giving them rigorously would take far too much time, but as we noted before, (i) and (ii) are simply a restatement of the direct substitution property. We have also already seen (iii), since limit law (xi) from Theorem 11 asserts precisely that root functions are continuous. ∎

For any set $S$ of numbers, we say that $f$ is *continuous on* $S$ if $f$ is continuous at $a$ for all elements $a$ of $S$. In particular, we say that $f$ is continuous on the interval $(c, d)$ if $f$ is continuous at $a$ for all $c < a < d$. (Similar definitions can be made for intervals of the form $(c, d]$, $[c, d)$, and $[c, d]$ by changing $<$ to $\leq$ accordingly.)

Knowing which types of functions are continuous now lets us determine many limits without straining our brain very much at all.

**Proposition 24.** *We have*

$$\lim_{x \to \pi/2} \frac{\sin x - 4\cos x}{3x} = \frac{2}{3\pi}.$$

*Proof.* We expand

$$\lim_{x \to \pi/2} \frac{\sin x - 4\cos x}{3x} = \frac{\lim_{x \to \pi/2} \sin x - 4 \cdot \lim_{x \to \pi/2} \cos x}{\lim_{x \to \pi/2} 3x}$$

by using the algebraic limit laws. In the numerator, we apply the fact that the sine and cosine functions are continuous, and in the denominator we use the direct substitution property of the polynomial $3x$. This yields

$$\lim_{x \to \pi/2} \frac{\sin x - 4\cos x}{3x} = \frac{\sin(\pi/2) + 4\cos(\pi/2)}{3\pi/2} = \frac{1 + 4 \cdot 0}{3\pi/2} = \frac{2}{3\pi}.$$ ∎

Often we wish to combine functions not just by adding or multiplying them, etc., but instead by *composing* them. For instance, the function $\sin(\pi/x)$ is the composition $f \circ g$ where $f(x) = \sin x$ and $g(x) = \pi/x$. For this operation we have the following theorem.

**Theorem 25.** *Suppose that $g(x)$ is continuous at $a$ and that $f(x)$ is continuous at $g(a)$. Then $f \circ g$ is continuous at $a$; in other words,*

$$\lim_{x\to a}(f\circ g)(x) = (f\circ g)(a) = f\big(g(a)\big).$$

*Proof.* This proof, which proceeds directly via the formal definition of the limit, is found in Appendix F of the textbook. ∎

This theorem shows that, erratic though its behaviour may be near the vertical line $x = 0$, the function $\sin(\pi/x)$ is continuous at all $a \neq 0$, since it is the composition of the sine function with a rational function, which is continuous on its domain. We can also use the theorem to compute limits that seem difficult to compute otherwise.

**Proposition 26.** *We have*

$$\lim_{x\to 1}\arccos\left(\frac{1}{\sqrt{x+1}}\right) = \frac{\pi}{4}.$$

*Proof.* First, we observe that $g(x) = 1/\sqrt{x+1}$ is continuous on its domain. To see this, note that by Theorem 25, since $\sqrt{x}$ and $x+1$ are both continuous, their composition $\sqrt{x+1}$ is, and by the quotient law so is $1/\sqrt{x+1}$ at all points except at $a = -1$.

To apply Theorem 25 again with $f(x)$ and $g(x)$, we must check that arccos is continuous at $g(1) = 1/\sqrt{1+1} = 1/\sqrt{2}$. Well, arccos is continuous on its domain, which is the interval $[-1, 1]$, and $1/\sqrt{2}$ falls within this interval. So by Theorem 25, we have

$$\lim_{x\to 1} f\big(g(x)\big) = f\big(g(a)\big) = \arccos\Big(\frac{1}{\sqrt{2}}\Big) = \frac{\pi}{4},$$

which is what we wanted to show. ∎

21.IX **The intermediate value theorem.** The remainder of this section is devoted to a very important theorem. It expresses the intuitive idea that for a continuous function to get from one value to another, it must pass through all values in between.

**Theorem 27** (*Intermediate value theorem*). *Let $f$ be a function that is continuous on some closed interval $[a, b]$ and either*

$$f(a) < c < f(b) \qquad \text{or} \qquad f(b) < c < f(a),$$

*then there is some $x$ in the open interval $(a, b)$ such that $f(x) = c$.*

The proof of this theorem is non-trivial and takes several lectures of a second-year math course to set up properly. In this class we'll simply use it as a black box.

An easy first application is to show that certain polynomials have roots.

**Proposition 28.** *The polynomial*

$$f(x) = 3x^4 - 3x^3 + 4x^2 - 9x + 4$$

*has a root.*

*Proof.* We wish to show that

$$3x^4 - 3x^3 + 4x^2 - 9x + 4 = 0$$

has a solution. There is a quartic formula (akin to the quadratic formula but for polynomials of degree 4), but it is terrifically complicated. One could also try to factorise the polynomial but it is difficult to figure out where to start. But we are just tasked with showing that there *exists* a root; we are not asked to find out what the root is exactly. So we can use the intermediate value theorem.

Note that the constant term of the polynomial is 4, so $f(0) = 4$. This means that at some point, the graph of the function is above the $x$-axis. Now let's try to find an input that makes $f$ negative. An easy thing to try is $f(1)$, since we just have to add up all the coefficients:

$$f(1) = 3 - 3 + 4 - 9 + 4 = -1$$

Since $f$ is a polynomial, it is continuous everywhere and in particular is continuous on the interval $[0, 1]$. By the intermediate value theorem with $a = 0$ and $b = 1$ then, since

$$f(0) = 4 > 0 > -1 = f(1),$$

there must be some $0 < x < 1$ such that $f(x) = 0$. This completes the proof. ∎

Here is a fun "real-world" application.

**Theorem 29** (*Temperature theorem*). *Show that at any given point in time, there is some location in the world that has the same temperature as the opposite point on the globe (assume that temperature varies continuously with position on the globe).*

*Proof.* For any point $x$ on the globe, let $f(x)$ be the temperature at that point. Pick any starting point $p$ and call the opposite point on the globe $-p$. (Strictly speaking, we're now working in 3-dimensional space. Think of the centre of the earth as the origin. Then the negative of any point is just its opposite point on the globe. Don't worry about this point too much. We can still define continuity for functions on 3-dimensional space, and the intermediate value theorem still holds on curves—the equivalent of intervals—in space.)

If $f(p) = f(-p)$, then we're done (the point we sought is just $p$). If $f(p) \neq f(p')$, then we have to be a bit more clever and invoke the intermediate value theorem. Let

$$g(x) = f(x) - f(-x)$$

for all points $x$ on the earth. Since $f$ is continuous, by the limit laws $g$ is continuous as well. For the point $p$ that we chose initially, we have

$$g(p) = f(p) - f(-p) \neq 0.$$

We don't know if $g(p)$ is positive or negative. We just know it's not zero. But now what about $g(-p)$? It turns out that

$$g(-p) = f(-p) - f\big(-(-p)\big) = f(-p) - f(p) = -g(p),$$

so whatever sign $g(p)$ had, $g(-p)$ has the opposite sign.

We're almost there. Draw any line on the surface of the earth (this is actually a curve, since the earth is round) between $p$ and $-p$. The function $g$ is continuous on this curve, so the intermediate theorem applies. At one endpoint of the curve, $-p$, the function $g$ has one sign, and at the other endpoint $g$ has the opposite sign. So by the intermediate value theorem, there must be some point $x$ on this curve with $g(x) = 0$. But what does this mean for $f$? By the original definition of $g$, we have

$$0 = g(x) = f(x) - f(-x),$$

so $f(x) = f(-x)$, and this point $x$ has the same temperature as its opposite point on the globe. ∎

This theorem applies not just to the temperature, but to any continuous function, and we didn't use the whole globe, but just a single circle on it. The general version of Theorem 29 for any function on a sphere in any dimension is called the Borsuk–Ulam theorem; it was first proved in 1933.

## 5. Derivatives

Equipped now with a more robust idea of what it means for a function to approach a value but not necessarily get there, we revisit the rate-of-change problem introduced in the first lecture, which is the broad motivating topic for this course as a whole.

Recall that we were considering a function $f$ and wanted to know the rate at which it is changing near some input $a$ (for now, assume that $f(a)$ is well defined). We convinced ourselves geometrically that this rate of change could be modelled as the slope of a line "tangent" to the graph of $f(a)$ at the point $\big(a, f(a)\big)$. We couldn't rigorously define this line at the time, but it seemed that its slope should be the value of the quotient

$$\frac{f(x+h) - f(x)}{h}$$

"when" $h = 0$, even though we knew that $h$ cannot be zero, being in the denominator.

Now that we are comfortable with limits, however, it is abundantly clear to us that we can overcome this impasse by declaring the slope to be

$$\lim_{h\to 0}\frac{f(a+h)-f(a)}{h},$$

provided this slope is defined. And of course, once we know the slope, we simply remember that this so-called "tangent line" is supposed to touch the function at the point $\big(a, f(a)\big)$, so we already know a point on the line. Knowing just one point as well as the slope determines the line completely. Hence the *tangent line* of a function $f$ at the point $P = \big(a, f(a)\big)$ can be defined to be the line through $P$ with slope

$$m = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h},$$

so long as this limit exists.

Here's an example demonstrating how to find the equation of a tangent line.

**Proposition 30.** *For the function*

$$f(x) = x^3,$$

*the tangent line passing through the point* $\big(-1, f(-1)\big) = (-1,-1)$ *is given by the equation*

$$y = 3x + 2,$$

*or alternatively*

$$3x - y + 2 = 0.$$

*Proof.* To find the slope of the tangent line at $a = -1$, we compute

$$m = \lim_{h\to 0}\frac{f(-1+h)-f(-1)}{h} = \lim_{h\to 0}\frac{(h-1)^3-(-1)^3}{h}.$$

Expanding $(h-1)^3$ and then simplifying gives us

$$m = \lim_{h\to 0}\frac{h^3-3h^2+3h-1-(-1)}{h} = \lim_{h\to 0}\frac{h^3-3h^2+3h}{h},$$

and since we're only interested in values of $h$ near 0 but not *at* 0, we can divide by $h$ on the top and bottom and apply the direct substitution property to get

$$m = \lim_{h\to 0}(h^2-2h+3) = 0-0+3 = 3.$$

So the slope of the line is 3 and the equation of the line must be of the form

$$y = 3x + b$$

for some number $b$. We solve for this $b$ by substituting one known point on the line, namely $(-1,-1)$. We have

$$-1 = 3(-1) + b,$$

whence $b = 2$ and we see that the tangent line can be expressed by the equation

$$3x - y + 2 = 0,$$

as claimed. ∎

23.IX **Secant lines.** Although one of the primary innovations of differential calculus is the ability to derive tangent lines, we take a brief detour to talk about secant lines and what they represent. Suppose $f(x)$ is a function and we graph it on the plane by setting $y = f(x)$. Then $y$ is a quantity that depends on $x$. If $x$ changes from $x_1$ to $x_2$, then the *change in* $x$ (also called the *increment* of $x$) is the quantity

$$\Delta x = x_2 - x_1.$$

The change in $y = f(x)$ that corresponds to this change in $x$ is, similarly, denoted

$$\Delta y = f(x_2) - f(x_1).$$

Then the quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the *average rate of change of* $y$ *with respect to* $x$ over the interval $[x_1, x_2]$. Geometrically, this is the slope of the it *secant line* that passes through the two points $\big(x_1, f(x_1)\big)$ and $\big(x_2, f(x_2)\big)$. We already saw above that the slope of the tangent line is the limit of the slopes of secant lines as the distance between the two points goes to 0. The majority of this course will be about the manifold applications of this definition.

But for some problems, the answer lies directly in the simpler calculation associated with the secant line. Take this one for instance. The fuel usage of a car is often measured in litres per hundred kilometres (so a smaller number means better fuel economy). Suppose that between kilometre 20 and kilometre 320 on Highway 40, your car goes from a full tank (40 litres) down to 19 litres of gasoline in that time. If we wish to know the average rate of change of $g$ with respect to $d$, then we divide

$$\frac{\Delta g}{\Delta d} = \frac{40 - 19}{320 - 20} = \frac{21 \text{ litres}}{300 \text{ kilometres}} = \frac{7 \text{ litres}}{100 \text{ kilometres}}$$

to conclude that the average rate of fuel consumption is 7 L/100km.

If we had a fuller description or a graph of the function of fuel usage with respect to time, then we'd be able to take secant lines cutting the graph at closer

and closer points. In the limit, we'd obtain a tangent line, whose slope indicates the fuel usage at a specific given point, but this idea of *instantaneous* rate of change of fuel with respect to distance has very little real-world meaning, so we omit it.

(Well, that's not true. The instananeous rate of change of fuel usage with respect to distance does have some real-world meaning, although the interpretation is perhaps more amusing than actually useful. A litre equals one thousandth of a cubic metre, and a kilometre is of course a thousand metres. So we see that the unit $\mathrm{L}/100\,\mathrm{km}$ equals $10^{-8}\,\mathrm{m}^2$ or $0.01\,\mathrm{mm}^2$—it measures area! This might be baffling the first time you see it, but the best way to interpret this, borrowing numbers from the previous example, is to imagine all 21 litres of fuel (which is three-dimensional) stretched out over the 300 km stretch of highway as a very long skinny rectangular prism. The average fuel usage, then, is the average cross-sectional area of this rectangular prism—namely 0.07 square millimetres in the example. If we now imagine the amount of fuel used varying smoothly as the driver manages the throttle, then the thickness of this imaginary long rectangular prism of liquid varies along the drive as well, and the instantaneous rate of change of fuel with respect to distance at any point is exactly the cross-sectional area of the rectangular prism at that point.)

**Instantaneous velocity.** A far more useful application of the concept of instantaneous rate of change is when calculating distance travelled over time. This is the notion of *speed* or *velocity*, which we are well acquainted with from a young age.

Let $s(t)$ be the position of some object at time $t$. We deduced earlier that, whatever the instantaneous velocity is at some time $a$, it can be approximated by taking averages

$$\frac{s(a+h)-s(a)}{h},$$

over smaller and smaller time intervals $h$. Now that we know what a limit is, we can simply *define* the *instantaneous velocity* at some time $t=a$ of an object with position function $s(t)$ to be

$$v(a)=\lim_{h\to 0}\frac{s(a+h)-s(a)}{h}. \tag{9}$$

This is best illustrated by an example. Ha Ling Peak in the Rocky Mountains is 2407 metres above sea level, and the town of Canmore below it is at roughly 1400 metres above sea level. (Let's say it's an even 1400 metres just for the sake of the example.) At the top of the peak there is a sign telling you not to throw rocks off the cliff as they could hit someone down below and cause grave injury. Suppose you could throw a rock off Ha Ling Peak with enough horizontal velocity to make it to town. Let's find out what its vertical velocity would be once it reached the ground, ignoring factors such as air drag, etc.

The elevation of the rock after $t$ seconds of free fall is modelled by the function

$$s(t)=2407-4.9t^2,$$

assuming its initial vertical velocity is zero, and it starts falling from 2407 metres above sea level. First, we find out how long it takes for the rock to reach the ground in town, which is at 1400 metres. We need to solve the equation

$$1400 = 2407 - 4.9t^2;$$

this rearranges to

$$t = \sqrt{\frac{1400}{4.9}} \approx 14.3$$

seconds.

So we want to calculate $v(14.3)$, where $v(a)$ is the instantaneous velocity at time $t = a$. First we need to derive a formula for $v$. Plugging our formula for $s$ into the definition (9) of $v$, we solve the resulting limit to obtain

$$\begin{aligned} v(a) &= \lim_{h\to 0} \frac{2407 - 4.9(a+h)^2 - (2407 - 4.9a^2)}{h} \\ &= \lim_{h\to 0} \frac{-4.9a^2 - 9.8ah - h^2 + 4.9a^2}{h} \\ &= \lim_{h\to 0} \frac{-9.8ah - h^2}{h} \\ &= \lim_{h\to 0} (-9.8a - h) \\ &= -9.8a. \end{aligned}$$

The negative sign here simply expresses that the rock will be moving downwards (losing elevation). From here we can simply plug in $a = 14.3$ to conclude that the vertical velocity of the rock right before it hits the ground is

$$v(14.3) = -9.8(14.3) = -140.5$$

metres per second. (Of course in real life this could not happen because the terminal velocity of the rock is likely much lower than this, and you would have to throw the rock at well over 100 metres per second to even reach the town in the first place. Also because you would never do this for fear of landing in Alberta jail.)

**The derivative.** In the situation above, the actual equation of the tangent line was not totally necessary to solve the corresponding motivating problem. We only needed to know the slope of this line. Hence for a given function $f(x)$ we have a special name for the slope

$$\lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \tag{10}$$

of the tangent line at the input $a$; it is called the *derivative* of the function $f$ at $a$, and denoted $f'(a)$. Sometimes it's useful to write $x = a + h$, so that

$$f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}.$$

This is a completely equivalent alternative definition of the derivative which we'll use from time to time, but more often than not, the definition (10) leads to simpler calculations.

Here's an example.

**Proposition 31.** *Let*

$$f(x) = x^2 - x + 1.$$

*We have* $f'(3) = 5$*, and more generally for every number* $a$ *we have*

$$f'(a) = 2a - 1.$$

*Proof.* If we do the derivation for general $a$, we can then just plug in $a = 3$ to get the answer of $2(3) - 1 = 5$. But let's do the case $a = 3$ first just to get a sense of the calculations, then tackle the general case after.

We have

$$\begin{aligned}
f'(3) &= \lim_{h\to 0} \frac{f(3+h) - f(3)}{h} \\
&= \lim_{h\to 0} \frac{(3+h)^2 - (3+h) + 1 - (3^2 - 3 + 1)}{h} \\
&= \lim_{h\to 0} \frac{3^2 + 6h + h^2 - 3 - h + 1 - 3^2 + 3 - 1)}{h} \\
&= \lim_{h\to 0} \frac{5h + h^2}{h} \\
&= \lim_{h\to 0} (5 + h) \\
&= 5 + 0 \\
&= 5,
\end{aligned}$$

and for the general case at $a$, we simply replace 3 with $a$ everywhere to obtain

$$\begin{aligned}
f'(a) &= \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} \\
&= \lim_{h\to 0} \frac{(a+h)^2 - (a+h) + 1 - (a^2 - a + 1)}{h} \\
&= \lim_{h\to 0} \frac{a^2 + 2ah + h^2 - a - h + 1 - a^2 + a - 1)}{h} \\
&= \lim_{h\to 0} \frac{2ah - h + h^2}{h} \\
&= \lim_{h\to 0} (2a - 1 + h) \\
&= 2a - 1
\end{aligned}$$

instead. ∎

25.IX **The derivative as a function.** The second calculation in the previous proposition completely characterised the values of $f'(a)$ for different values of $a$. This, along with the notation $f'$ we employed, suggests that $f'$ is itself a function—plug in different inputs $a$ and you get different outputs. We like to indicate inputs of functions by $x$, so substituting $x$ for $a$ in the definition (10) of the derivative, we obtain

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}, \tag{11}$$

wherever this limit exists. Hence taking the derivative or *differentiating* the function $f$ resulted in another function, $f'$.

Let us calculate the derivatives of several simple functions, to get a sense of this idea.

**Proposition 32.** *The derivative of*

$$f(x) = 2x^3 - x$$

*is*

$$f'(x) = 6x^2 - 1.$$

*Proof.* We simply compute

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{2(x+h)^3 - (x+h) - (2x^3 - x)}{h} \\
&= \lim_{h\to 0} \frac{2(x^3 + 3x^2h + 3xh^2 + h^3) - x - h - 2x^3 + x}{h} \\
&= \lim_{h\to 0} \frac{2x^3 + 6x^2h + 6xh^2 + 2h^3 - h - 2x^3}{h} \\
&= \lim_{h\to 0} \frac{6x^2h + 6xh^2 + 2h^3 - h}{h} \\
&= \lim_{h\to 0} (6x^2 + 6xh + 2h^2 - 1) \\
&= 6x^2 - 1. \quad \blacksquare
\end{aligned}$$

Recall that the derivative $f'(x)$ is only defined to exist at points $x$ for which the limit (32) exists. This sometimes means that the domain of a derivative is smaller than the domain of the original function.

**Proposition 33.** *The function*

$$f(x) = \sqrt{x}$$

*is defined on* $[0, \infty)$*, but its derivative, which is given by the formula*

$$f'(x) = \frac{1}{2\sqrt{x}},$$

*is only defined on* $(0, \infty)$*.*

*Proof.* It is clear that $\sqrt{x}$ is well defined for all $x \geq 0$. We now calculate

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h \to 0} \frac{x + h - x}{h\big(\sqrt{x+h} + \sqrt{x}\big)} \\ &= \lim_{h \to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}}, \end{aligned}$$

as claimed. But we see that this is undefined for $x = 0$ since we cannot have 0 in the denominator, so we conclude that 0 is not in the domain of $f'$, which proves the final assertion in the proposition. ∎

**Notes on notation.** Suppose we use the notation $y = f(x)$ to show that the variable $y$ depends on the variable $x$. Then some common alternative ways to denote the derivative $f'(x)$ are

$$f'(x) = \frac{dy}{dx} = \frac{d}{dx} f(x) = y' = \frac{df}{dx} = Df(x) = D_x f(x) = \dot{y},$$

sorted from left to right in rough descending order of how often you'll see these notations used in these notes.

The reason that there are so many different ways of writing down the derivative symbolically is largely historical; the theory of differential calculus was built up slowly over centuries, and the notations mathematicians came up with along the way were well suited for different contexts and purposes. Differential calculus was first invented in the late 1600s by Isaac Newton and Gottfried Wilhelm Leibniz independently (Leibniz's account was published first, but Newton's private notes predated those of Leibniz by several years). Leibniz used the notation $dy/dx$, while Newton favoured the "flyspeck" notation $\dot{y}$. We will not use the latter notation very much but it appears often in physics literature. The *prime notation* $f'(x)$ is probably the most popular notation in modern times. It was first used by Leonhard Euler but popularised by Joseph-Louis Lagrange, and first appeared in print in 1749.

Returning to the Leibniz notation $dy/dx$, we stress that this should not be regarded as a ratio; it is simply another way of writing $f'(x)$. It is reminiscent of the expression $\Delta y/\Delta x$ for the rate of change, which is apt because

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.$$

The symbol $d/dx$ is called the *differentiation operator*, because you can apply it to a function $f$ to obtain its derivative $(df/dx)$. (Recall that the operation of taking a derivative is called differentiation.) The term *operator* refers to a special type of function that takes as input a function and outputs another function, which is what $d/dx$ does, in an abstract sense. When we evaluate the function $f'$ at some number $a$, it is easy to write $f'(a)$ for this value. It is a little more cumbersome to write this number if we're using Leibniz notation; we write instead

$$\left.\frac{dy}{dx}\right|_{x=a} \qquad \text{or} \qquad \left.\frac{dy}{dx}\right]_{x=a}.$$

**Differentiability.** A function $f$ is said to be *differentiable at some number $a$* if $f'(a)$ (which is defined in terms of a limit) exists. It is said to be *differentiable* on a set $S$ if $f$ is differentiable at every number in $S$. (In practice, the set $S$ is most often taken to be an open interval $(a, b)$, an interval of the form $(a, \infty)$ or $(-\infty, b)$, or else the whole real line. If it's differentiably on the whole real line, we might say $f$ is *differentiable everywhere*.)

We already saw in Proposition 33 that the function $\sqrt{x}$, though it is defined on $[0, \infty)$, is only differentiable on $(0, \infty)$. We now proceed to consider a couple more examples.

**Proposition 34.** *The function*

$$f(x) = |x|$$

*is not differentiable at $x = 0$. However, it is differentiable at every other real number.*

*Proof.* We wish to calculate

$$f'(0) = \lim_{h \to 0} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0} \frac{|h|}{h}.$$

But we have encountered this limit on the right-hand side before! It was the subject of Proposition 3, and our very first example of a limit that does not exist. Thus $f'(0)$ does not exist.

To prove the second assertion, assume that $x \neq 0$ and show that $f'(x)$ does exist. This is a simple exercise and we leave it to the reader [*Hint:* Split up the proof into one case where $x > 0$ and the other case where $x < 0$.] ∎

Graphically, the function $|x|$ does not have a derivative at 0 because the slope of the tangent line near 0 jumps drastically from being $-1$ for negative $x$ to 1 for positive $x$. Anytime you see such a "kink" in a graph, you'll know the graph is not differentiable there.

But that is not the only way a function can fail to be differentiable. Consider the function

$$f(x) = \begin{cases} \sqrt{x}, & \text{if } x \geq 0; \\ -\sqrt{-x}, & \text{if } x < 0. \end{cases}$$

This function is continuous everywhere and if you graph it you'll see that the picture is "smooth" everywhere; there are no kinks. And yet we saw in Proposition 33 that its derivative did not exist at 0 because the limit defining the derivative approaches infinity at $a = 0$. (This was strictly speaking only for the positive half of $f$, but you can convince yourself that the negative half does the same thing near 0 as well.) In other words, this example is not differentiable at 0 because its *tangent line is vertical* there.

There is an important point here to be made about the relationship between differentiability and continuity. Both $|x|$ and the function $f$ from the previous paragraph are continuous at 0 but not differentiable there. Therefore continuity at a point does not guarantee differentiability at that point. However, the opposite implication *does* hold.

**Theorem 35.** *Suppose $f$ is a function and $a$ is a number such that $f(a)$ is defined. If $f$ is differentiable at $a$, then it is continuous at $a$.*

*Proof.* Assume that $f'(a)$ exists; that is, the limit

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

equals some well defined number. Then we have

$$\begin{aligned} \lim_{x \to a} f(x) &= \lim_{x \to a} f(x) - f(a) + f(a) \\ &= \lim_{x \to a} \left( \frac{f(x) - f(a)}{x - a} \cdot (x - a) \right) + f(a) \\ &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \to a} (x - a) + f(a) \\ &= f'(a) \cdot 0 + f(a) \\ &= f(a), \end{aligned}$$

which is exactly the criterion for $f$ to be continuous at $a$. ∎

Recall the function $\sin(\pi/x)$, which does not have a limit as $x \to 0$. Using the squeeze theorem, we saw that if we modify the function slightly by multiplying $x$, the resulting function $x \sin(\pi/x)$, while still not *defined* at 0, does go to 0 as $x \to 0$. So if we artifically adjoin the point $(0, 0)$ to this graph, then the resulting functionl is continuous. However, it is not differentiable, as we now proceed to show.

**Proposition 36.** *The function*

$$f(x) = \begin{cases} x\sin(\pi/x), & \text{if } x \neq 0; \\ 0, & \text{if } x = 0 \end{cases}$$

*is not differentiable at* 0.

*Proof.* We compute

$$\begin{aligned} f'(0) &= \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h\to 0} \frac{(0+h)\sin\big(\pi/(0+h)\big) - 0}{h} \\ &= \lim_{h\to 0} \frac{h\sin(\pi/h)}{h} \\ &= \lim_{h\to 0} \sin\Big(\frac{\pi}{h}\Big), \end{aligned}$$

and we have already seen that this limit does not exist. ∎

To illustrate that the matter of differentiability is fairly subtle, consider that if we multiply $\sin(\pi/x)$ by $x^2$ instead of $x$ (and artificially adjoin $(0,0)$ as before, then the result *is* differentiable at 0.

**Proposition 37.** *The function*

$$f(x) = \begin{cases} x^2\sin(\pi/x), & \text{if } x \neq 0; \\ 0, & \text{if } x = 0 \end{cases}$$

*is both continuous and differentiable at* 0.

*Proof.* To prove that $f(x)$ is continuous at 0, we need to show that

$$\lim_{x\to 0} x^2 \sin\Big(\frac{\pi}{x}\Big) = 0.$$

This follows from the squeeze theorem and the fact that $-x^2 \leq f(x) \leq x^2$ for all numbers $x$. (Fill in the details here as an exercise.)

To show that $f$ is differentiable at 0, we just need to show that $f'(0)$ exists. So we compute

$$\begin{aligned} f'(0) &= \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h\to 0} \frac{(0+h)^2\sin\big(\pi/(0+h)\big) - 0}{h} \\ &= \lim_{h\to 0} \frac{h^2\sin(\pi/h)}{h} \\ &= \lim_{h\to 0} h\sin\Big(\frac{\pi}{h}\Big), \end{aligned}$$

but this last limit equals 0, by Proposition 21. ∎

**Higher derivatives.** We said earlier that the operator $d/dx$ takes in a function and spits out another function (which may have a different domain than the original one). It is natural, then, to consider applying the operator again to the resulting function. If the original function is $f$, then its derivative is denoted by $df/dx$ or $f'$, and taking the derivative again gives us *second derivative* of $f$, denoted by

$$\frac{d^2}{dx^2}f(x) \qquad \text{or} \qquad f''(x).$$

If we write $y = f(x)$, which we often do when considering the graph of the function $f$, then we can also denote the second derivative by $d^2y/dx^2$ or $y''$.

We can keep repeating the process, obtaining the third derivative, fourth derivative, and so on. We denote the $n$th derivative of $f$ by

$$\frac{d^n}{dx^n}f(x) \qquad \text{or} \qquad f^{(n)}(x),$$

because writing the $'$ symbol $n$ times can become tedious (though it's not uncommon to see $f'''(x)$ instead of $f^{(3)}(x)$).

There are many uses for higher derivatives, which we'll see as the course progresses. At this juncture we'll just note one function whose second derivative has a very intuitive interpretation. This is the position function $s(t)$ of a moving object with respect to time. We already convinced ourselves earlier that its derivative $s'(t)$ describes the object's velocity with respect to time, since velocity is the rate of change of position. Taking another derivative tells us how much the velocity changes with respect to time, so $s''(t)$ is the object's *acceleration*.

For instance, the elevation of the rock in our Ha Ling Peak example was described by the function

$$s(t) = 2407 - 4.9t^2,$$

and we calculated its velocity function

$$v(t) = s'(t) = -9.8t.$$

Taking the derivative once more (we'll spare the limit calculations, because soon we'll develop strategies to compute derivatives very quickly) gives us the acceleration

$$a(t) = s''(t) = -9.8,$$

which is just a constant, namely the constant of acceleration due to gravity.

This discussion concludes the section we've devoted to defining the derivative and various other notions pertaining to it. It also concludes the first third of this course, which introduced a number of foundational topics, namely limits, continuity, and differentiation. In the next third of the course, we build up a toolbox of techniques that will allow us to quickly calculate derivatives of a large class of functions that are commonly used in science and engineering.

# II. TECHNIQUES

*Sit* a *quantitas data constans,*
*erit* da *æqualis* 0,
*et* d$\overline{\text{ax}}$ *æqualis* adx*:*
*si sit* y *æqualis* v
*(seu ordinata quævis curvæ* Y*,*
*æqualis cuivis ordinatæ respondenti* V*)*
*erit* dy *æqualis* dv.

— GOTTFRIED WILHELM LEIBNIZ, *Nova methodus pro maximis et minimis* (1684)

## 6. Algebraic rules for differentiation

28.IX Early in our study of limits we proved simple algebraic laws like $\lim(f+g) = \lim f + \lim g$. The goal of this section is to repeat this exercise for derivatives, though as we shall see, some of the algebraic rules for differentiation are not as simple as their counterparts in the limit world.

We begin with the easy stuff. A constant function has a completely flat graph; it does not change with respect to its input. Hence its derivative should be 0 everywhere. Let us prove this simple fact.

**Theorem 38.** *Let $c$ be any number and consider it to be a constant function of $x$ (no matter what number $x$ it is given, the function just outputs the number $c$). Then*

$$\frac{d}{dx}c = 0.$$

*Proof.* Let $f(x) = c$ for all $x$. For any number $x$,

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{c-c}{h} = 0. \quad \blacksquare$$

Next, we record the sum, difference, and constant multiple rules for differentiation, which follow directly from the corresponding limit laws.

**Theorem 39** (*Sum, difference, and constant multiple rules*). *Suppose that $f$ and $g$ are both differentiable at some number $x$. Then the function $f+g$ and $f-g$ are also differentiable at $x$, and we have*

$$(f+g)'(x) = f'(x)+g'(x) \qquad \text{as well as} \qquad (f-g)'(x) = f'(x)-g'(x).$$

*Furthermore, for any constant $c$, the function $cf$ is differentiable at $x$, and*

$$(cf)'(x) = cf'(x).$$

*Proof.* For the sum rule, we simply expand

$$\begin{aligned}(f+g)'(x) &= \lim_{h\to 0}\frac{(f+g)(x+h)-(f+g)(x)}{h}\\ &= \lim_{h\to 0}\frac{f(x+h)+g(x+h)-f(x)-g(x)}{h}\\ &= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} + \lim_{h\to 0}\frac{g(x+h)-g(x)}{h}\\ &= f'(x)+g'(x),\end{aligned}$$

where the third equality is a consequence of the sum law for limits. The difference and constant multiple rules are proved similarly. Try them as an exercise. $\blacksquare$

The sum law in particular can be used to prove the power rule for monomials of the form $x^n$, where $n$ is a positive integer.

**Theorem 40** (*Power rule*). *Let $n \geq 1$ be an integer and let*

$$f(x) = x^n.$$

*Then for any number $a$, we have*

$$f'(a) = na^{n-1}.$$

*Proof.* This proof will make use of the identity

$$x^n - a^n = (x-a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-3}x^2 + a^{n-2}x + a^{n-1}).$$

It can be verified by simply multiplying out the right-hand side. Now we expand

$$\begin{aligned} f'(a) &= \lim_{x \to a} \frac{x^n - a^n}{x - a} \\ &= \lim_{x \to a} \big(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \cdots + a^{n-3}x^2 + a^{n-2}x + a^{n-1}\big). \end{aligned}$$

Note that there are exactly $n$ terms in the limit, and in each term, the power of $a$ and the power of $x$ together add up to $n-1$. So by applying the sum law for limits to distribute the limit over the sum before taking $x \to a$ in each term, we obtain

$$f'(a) = na^{n-1},$$

which is what we wanted. ∎

Using the theorems we've proved so far in this section, we can very easily recompute the derivative of the function $x^2 - x + 1$ that appeared in Proposition 31. We simply have

$$\frac{d}{dx}(x^2 + x + 1) = \frac{d}{dx}x^2 - \frac{d}{dx}x + \frac{d}{dx}1 = 2x - 1,$$

which we proved with greater difficulty earlier. Other polynomials are dealt with similarly. Let's see one more example.

Suppose we want to find all points on the curve

$$y = -\frac{x^3}{3} + 9x + 5$$

at which the tangent line is parallel to the line $y = 5x$ (if any such points exist). In other words, where does this curve have a tangent line with a slope of 5? We compute the derivative

$$\frac{dy}{dx} = 9 - x^2 = (3 - x)(3 + x).$$

From here it's easy to see that this equals 5 when $x = 2$ or $x = -2$. (Alternatively you could solve for $9 - x^2 = 5$, which is equivalent to $x^2 = 4$, which has the same solutions.) To find the exact points on the curve, we now plug these points into the original function. When $x = 2$, we have

$$y = -\frac{2^3}{3} + 9 \cdot 2 + 5 = 23 - \frac{8}{3} = \frac{61}{3},$$

and when $x = -2$ we have

$$y = -\frac{(-2)^3}{3} + 9 \cdot (-2) + 5 = \frac{8}{3} - 13 = \frac{-31}{3}.$$

We conclude that the two points on the curve $y = -x^3/3 + 9x + 5$ at which the tangent line has slope equal to 5 are $(2, 61/3)$ and $(-2, -31/3)$.

Returning to the formula given for the power rule, you may have noticed something interesting. In Proposition 33, we showed that

$$\frac{d}{dx} x^{1/2} = \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}} = \frac{1}{2} x^{-1/2}.$$

One might speculate, then, that the power rule holds even when $n$ is not a positive integer. This is in fact true, but we don't have the tools to prove it yet, so we defer it until later.

**The product and quotient rules.** The next algebraic rules to consider are multiplication and division. Because $\lim fg = (\lim f)(\lim g)$ and $\lim f/g = (\lim f)/(\lim g)$ wherever these limits (and the division) make sense, you might guess that we have something like $(fg)' = f' \cdot g'$. This turns out to be off the mark.

To derive the correct answer, you need to draw a rectangle. Imagine $f$ and $g$ as quantities depending on some independent variable $x$. To make your life easy, assume both $f$ and $g$ are positive. Draw a rectangle whose length is $f$ and whose height is $g$, so the area of this rectangle is exactly the quantity $fg$ we want to study. In particular, we want to know how much its area changes if you perturb $x$ very slightly. Suppose that if you increase $x$ very slightly, the value of $f$ increases very slightly and the value of $g$ increases very slightly as well. Call these increases $\Delta f$ and $\Delta g$, respectively. (In general, $f$ could actually decrease, or $g$ could decrease, or both, but this is just a sketch for the sake of intuition, so ignore these possibilities for now.) What has happened to the area of the rectangle; in other words, what is $\Delta(fg)$? Well, three new rectangles have been added to it whose areas are $f \cdot \Delta g$, $g \cdot \Delta f$, and $(\Delta f)(\Delta g)$. Leibniz reasoned that the area $(\Delta f)(\Delta g)$, because it is "infinitely smaller" than the other terms, can simply be ignored. Hence he claimed that

$$\Delta(fg) = f\Delta g + g\Delta f,$$

whence

$$\frac{\Delta(fg)}{\Delta x} = f\frac{\Delta g}{\Delta x} + g\frac{\Delta f}{\Delta x},$$

and since $d(fg)/dx$ is the limit of $\Delta(fg)/\Delta x$ as $\Delta x \to 0$, we can simply replace all instances of $\Delta$ with $d$ above to obtain

$$\frac{d}{dx}(fg) = f\frac{dg}{dx} + g\frac{df}{dx}.$$

Leibniz was correct, though his "proof" does not hold up to modern standards. (If we are throwing away the quantity $(\Delta f)(\Delta g)$ simply because its area is "infinitely small" as $\Delta x \to 0$, why not throw away the other thin rectangles too?—these areas also go to 0.) Let us now restate the product rule in alternative notation (using $f'$ and $g'$) and also give a rigorous proof with our modern definition of derivative.

**Theorem 41** (*Product rule*). *If $f$ and $g$ are both differentiable at $x$, then the product $fg$ is also differentiable at $x$, and*

$$(fg)'(x) = f(x)g'(x) + g(x)f'(x).$$

*Proof.* There is only one step in this proof that is not entirely obvious. First we expand

$$\begin{aligned}(fg)'(x) &= \lim_{h\to 0} \frac{(fg)(x+h) - (fg)(x)}{h} \\ &= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}\end{aligned}$$

The nonobvious thing to do is to add $-g(x)f(x+h) + g(x)f(x+h)$ into the numerator. This will help us factor things out and make the formulas for $f'(x)$ and $g'(x)$ both appear. Picking up where we left off, we have

$$\begin{aligned}(fg)'(x) &= \lim_{h\to 0} \frac{f(x+h)g(x+h) - g(x)f(x+h) + g(x)f(x+h) - f(x)g(x)}{h} \\ &= \lim_{h\to 0} \frac{f(x+h)\big(g(x+h) - g(x)\big) + g(x)\big(f(x+h) - f(x)\big)}{h} \\ &= \Big(\lim_{h\to 0} f(x+h)\Big) \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} + g(x) \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \Big(\lim_{h\to 0} f(x+h)\Big) g'(x) + g(x)f'(x).\end{aligned} \tag{12}$$

The last thing to deal with is the limit $\lim_{h\to 0} f(x+h)$. Since $f$ is differentiable at $x$, it is also continuous at $x$ by Theorem 35. So

$$\lim_{h\to 0} f(x+h) = f(x),$$

and substituting this into (12) proves the theorem. ∎

30.IX Next we set our sights on proving a quotient rule, that is, a formula for $(f/g)'$. Because we can write a quotient $f/g$ as $f \cdot (1/g)$, supposing we can figure out what the derivative of $1/g$ is, we can use the product rule to compute what the quotient rule should be.

The formula for the derivative of the reciprocal of a function is more complicated than you might expect, though if you have taken calculus before in high school and know the derivative of $1/x$ as well as the chain rule, then you should understand why this formula makes sense. If this is your first time taking calculus, don't worry, we'll learn those other things soon, and you can revisit this at a later stage. For now, let's just get on with the theorem and its proof.

**Theorem 42.** *Suppose that $g$ is differentiable at $x$ and that $g(x) \neq 0$, so that $1/g(x)$ is defined. Then the function $1/g$ is differentiable at $x$ as well, and*

$$\left(\frac{1}{g}\right)'(x) = \frac{-g'(x)}{g(x)^2}.$$

*Proof.* Though the formula is a little bit gruesome, the proof is not. Expanding, putting everything over a common denominator, and then rearranging yields

$$\begin{aligned}\left(\frac{1}{g}\right)'(x) &= \lim_{h\to 0} \frac{1/g(x+h) - 1/g(x)}{h} \\ &= \lim_{h\to 0} \frac{g(x) - g(x+h)}{hg(x+h)g(x)} \\ &= -\lim_{h\to 0} \frac{g(x+h) - g(x)}{h} \cdot \lim_{h\to 0} \frac{1}{g(x+h)g(x)} \\ &= -g'(x) \cdot \lim_{h\to 0} \frac{1}{g(x+h)g(x)}.\end{aligned} \tag{13}$$

From here we once again use Theorem 35 to see that $g$ is continuous at $x$, so that

$$\lim_{h\to 0} \frac{1}{g(x+h)g(x)} = \frac{1}{g(x)^2},$$

and substituting this fact into (13) now yields the claimed identity. ∎

Now that we have the reciprocal rule, a general quotient rule for functions $f$ and $g$ can be derived simply by applying the product rule with the functions $f$ and $1/g$.

**Theorem 43** (*Quotient rule*). *Suppose that $f$ and $g$ are differentiable at $x$, and assume that $g(x) \neq 0$, so that $f/g$ is defined at $x$. Then $f/g$ is differentiable at $x$, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}.$$

*Proof.* Since $f/g = f \cdot (1/g)$, we use the product rule to obtain

$$\left(f \cdot \frac{1}{g}\right)'(x) = f(x)\left(\frac{1}{g}\right)'(x) + \left(\frac{1}{g}\right)(x)f'(x).$$

By Theorem 42, we have

$$\left(\frac{1}{g}\right)'(x) = \frac{-g'(x)}{g(x)^2};$$

substituting this into (14) gives us

$$\begin{aligned}\left(f \cdot \frac{1}{g}\right)'(x) &= f(x)\left(\frac{-g'(x)}{g(x)^2}\right) + \frac{f'(x)}{g(x)} \\ &= \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{g(x)^2} \\ &= \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2},\end{aligned}$$

which is exactly what we wanted. ∎

We are now able to find derivatives of any rational function. Here's an example. Suppose we are asked to find the points on the curve

$$y = \frac{x}{x^2+1}$$

at which the tangent line to the curve is is horizontal, i.e., has slope equal to 0. Since $x^2$ is always nonnegative, the denominator here is always nonzero, so $y$ is defined for all values of $x$. By the quotient rule, the derivative of $y$ is

$$y' = \frac{(x^2+1) - x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}.$$

This equals 0 when $x^2 = 1$, or when $x$ equals either 1 or $-1$. When $x = 1$, we have

$$y = \frac{1}{1^2+1} = \frac{1}{2},$$

and when $x = -1$, we have

$$y = \frac{-1}{(-1)^2+1} = \frac{-1}{2},$$

so we see that the two points on the curve $y = x/(x^2+1)$ whose tangent lines are horizontal are $(1, 1/2)$ and $(-1, -1/2)$.

## 7. Derivatives of the exponential and trigonometric functions

We now have a great number of rules that allow us to find the derivatives of algebraic expressions like $f + g$ and $fg$, assuming we know the derivatives of $f$ and $g$. But our inventory of basic functions whose derivative we know is still rather meagre. We only know how to deal with polynomials and rational functions, and adding, subtracting, multiplying and dividing polynomials or rational

functions will only give us more rational functions. So we'll have to go back to basics in order to build up our stockpile of functions.

The first class of functions we look at is exponential functions like $2^x$ or $10^x$. More generally, let $b$ be any number; we shall take it for granted that the quantity $b^x$ varies smoothly as one changes the inputs $x$. So we can consider the function

$$f(x) = b^x$$

and ask how it changes as $x$ varies. Using the formula $b^{x+y} = b^x b^y$, we compute

$$f'(x) = \lim_{h\to 0} \frac{b^{x+h} - b^x}{h} = \lim_{h\to 0} \frac{b^x b^h - b^x}{h} = \lim_{h\to 0} \frac{b^x(b^h - 1)}{h} = b^x \lim_{h\to 0} \frac{b^h - 1}{h}.$$

The limit on the right-hand side doesn't depend on $x$ at all, and we see that the original function $f(x) = b^x$ has reappeared on the right-hand side. Plugging in $x = 0$, we obtain the identity

$$\frac{d}{dx} b^x \bigg|_{x=0} = f'(0) = \lim_{h\to 0} \frac{b^h - 1}{h},$$

which tells us that

$$\frac{d}{dx} b^x = b^x \cdot \frac{d}{dx} b^x \bigg|_{x=0}, \tag{14}$$

assuming the derivative exists. This means that the derivative of an exponental function is proportional to the function itself; the slope of the tangent line is some constant (namely the derivative of the function at zero) times the function's height at that point.

In light of (14), it is natural to wonder if there is some base $b$ such that the derivative

$$\lim_{h\to 0} \frac{b^h - 1}{h}$$

equals 1; if this is true for some $b$, then the derivative of $b^x$ would be $b^x$ again, which seems like a pleasant thing to have. To see that such a base does exist, note that if you take $b = 2$ and estimate the resulting limit (44) by taking smaller and smaller values of $h$, the limit seems to be something less than 1 (in fact, it is roughly 0.693). Likewise, if you take $b = 3$ and do the same thing, the limit seems to be larger than 1 (in fact, it is roughly 1.099). So by the intermediate value theorem, there must be some number between 2 and 3, call it $e$, with

$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1.$$

By taking better approximations one can compute the value

$$e \approx 2.71828\ldots$$

with better precision, but the numerical value of $e$ doesn't come up in this class very often. A more important observation to record at this point is that this number $e$ satisfies

$$\frac{d}{dx}e^x = e^x.$$

We'll prove more facts about $e$ and its associated exponential function $e^x$ (which is so important that we call it *the* exponential function) as the course progresses. We sometimes write $\exp(x)$ instead of $e^x$, so that we can talk about the *function* exp and use the "prime" notation more effectively. (We can write, for instance, $\exp' = \exp$.)

02.X For now let's see a more complicated example involving exponentials. If nothing else, this will help us practise using the product and quotient rules from earlier.

**Proposition 44.** *Let*

$$f(x) = \frac{e^{2x} - 1}{2e^x},$$

*which is defined for all numbers $x$. We have*

$$f'(x) = \frac{e^{2x} + 1}{2e^x}.$$

*Proof.* We would like to use the quotient rule, but first we should deal with the fact that there is an expression in the formula for $f(x)$ we don't immediately know the derivative of, namely $e^{2x}$. We'll learn how to properly deal with function compositions such as these soon, but for now, just note that $e^{2x} = e^x e^x$ and invoke the product rule to obtain

$$\frac{d}{dx}e^{2x} = \frac{d}{dx}(e^x e^x) = e^x e^x + e^x e^x = 2e^{2x}. \tag{15}$$

Now we're in a better position to invoke the quotient rule on $f$ itself. We compute

$$\begin{aligned} f'(x) &= \frac{2e^x \cdot (d/dx)(e^{2x} - 1) - (e^{2x} - 1)(d/dx)(2e^x)}{(2e^x)^2} \\ &= \frac{2e^x \cdot 2e^{2x} - (e^{2x} - 1)2e^x}{4e^{2x}} \\ &= \frac{2e^x(2e^{2x} - e^{2x} + 1)}{4e^{2x}} \\ &= \frac{e^{2x} + 1}{2e^x}, \end{aligned}$$

which is exactly the claim. ∎

So the derivative $f'$ of $f(x) = (e^{2x} - 1)/2e^x$ admits an expression that is very similar to the formula for $f$ itself. If you have some spare time and want to practise the quotient rule some more, try finding $f''(x)$. You might notice a

pattern that will make it (very) easy to determine $f^{(3)}$ and so on as well. We'll revisit these functions at a later date.

***Formally defining the exponential.** This small subsection is intended for the reader who is unhappy with the fact that we have swept the continuity of $b^x$ under the rug. The problem with trying to prove such a thing now is that we don't yet have a proper definition of what the expression $b^x$ means when $x$ is not a rational number. If $x$ is rational—say $x = p/q$ for positive integers $p$ and $q$—then $b^x = \sqrt[q]{b^p}$, and $b^p$ is simply the product of $b$ with itself $p$ times. But this breaks apart if $x$ is irrational. For example, if $x = \pi$, what does it mean to multiply $x$ by itself "$\pi$ times?" The reason we cannot formally resolve this yet is that a precise definition of exponentials requires the concept of the integral, which is only introduced in MATH 141. Fortunately, most of you will take this next semester, but if you have never seen integrals before, the following paragraph will be gibberish to you.

To define $b^x$ for general real values of $b$ and $x$, we start by defining the (natural) logarithm of any $x > 0$ as the value of the definite integral

$$\ln x = \int_1^x \frac{1}{t}\, dt.$$

It can be shown that this is continuous, and there is a unique number $x$ that makes this integral $x$. We call this number $e$, and if we define exp as the inverse of ln, which exists because ln as we have defined it is one-to-one, then we have $\exp(1) = e$. (This is the same $e$ that we defined earlier in the section.) From here, one can prove that

$$\exp(x) = \big(\exp(1)\big)^x = e^x,$$

and finally we *define*

$$b^x = \exp\big(\ln(b^x)\big) = e^{x \ln b}.$$

By this definition, it is clear that this is a continuous function of $x$, since ln is and therefore its inverse exp must be. From here you can rederive all the facts about exponentials and logarithms that you already know.

**Trigonometric derivatives.** So much for exponentials. The rest of this section is devoted to finding formulas for the derivatives of the six basic trigonometric functions

$$\sin, \quad \cos, \quad \tan, \quad \csc, \quad \sec, \quad \text{and} \quad \cot.$$

Because the latter four functions can all be expressed as quotients involving sin and cos, we only need to differentiate sin and cos from scratch; the rest of the formulas will then follow from the quotient rule.

Differentiating the sine and cosine functions will require us to recall some important facts from trigonometry class. First, note that sin is odd and cos is even; in other words,

$$\sin(-\theta) = -\sin\theta \qquad \text{and} \qquad \cos(-\theta) = \cos\theta.$$

Next, by drawing a unit circle and invoking the Pythagorean theorem, we see that

$$\sin^2\theta + \cos^2\theta = 1$$

for all $\theta$. It is a bit harder to prove the *addition formulas*

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha \tag{16}$$

and

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta; \tag{17}$$

if you want to practise your trigonometric skills, these facts are proved in Exercises 89 to 91 of Appendix D of the book. (In those exercises, you actually end up proving

$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha$$

instead of (16), but from this subtraction formula one can recover (16) by substituting $-\beta$ instead of $\beta$, and remembering that the sine function is odd.)

In addition to the aforementioned trigonometric propositions, we shall also need the following two special limits.

**Lemma 45.** *We have*

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1 \tag{18}$$

*and*

$$\lim_{\theta\to 0}\frac{\cos\theta - 1}{\theta} = 0. \tag{19}$$

(A *lemma* is a special name for a proposition that is not seen as overly important in and of itself, but is a vital step in the proof of a more useful theorem.)

*Proof.* We prove these statements in order. First, we calculate the right-hand limit $\lim_{\theta\to 0^+}\sin\theta/\theta$. We shall do this using the squeeze theorem. (This proof is significantly easier if you have access to a piece of paper on which to draw. Alternatively, refer to Figure 6 in Section 3.3 of the textbook.)

Assume that $0 < \theta < \pi/2$, which we can do since we want to take $\theta$ to 0. Draw a line segment $AC$ along the $x$-axis of length 1, and draw a line segment of length 1 $AB$ above it so that the angle $BAC$ is $\theta$. Enclosing this angle with an arc of a circle of radius 1, we now have a slice of a circle. To calculate the area of this slice, we note that the arc of the circle has length $\theta$, so its length is $\theta/(2\pi)$ proportion of the full circumference of the circle. Thus the area of the slice must be $\theta/(2\pi)$ times the area of a full circle with radius 1, which is $\pi$. In other words, the area of the slice is $\theta/2$.

Now draw a line from $B$ down to the $x$-axis so that it intersects $AC$ perpendicularly. Call this intersection point $D$. The right triangle $ABD$ lies entirely within the slice of the circle we considered above. The area of this right triangle is easy to calculate, since it is a right triangle with hypotenuse 1 and angle $\theta$.

Thus $AD$ has length $\cos\theta$ and $BD$ has length $\sin\theta$, and the triangle $ABD$ has area $(\sin\theta\cos\theta)/2$. Since this triangle lies completely within the circle slice, we have

$$\frac{\sin\theta\cos\theta}{2} \leq \frac{\theta}{2}. \tag{20}$$

Lastly, draw a vertical line extending up from the point $C$ (perpendicular to the $x$-axis). At the same time, extend the line $AB$ so that it intersects with this new vertical line; call this intersection point $E$. The triangle $ACE$ completely contains the circle slice from earlier. Moreover, the length of $AC$ is 1 and we have a right angle at $C$, so the length of $CE$ must be $\tan\theta$. This shows that the area of $ACE$ is $\tan\theta/2$, and hence

$$\frac{\theta}{2} \leq \tan\theta, \tag{21}$$

since this bigger triangle contains the circle slice. Combining the inequalities (20) and (21) and dividing through by 2, we obtain

$$\sin\theta\cos\theta \leq \theta \leq \tan\theta.$$

Now, dividing by $\sin\theta$ yields

$$\cos\theta \leq \frac{\theta}{\sin\theta} \leq \frac{\tan\theta}{\sin\theta} = \frac{1}{\cos\theta},$$

and taking the reciprocal of the whole inequality gives us

$$\frac{1}{\cos\theta} \geq \frac{\sin\theta}{\theta} \geq \cos\theta.$$

By the squeeze theorem, we must have

$$\lim_{\theta\to 0^+} \frac{1}{\cos\theta} \geq \lim_{\theta\to 0^+} \frac{\sin\theta}{\theta} \geq \lim_{\theta\to 0^+} \cos\theta,$$

but both the left- and right-hand sides go to 1, so $\sin\theta/\theta \to 1$ as well, as $x \to 0$ from the right.

To see that the limit is also 1 from the left, note that

$$\frac{\sin(-\theta)}{-\theta} = \frac{-\sin(-\theta)}{\theta} = \frac{\sin\theta}{\theta},$$

since sin is an odd function. In short, the function $\sin\theta/\theta$ is even, so its one-sided limits are equal. This proves (18).

It still remains to prove (19); fortunately, this can be derived from (18) without much further ado. We expand and multiply by the conjugate of $\cos\theta - 1$, namely $\cos\theta + 1$, on the top and bottom to obtain

$$\begin{aligned}\lim_{\theta\to 0} \frac{\cos\theta - 1}{\theta} &= \lim_{\theta\to 0} \frac{\cos\theta - 1}{\theta} \cdot \frac{\cos\theta + 1}{\cos\theta + 1} \\ &= \lim_{\theta\to 0} \frac{\cos^2\theta - 1}{\theta(\cos\theta + 1)}.\end{aligned}$$

From here, we use the fact that $\sin^2\theta + \cos^2\theta = 1$ to write

$$\begin{aligned}\lim_{\theta\to 0}\frac{\cos^2\theta - 1}{\theta(\cos\theta+1)} &= \lim_{\theta\to 0}\frac{\sin^2\theta}{\theta(\cos\theta+1)}\\ &= \lim_{\theta\to 0}\frac{\sin\theta}{\theta}\cdot\lim_{\theta\to 0}\frac{\sin\theta}{\cos\theta+1}\\ &= 1\cdot\frac{0}{1+1}\\ &= 0,\end{aligned}$$

where the third equality follows from (18). ∎

We are now able to find formulas for all six trigonometric functions. We state this as one megatheorem.

**Theorem 46** (*Derivatives of trigonometric functions*). *We have*

*i)* $\sin' = \cos$;

*ii)* $\cos' = -\sin$;

*iii)* $\tan' = \sec^2$;

*iv)* $\csc' = -\csc\cdot\cot$;

*v)* $\sec' = \sec\cdot\tan$; *and*

*vi)* $\cot' = -\csc^2$.

*wherever these derivatives exist. (The derivatives* $\sin'$ *and* $\cos'$ *exist everywhere;* $\tan'$ *and* $\sec'$ *are defined wherever* $\cos\neq 0$*, that is, everywhere except at inputs of the form* $n\pi/2$ *for all odd integers* $n$*; and* csc *and* cot *are defined wherever* $\sin\neq 0$*, so they are defined everywhere except at integer multiples of* $\pi$*.)*

*Proof.* By the addition formula (16) for sine, we have

$$\begin{aligned}\sin'(x) &= \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h}\\ &= \lim_{h\to 0}\frac{\sin x\cos h+\sin h\cos x-\sin x}{h}\\ &= \lim_{h\to 0}\frac{\sin x(\cos h-1)+\sin h\cos x}{h}\\ &= \sin x\lim_{h\to 0}\frac{\cos h-1}{h}+\cos x\lim_{h\to 0}\frac{\sin h}{h}\\ &= \sin x\cdot 0+\cos x\cdot 1\\ &= \cos x\end{aligned}$$

for all numbers $x$, where the fifth line follows from the special limits computed

in Lemma 45. Likewise, by the addition formula (17) for cosine,

$$\begin{aligned}
\cos'(x) &= \lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h} \\
&= \lim_{h\to 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\
&= \lim_{h\to 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\
&= \cos x \lim_{h\to 0} \frac{\cos h - 1}{h} - \sin x \lim_{h\to 0} \frac{\sin h}{h} \\
&= \cos x \cdot 0 - \sin x \cdot 1 \\
&= -\sin x
\end{aligned}$$

for all $x$, where again we have invoked Lemma 45 in the fifth line. This proves both (i) and (ii).

To prove (iii), we apply the quotient rule to the identity $\tan = \sin / \cos$. Wherever the cosine function is not 0, we have

$$\tan' = \left(\frac{\sin}{\cos}\right)' = \frac{\cos \cdot \sin' - \sin \cdot \cos'}{\cos^2} = \frac{\cos^2 - \sin \cdot (-\sin)}{\cos^2} = \frac{1}{\cos^2} = \sec^2,$$

where we have once again used the Pythagorean identity $\cos^2 + \sin^2 = 1$.

Because the cosecant, secant, and cotangent functions are defined as reciprocals of the sine, cosine, and tangent functions, respectively, we shall use the reciprocal rule $(1/g)' = -g'/g^2$ given by Theorem 42 to prove (iv), (v), and (vi). For the cosecant function, we compute

$$\csc' = \left(\frac{1}{\sin}\right)' = \frac{-\sin'}{\sin^2} = \frac{-\cos}{\sin^2} = -\csc \cdot \cot,$$

which settles (iv). Finally we have

$$\sec' = \left(\frac{1}{\cos}\right)' = \frac{-\cos'}{\cos^2} = \frac{-(-\sin)}{\cos^2} = \sec \cdot \tan$$

and

$$\cot' = \left(\frac{1}{\tan}\right)' = \frac{-\tan'}{\tan^2} = \frac{-\sec^2 \cdot \cos^2}{\sin^2} = -\csc^2,$$

which proves the rest of the theorem. ∎

Depending on how much you trust your memory, you may be able to store all these formulas safely in your brain. At minimum, you should commit the derivatives of sine and cosine to memory, along with the quotient and reciprocal rules. Then you can rederive the rest of the trigonometric derivatives in a matter of seconds. Whatever strategem you employ, a good sanity check to keep in mind

is that the "co-" functions cos, csc, and cot are the ones whose derivatives have a negative sign in their formulas.

We end off this section by observing a curious pattern. The derivative of sin is cos and the derivative of cos is $-\sin$. Hence $\sin''' = -\cos$ and $\sin^{(4)}$ is sin. Repeating this process again, we see that $\sin^{(n)} = \sin$ whenever $n$ is a multiple of 4. This allows us to compute huge higher derivatives of the sine function (and the cosine function) without having to actually take the derivative so many times. For instance, say we want to know the 439th derivative of sin. By long division, we have

$$439 = 436 + 3 = 109 \cdot 4 + 3.$$

The number 436 is a multiple of 4, meaning that $\sin^{(436)}$ is just sin. Taking the derivative three more times, we get $\sin^{(437)} = \cos$, $\sin^{(438)} = -\sin$, so $\sin^{(439)} = -\cos$.

## 8. Compositions and inverses

07.IX We are now equipped to differentiate rational functions, the exponential function $e^x$, and trigonometric functions, as well as all sums, products, and quotients thereof. However, there are still plenty of functions we don't know how to deal with. For instance, we got lucky with the function $e^{2x}$, which we saw in Proposition 44, since this could be factored into $e^x e^x$ and then differentiated using the product rule. But what if we'd had $e^{x^2}$ instead? Or $\sin^2(e^x)$? These do not admit easy factoring tricks, and yet it seems like we should know how to differentiate them, given that we know how to differentiate the functions $\sin x$, $e^x$, and $x^2$. More generally, suppose we know how to differentiate the functions $f(x)$ and $g(x)$. Can we derive a formula for the derivative of $f \circ g$ (defined by $(f \circ g)(x) = f\big(g(x)\big)$)?

Let's return to the example of $e^{2x}$ that we had earlier. Since $(d/dx)e^x = e^x$, by just naïvely substituting $2x$ in place of $x$, we might think that $(d/dx)e^{2x} = e^{2x}$. But this is not the case, as shown by the formula (15) we derived in the last section. To organise our work properly, let us switch letters when discussing the input to the exponential function. We know that $(d/du)e^u = e^u$, which gives us a description of how the function exp changes if you tweak its input $u$. But $u = 2x$ in our example above, and we want to know how the entire function $e^{2x}$ reacts when we perturb $x$, not $u$. We can solve this by calculating the derivative $(du/dx)2x = 2$, which tells that when $x$ changes by a little bit, $u$ changes by twice as much. So whatever $(d/du)e^u$ was, the derivative $(d/dx)e^{2x}$ should be twice as much. Hence we have

$$\frac{d}{dx}e^{2x} = 2e^u = 2e^{2x},$$

which exactly matches what we derived in (15).

How can we generalise this? Well, say $f$ is a function of $u$ and $u$ is a function of $x$. Suppose further that we know $df/du$ and $du/dx$. Then to find out how

much $f$ responds to a change in $x$ we need to multiply how much $u$ responds to $x$ by how much $f$ responds to $u$. In other words.

$$\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx}. \tag{22}$$

(You can remember this formula by pretending that it was derived by simply "multiplying by $dx$" in the numerator and denominator.) This phenomenon is known as the *chain rule*. Though we shall now restate it as a theorem more formally in modern function notation with proper hypotheses, the reader may choose to commit (22) to memory.

**Theorem 47** (*Chain rule*). *Suppose that $f$ and $g$ are functions and $a$ is a number such that such that $g$ is differentiable at $a$ and $f$ is differentiable at $g(a)$. Then the function composition $f \circ g$ is differentiable at $a$, and*

$$(f \circ g)'(a) = f'\big(g(a)\big)g'(a). \tag{23}$$

Let us first expand out the definitions of the three derivatives that appear in (23) to really understand what we're trying to prove. It can be rewritten

$$\lim_{x\to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{x-a} = \lim_{x\to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{g(x)-g(a)} \cdot \lim_{x\to a} \frac{g(x)-g(a)}{x-a}.$$

At first, proving this seems extremely easy. Can we not just multiply the left-hand side by $g(x)-g(a)$ in both the numerator and denominator and then apply the product law for limits to obtain the right-hand side? The answer is yes if $g(x)-g(a)$ is always nonzero for $x$ sufficiently close to $a$, but no in general, since even if $g$ is differentiable at $a$, it may be the case that $g(x) - g(a)$ is zero for some $x \neq a$ close to $a$. (Even if we only consider $x$ very close to $a$, it could happen that this occurs for $x$ *arbitrarily* close to $a$; take for example the case of $g(x) = x^2 \sin(\pi/x)$ and $a = 0$.)

The real proof of the chain rule is constructed much more carefully in order to get around this obstruction. It is therefore fairly complicated—the formal definition of a limit has a starring role in this proof—and may be regarded as optional for this course. (That said, the reader is encouraged to skim through it at least once. Most of the proof's difficulty lies in precisely expanding out definitions and organising variables, both mathematical skills that are broadly useful in this course and beyond.)

**Proof.* We assumed that $f'$ is differentiable at $g(a)$, so $f'\big(g(a)\big)$ is defined, and we define a new function $s$ by

$$s(x) = \begin{cases} \dfrac{f\big(g(x)\big) - f\big(g(a)\big)}{g(x)-g(a)}, & \text{if } g(x) \neq g(a); \\ f'\big(g(a)\big), & \text{otherwise.} \end{cases}$$

Note that $s(a)$ is simply $f'\big(g(a)\big)$, since $g(a) - g(a) = 0$. The first (and hardest) step in this proof is to show that $s$ is continuous at $a$, i.e., that

$$\lim_{x \to a} s(x) = s(a) = f'\big(g(a)\big).$$

We prove this directly from the definition of a limit. Let $\epsilon > 0$. We wish to find $\delta$ such that for all $x$ with $0 < |x - a| < \delta$ one has $\big|s(x) - f'(g(a))\big| < \epsilon$. Writing out the definition of $f'\big(g(a)\big)$, we have

$$\lim_{h \to 0} \frac{f\big(g(a) + h\big) - f\big(g(a)\big)}{h} = f'\big(g(a)\big).$$

By the precise definition of a limit, this means that there exists $\delta_1 > 0$ such that for all $h$ with $0 < |h - 0| < \delta_1$, we have

$$\left| \frac{f\big(g(a) + h\big) - f\big(g(a)\big)}{h} - f'\big(g(a)\big) \right| < \epsilon. \tag{24}$$

Another thing we know is that $g$ is differentiable at $a$. Hence by Theorem 35, the function $g$ is also continuous at $a$, meaning that

$$\lim_{x \to a} g(x) = g(a).$$

By the precise definition of a limit once again, we know that there is $\delta_2 > 0$ such that for all $x$ with $0 < |x - a| < \delta_2$, we have

$$\big|g(x) - g(a)\big| < \delta_1.$$

(The limit definition says "for all $\epsilon$," and we have chosen to put $\epsilon = \delta_1$ here.)

Set $\delta = \delta_2$ and assume that $0 < |x - a| < \delta$. If $g(x) = g(a)$, then $s(x) = f'\big(g(a)\big)$ and we are done. If $g(x) \neq g(a)$, then we know that

$$0 < \big|g(x) - g(a)\big| < \delta_1,$$

so we can let $h = g(x) - g(a)$ in (24) to obtain

$$\left| \frac{f\big(g(a) + g(x) - g(a)\big) - f\big(g(a)\big)}{g(x) - g(a)} - f'\big(g(a)\big) \right| < \epsilon.$$

But now we notice (after simplifying) that this is precisely the statement

$$\big|s(x) - f'(g(a))\big| < \epsilon.$$

So $s(x) \to f'\big(g(a)\big) = s(a)$ as $x \to a$ and we conclude that $s$ is continuous at $a$.

With this fact in hand, the rest of the proof is easy. So long as $x \neq a$, we have

$$\frac{f\big(g(x)\big) - f\big(g(a)\big)}{x-a} = s(x)\frac{g(x)-g(a)}{x-a}.$$

(If $g(x) \neq g(a)$, then expand out the definition of $s(x)$ and cancel out terms to prove this. If $g(x) = g(a)$, then both sides equal 0.) So taking the limit on both sides of this equality, we have

$$\begin{aligned}(f\circ g)'(a) &= \lim_{x\to a}\frac{f\big(g(x)\big) - f\big(g(a)\big)}{x-a}\\ &= \lim_{x\to a} s(x)\frac{g(x)-g(a)}{x-a}\\ &= \lim_{x\to a} s(x)\cdot\lim_{x\to a}\frac{g(x)-g(a)}{x-a}\\ &= s(a)\cdot g'(a)\\ &= f'\big(g(a)\big)\cdot g'(a),\end{aligned}$$

which at last proves the theorem. ∎

Let us try our hand at an example. Consider the function $\sin(2x)$. If we let $f(x) = 2x$, then this is the composition $\sin\circ f$. By the chain rule, we have

$$\frac{d}{dx}\sin(2x) = (\sin\circ f)'(x) = \sin'\big(f(x)\big)f'(x) = \cos(2x)\cdot 2. \tag{25}$$

This shows that if we can remember the *double angle formula* for sine, which is

$$\sin(2x) = 2\sin x\cos x,$$

we can easily recover the double angle formula for cosine using the product rule. We have

$$\frac{d}{dx}\sin(2x) = \frac{d}{dx}2\sin x\cos x = 2(\cos^2 x - \sin^2 x). \tag{26}$$

So by equating (25) and (26), we see that

$$\cos(2x) = \cos^2 x - \sin^2 x.$$

Here's another fun trigonometric example. By using the chain rule, we can give a "proof" that

$$(\sin\theta)^2 + (\cos\theta)^2 = 1$$

for all $\theta$. Here's how this "proof" goes. The derivative of the left hand side is

$$\frac{d}{d\theta}\Big((\sin\theta)^2 + (\cos\theta)^2\Big) = 2\sin\theta\cos\theta + 2\cos\theta(-\sin\theta) = 0,$$

by the chain rule applied to both terms. This doesn't immediately tell us the function is equal to 1, but it does tell us that it's *constant*; that is, the expression $(\sin\theta)^2 + (\cos\theta)^2$ is the same, no matter what $\theta$ we plug in. So let's pick an easy choice of $\theta$, like 0. We have $\sin\theta = 0$ and $\cos\theta = 1$, so the function is equal to 1 at $\theta = 0$ and must therefore be equal to 1 for all $\theta$.

Why is "proof" written in quotation marks? Should you be sceptical about the logic used in the previous paragraph? Don't worry, there's no issue with the logic there. The problem is that we used the formulas for the derivatives of sin and cos in this "proof" of (27), but to prove those derivative formulas we used this very equation! If you use a statement $P$ to prove a statement $Q$, then you cannot use statement $Q$ to prove statement $P$. This would be an example of *circular reasoning*, which is the proof equivalent of pulling oneself up by one's bootstraps: a logical fallacy, to say the least. If you want to show that $Q$ is true using $P$, then you'd better have a completely separate proof that $P$ is true, without using $Q$ anywhere in it. Luckily in the case above, we already have a completely separate proof of (27) that doesn't use derivatives; simply draw a right triangle with one of the (other) angles equal to $\theta$, and invoke the Pythagorean theorem.

**General exponential functions.** The chain rule also allows us to figure out the derivative of general exponential functions $b^x$ (for any number $b > 0$) just from knowing that $(d/dx)e^x = e^x$. Recall that the inverse of the function $e^x$ is $\ln x$; that is, if $e^x = y$, then $\ln y = x$.

**Theorem 48.** *For any positive number $b$, we have*

$$\frac{d}{dx} b^x = (\ln b) b^x.$$

*Proof.* Since $b > 0$, the logarithm $\ln b$ of $b$ is a well defined number, and we have $b = e^{\ln b}$. So

$$b^x = (e^{\ln b})^x = e^{(\ln b)x}.$$

Letting $f(x) = (\ln b)x$, the expression on the right-hand side here is simply the composition $\exp \circ f$ applied to $x$. Hence the chain rule gives

$$\frac{d}{dx} b^x = (\exp \circ f)'(x) = \exp'\big(f(x)\big) f'(x) = e^{(\ln b)x} \cdot (\ln b).$$

But recall that $e^{(\ln b)x}$ is just $b^x$, so substituting that back in, we arrive at the theorem statement. ∎

The sudden appearance of the function ln in our lives might make us wonder what the derivative of ln is. We'll do better than just computing this derivative—the function ln is the inverse of exp, a function we're already acquainted with, so we'll give a general rule for computing the derivatives of *inverses* of functions whose derivative we already know.

**Inverse functions.** This is the last "fundamental" differentiation formula that we'll cover. Since we know how to differentiate $e^x$, we ought to be able to differentiate its inverse $\ln x$—the slope of a tangent line on the graph of ln should be related in some way to the slope of a tangent line on the graph of exp. Likewise, we figured out from scratch that $(d/dx)\sqrt{x} = 1/(2\sqrt{x})$, but we should learn how to differentiate $n$th roots like $\sqrt[3]{x}$, $\sqrt[4]{x}$, and so on. This should follow from our knowledge of the derivatives of $x^3$ and $x^4$, etc.

Let's remind ourselves what the inverse of a function is. Suppose $f$ is defined on some domain $D$ and $R$ is the set of possible values it outputs—this set is called its range. (It's been a long time since we have defined it, so you might well have forgotten that definition.) We say that $f$ is *one-to-one* if every input $x$ has a unique output; formally, this means that whenever we have $f(x_1) = f(x_2)$ we must have $x_1 = x_2$. You can think about this as passing the "horizontal line test." If a horizontal line passes $f$ at two different points $x_1$ and $x_2$, then $f(x_1) = f(x_2)$ for different $x_1$ and $x_2$, so $f$ is *not* one-to-one. But if the opposite of this occurs, and every horizontal line passes through $f$ at at most one point, then we say that it passes the horizontal line test and it is one-to-one.

If this rings a bell and reminds you of the "vertical line test" for checking if a (purported) function is even well defined, then you might have some idea as to how the property of being one-to-one relates to inverses. If a function is one-to-one, every $x$ in its domain gets mapped to a unique $y$ in its range. So given some $y$ in the range, there is only one $x$ that could possibly have been fed into $f$ in order to receive output $y$. This means that if we define a function $g$ whose domain is the range of $f$ by setting, for all $y$,

$$g(y) = \text{the unique } x \text{ such that } f(x) = y,$$

then the range of $g$ is the domain of $f$, and we have $f\big(g(y)\big) = y$ for all $y$ in the domain of $g$, and $g\big(f(x)\big) = x$ for all $x$ in the domain of $f$. This function $g$ is usually denoted by $f^{-1}$, and is called the *inverse* of $f$.

If we draw $f$ as a graph on the $(x, y)$ plane, then since $f^{-1}$ reverses the roles of $x$ and $y$ (we have $f^{-1}(x) = y$ whenever $f(x) = y$, the graph of $f^{-1}$ can be obtained from the graph of $f$ by *reflecting it across the line* $y = x$. Think about this with some inverse pairs you already know, like $f = \exp$ and $f^{-1} = \ln$, or $f(x) = x^2$ and $f^{-1}(x) = \sqrt{x}$.

16.x This graphical interpretation allows us to intuitively reason about what the derivative of $f^{-1}$ should be. Draw a smooth curve in the plane, representing the graph of $f$ in some interval, and then pick a point on the curve. This is the point $\big(a, f(a)\big)$ for some number $a$. Draw the tangent line to the curve at that point; what we learned about derivatives tells us that this line has slope $f'(a)$.

Now reflect the whole picture across the line $y = x$. The new curve is the graph of $f^{-1}$, and the reflected counterpart to the point on the curve we had is $\big(f(a), a\big)$. The tangent line has also been reflected. What is its slope? Well, whatever the "rise over run" was for the original line has been flipped on its head! The slope of this new line is the "run over rise" of the original line—in

other words, *the slope of the tangent line to the curve $f^{-1}$ at the point $\big(f(a), a\big)$ is $1/f'(a)$*. Writing this out in terms of the derivative gives us the formula

$$(f^{-1})'\big(f(a)\big) = \frac{1}{f'(a)}. \tag{27}$$

But it's a bit cumbersome to express the input of $f^{-1}$ in terms of the output of $f$, so let's give the number $f(a)$ its own letter, say $b$, and we know that since $f^{-1}$ is the inverse of $f$, we must have $f^{-1}(b) = a$. This allows us to express the following theorem in a format that's a bit more useful.

**Theorem 49** (*Derivatives of inverses*). *Suppose that $f$ is one-to-one on some domain containing the number $a$, so that if we let $b = f(a)$, then the inverse function $f^{-1}$ is defined at $b$ (and we have $f^{-1}(b) = a$). If $f$ is differentiable at $a$ and $f'(a) \neq 0$, then $f^{-1}$ is differentiable at $b$, and its derivative is given by the formula*

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'\big(f^{-1}(b)\big)}.$$

We'll now proceed to give a proof in terms of the formal definition of the derivative in terms of limits, but it is marked as optional, not because it's long but because it's a bit finicky with the precise definitions related to inverses. As before, try to skim the proof at least once, but if you understand the intuitive graphical reasoning given above for why Theorem 49 is true, that should be enough to carry you through this course.

**Proof.* We'll write $g = f^{-1}$ just to simplify notation in this proof; so we have $f(a) = b$ and $g(b) = a$, and we want to show that $g'(b) = 1/f'\big(g(b)\big)$. For every small $h$, there is a unique $k_h$ such that

$$g(b + h) = a + k_h \tag{28}$$

(we subscript the $k$ with $h$ to remind ourselves that the value of this number $k_h$ depends on $h$). We can rearrange (28) to $k_h = g(b + h) - a$, and since $g$ is continuous, we have

$$\lim_{h\to 0} k_h = \lim_{h\to 0} g(b + h) - a = g\Big(\lim_{h\to 0}(b + h)\Big) - a = g(b) - a = 0.$$

In other words, $k_h$ goes to 0 as $h$ goes to 0.

Since $f$ is the inverse of $g$, we have $f(a+k_h) = b+h$, which we can rearrange to obtain

$$h = f(a + k_h) - b = f(a + k_h) - f(a).$$

This allows us to expand

$$g'(b) = \lim_{h\to 0} \frac{g(b + h) - g(b)}{h} = \lim_{h\to 0} \frac{a + k_h - a}{f(a + k_h) - f(a)} = \lim_{h\to 0} \frac{k_h}{f(a + k_h) - f(a)}.$$

But this just looks like the definition of a derivative upside down, and we already observed that $k_h$ goes to 0 as $h$ goes to 0. We conclude that

$$g'(b) = \lim_{k_h \to 0} \frac{k_h}{f(a+k_h) - f(a)} = \frac{1}{f'(a)},$$

which proves the theorem. ∎

Note that if we already knew that $f^{-1}$ is differentiable at $f(a)$ and we just wanted to find out what that derivative is, then we could have invoked the chain rule. For all $x$ in the domain of $f^{-1}$, we have

$$x = f\big(f^{-1}(x)\big).$$

Taking the derivative of both sides (using the chain rule on the right-hand side) yields

$$1 = f'\big(f^{-1}(x)\big)(f^{-1})'(x),$$

which, after rearranging, is exactly (27). But of course in order to use the chain rule we had to already know that $f^{-1}$ is differentiable, which leads us towards having to prove Theorem 49 properly anyway.

This formula given by Theorem 49 for the derivatives of inverses allows us to find the derivatives of a great deal more functions.

**Roots.** First we obtain a formula for the derivative of $n$th roots, where $n$ is a positive integer.

**Theorem 50.** *For any positive integer $n$, we have*

$$\frac{d}{dx}\sqrt[n]{x} = \frac{\sqrt[n]{x}}{nx} = \frac{1}{n}x^{1/n-1}.$$

*Proof.* Let $f(x) = x^n$. By Theorem 40 (the power rule) we know that $f'(x) = nx^{n-1}$. Then we have $f^{-1}(x) = \sqrt[n]{x}$, since $\sqrt[n]{x^n} = x$ for all $x$. So by Theorem 49, we have

$$\frac{d}{dx}\sqrt[n]{x} = \frac{1}{f'\big(f^{-1}(x)\big)} = \frac{1}{f'\big(\sqrt[n]{x}\big)} = \frac{1}{n\big(\sqrt[n]{x}\big)^{n-1}} = \frac{1}{nx^{(n-1)/n}},$$

which rearranges to the theorem statement, since $(n-1)/n = 1 - 1/n$. ∎

Note that plugging $n = 2$ into this theorem gives us

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2}x^{1/2-1} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}},$$

which matches what we derived (from first principles) as Proposition 33. We can now take the derivative of $x^r$ whenever $r$ is rational by expressing $r$ as $p/q$ for some integers $p/q$, writin $x^r = x^{p/q} = \sqrt[q]{x^p}$, and then invoking the chain rule. This gives the correct answer (try it and see what you get), but this method is messy and we'll soon develop a much easier way to get the same formula for general powers (including irrational powers!) very soon.

**Logarithms.** Let $b$ be any positive number. If $b^x = y$, then we have $\log_b y = x$. In other words, the inverse of the function $f(x) = b^x$ is $f^{-1}(x) = \log_b(x)$. Since we know the derivative of $e^x$, we can easily calculate the derivative of its inverse function $\ln = \log_e$ by using Theorem 49.

**Theorem 51.** *We have*

$$\frac{d}{dx}\ln x = \frac{1}{x}.$$

*Proof.* Let $f(x) = e^x$, so that $f^{-1}(x) = \ln x$. Then $f'(x)$ is also $e^x$, so by Theorem 49, we have

$$\ln'(x) = \frac{1}{f'\big(f^{-1}x\big)} = \frac{1}{e^{\ln x}} = \frac{1}{x},$$

and we are done. ∎

The function ln is not defined for negative $x$, but perhaps you might be asked to find $\ln(-x)$ in this case, which is allowed. Then by the chain rule we have

$$\frac{d}{dx}\ln(-x) = \frac{1}{-x}\cdot(-1) = \frac{1}{x},$$

which is the same formula that we had for $\ln'(x)$ in the case that $x$ is positive. We can sum this up by writing

$$\frac{d}{dx}\ln|x| = \frac{1}{x},$$

a formula that covers both positive and negative cases at once (but not $x = 0$, of course, since $\ln 0$ is not defined).

Using the formula $(d/dx)\ln x = 1/x$ and the fact that $\log_b(x) = (\ln x)/(\ln b)$ for any $b > 0$, it is very easy to show (do it!) that the derivative of a general logarithmic function $\log_b(x)$ is given by

$$\frac{d}{dx}\log_b|x| = \frac{1}{x\ln b}.$$

**Inverse trigonometric functions.** Our formula for the derivatives of inverses naturally leads us to the calculation of the derivatives of the inverse trigonometric functions. Recall that if we restrict its domain to the interval $[-\pi/2, \pi/2]$, then the sine function is one-to-one, so it has an inverse, called arcsin or $\sin^{-1}$. The other inverse trigonometric functions are defined similarly, and denoted arccos, arctan, arccsc, arcsec, and arccot, respectively. Visit Section 1.5 of the textbook for a refresher on these functions and the correct restrictions of their domains. (Instead of the "arc" notation, you might see the $^{-1}$ notation—both are valid. In fact, the latter three functions mentioned above are possibly *more often* denoted by $\csc^{-1}$, $\sec^{-1}$, and $\cot^{-1}$, respectively, on the rare occasion that they are even mentioned, but in these notes we'll use the "arc" versions.)

Now that we have Theorem 49, determining the derivatives of the six inverse trigonometric functions becomes a routine (but instructive) exercise in trigonometry.

**Theorem 52.** *We have*

*i)* $\dfrac{d}{dx}\arcsin x = \dfrac{1}{\sqrt{1-x^2}}$;

*ii)* $\dfrac{d}{dx}\arccos x = -\dfrac{1}{\sqrt{1-x^2}}$;

*iii)* $\dfrac{d}{dx}\arctan x = \dfrac{1}{1+x^2}$;

*iv)* $\dfrac{d}{dx}\operatorname{arccsc} x = -\dfrac{1}{x\sqrt{x^2-1}}$;

*v)* $\dfrac{d}{dx}\operatorname{arcsec} x = \dfrac{1}{x\sqrt{x^2-1}}$; *and*

*vi)* $\dfrac{d}{dx}\operatorname{arccot} x = -\dfrac{1}{1+x^2}$;

*Proof.* We prove (i), (iii), and (v); the other three are left to you to do in private. In each case, our invocation of Theorem 49 will leave us with the composition of a trigonometric function and an inverse trigonometric function. Whenever you see such a beast you will always be able to simplify it by reasoning with a suitably constructed right triangle.

For (i), Theorem 49 gives us

$$\arcsin'(x) = \frac{1}{\sin'(\arcsin x)} = \frac{1}{\cos(\arcsin x)}. \tag{29}$$

To simplify this, we must draw a right triangle with angle $\theta = \arcsin x$. This means that $\sin\theta = x$, so the opposite divided by the hypotenuse must equal $x$. An easy choice is to make the opposite $x$ and the hypotenuse 1. By the Pythagorean theorem, the adjacent is therefore $\sqrt{1-x^2}$, and

$$\cos(\arcsin x) = \cos\theta = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}.$$

Plugging this into (29) yields exactly the claimed formula in (i).

For (iii), we begin with

$$\arctan'(x) = \frac{1}{\tan'(\arctan x)} = \frac{1}{\sec^2(\arctan x)}.$$

Draw a right triangle with angle $\theta$, opposite $x$, and adjacent 1, so that $\tan\theta = x$, and so $\arctan x = \theta$. Then the Pythagorean theorem tells us that the hypotenuse is $\sqrt{1+x^2}$, so

$$\cos\theta = \frac{1}{\sqrt{1+x^2}},$$

whence

$$\sec(\arctan x) = \sec\theta = \sqrt{1+x^2}$$

and

$$\arctan'(x) = \frac{1}{\sec^2(\arctan x)} = \frac{1}{1+x^2}.$$

Finally, for (v) we expand

$$\operatorname{arcsec}'(x) = \frac{1}{\sec(\operatorname{arcsec} x)\tan(\operatorname{arcsec} x)} = \frac{1}{x\tan(\operatorname{arcsec} x)}. \tag{30}$$

Draw a right triangle with angle $\theta$, hypotenuse $x$, and adjacent 1. This makes $\cos\theta = 1/x$, so $\sec\theta = x$ and thus $\theta = \operatorname{arcsec} x$. We observe that the opposite must be $\sqrt{x^2-1}$ by the Pythagorean theorem, so

$$\tan(\operatorname{arcsec} x) = \tan\theta = \frac{\sqrt{x^2-1}}{1} = \sqrt{x^2-1}.$$

Plugging this into (30) gives us exactly the claim of (v). ∎

As with the derivatives of the trigonometric functions, you may elect not to commit all of these derivative formulas to memory if you trust your ability to reprove them on the spot. To aid in checking your work if you prefer to go by this route, remember that once again it is the functions with "-co-" in their names that have negative signs in their derivatives.

19.x ***The Mercator projection.** Now that we know how to take the derivative of the logarithm, we are able to understand one of the most important mathematical problems of the seventeenth century, the century during which much of calculus was developed.

Early mapmakers spaced lines of latitude and lines of longitude equally on their maps. This caused issues with maritime navigation over large distances: travelling at a forty-five degree angle on the map didn't exactly correspond with going north-east. Early sailors (and their precious cargo) were liable to get lost. In 1569, Gerardus Mercator drew a map that seemingly fixed this problem. Lines that were at (say) a right angle to each other in real life were drawn as right angles on the map.

It became clear that Mercator's projection was the best projection to use for maritime navigation, and all cartographers of the time wanted to map out newly explored areas of the world with this new projection. But Mercator had been vague with how exactly to perform the projection. All he'd written was that the map had had to be stretched more and more as you got closer to the poles. This is obvious to anyone who has ever beheld a sphere. How far exactly? was the big question.

In 1599, Edward Wright mathematically reverse-engineered Mercator's projection. His reasoning was as follows. Scale all our units so that the radius of the earth is 1; then the length of the equator is $2\pi$. Moving up along the earth's surface to a latitude of $\theta$ (in real life we are used to expressing latitude in degrees, but for this discussion, $\theta$ will be assumed to be in radians), the line of latitude

$\theta$ is a circle as well. It is a smaller circle than the equator, but on the map it will have the same length, so that the map is a rectangle. Hence this circle must be scaled up by some factor. To find out how much, note that the radius of this circle is $\cos\theta$ (draw it, remembering that the radius of the earth is set to be 1). So the line of latitude $\theta$ is a circle of length $2\pi\cos\theta$, and to scale it up to $2\pi$ like the equator on our map, we have to multiply horizontal distances by $1/\cos\theta = \sec\theta$.

Imagine you are an early European explorer who has just arrived on Montreal Island. You want to know where to place it on your partially filled-in world map. You have done some diligent tracking of distances, and estimate that you are 3500 miles west of the prime meridian. (Let us gloss over the calculation of longitude, which was actually extremely difficult in the old days without precise timekeeping technology.) You know that the equator is roughly 25000 miles long. If Montreal had been on the equator, you'd have placed it $3500/25000 = 14\%$ of the equator's distance left of the prime meridian on the map. But you also know that the latitude of Montreal is around 45 degrees, or $\pi/4$ radians, so the scale factor is $\sec(\pi/4) = \sqrt{2}$, and you should instead place it at $\sqrt{2}\cdot 14\% \approx 20\%$ of the way around the globe at the 45th parallel.

So calculating the $x$-coordinate of a point on the map isn't too hard. But knowing how far up the map (at what $y$-coordinate) to draw Montreal is much more difficult, and is the part that Edward Wright was the first to mathematically describe in 1599. Here's the key insight. To preserve angles, all small shapes have to be preserved. A square farm in the world should be square on the map. We already know that if you move a small distance eastward or westward at latitude $\theta$, then $x$ should change by that distance times $\sec\theta$. Hence small northward distances at latitude $\theta$ should be scaled by $\sec\theta$ as well. In other words, if we increase $\theta$ by some small amount, call it $d\theta$, then the $y$ coordinate should change by $\sec\theta\cdot d\theta$. So $dy = \sec\theta\cdot d\theta$ and we have

$$\frac{dy}{d\theta} = \sec\theta.$$

In modern parlance, $y$ should be a function of $\theta$ whose derivative is $\sec\theta$.

*What is a function of that changes by* $\sec\theta$ *as you tweak it slightly near* $\theta$*?* This was one of the famous mathematical problems of the age, as solving it would immediately make it possible to draw accurate maps, and thus conquer the world. Wright calculated a big table of values of $y$ for various $\theta$, which was a useful enough for the time. But he wouldn't have been able to write down a formula, as it involves the logarithm, which wasn't invented until 1614 by John Napier. People started compiling tables of values of logarithms (calculated by hand!) as they were useful for many different applications. A book came out in 1620 compiling values of logarithms of trigonometric functions.

In 1645 Henry Bond noticed a curious coincidence. Wright's table of $y$ values for various $\theta$ in Mercator's projection seemed to appear in the table for the logarithm of the tangent function, but shifted by a bit. In particular, it seemed to him that $y(\theta) = \ln\big(\tan(\theta/2+\pi/4)\big)$. This caused a big stir after it

was published, but was still only a conjecture. Remember, the first calculus texts only appeared in the 1670s, so people in the 1640s the time hadn't even developed the notation to write the conjecture down as $(d/d\theta)\ln\big(\tan(\theta/2+\pi/4)\big)=\sec\theta$, let alone prove it using the chain rule. Some rich people offered prize money to anyone who could prove the conjecture. In fact, one of the first inventors of calculus, Isaac Newton, wrote in a 1665 letter that he was aware of the problem, This is smack in the middle of the time period when he was also writing his first private notes on what was to become calculus.

In the end, James Gregory was the first to prove the conjecture in 1668. His explication was horribly complicated, and a simpler method was published by Isaac Barrow in 1670. Both of these proofs, predating calculus notation by several years, would have been written out more or less in words. With our modern understanding of calculus, the proof simply follows from two applications of the chain rule and the identity $\sin(2x)=2\sin x\cos x$.

**Theorem 53.** *On the interval* $(0,\pi/2)$*, the derivative of*

$$y(\theta)=\ln\left(\tan\left(\frac{\theta}{2}+\frac{\pi}{4}\right)\right)$$

*is* $\sec\theta$.

*Proof.* By the chain rule, we have

$$\begin{aligned}
y'(\theta)&=\frac{1}{\tan(\theta/2+\pi/4)}\cdot\tan'\left(\frac{\theta}{2}+\frac{\pi}{4}\right)\\
&=\frac{1}{\tan(\theta/2+\pi/4)}\cdot\sec^2\left(\frac{\theta}{2}+\frac{\pi}{4}\right)\cdot\frac{1}{2}\\
&=\frac{\cos(\theta/2+\pi/4)}{\sin(\theta/2+\pi/4)}\cdot\frac{1}{2\cos^2(\theta/2+\pi/4)}\\
&=\frac{1}{2\sin(\theta/2+\pi/4)\cos(\theta/2+\pi/4)}\\
&=\frac{1}{\sin(\theta+\pi/2)}\\
&=\frac{1}{\cos\theta},
\end{aligned}$$

which is exactly $\sec\theta$. ∎

So to finish off our example of trying to place Montreal on the world map, we need only plug in the latitude $\theta=\pi/4$ into this equation, and calculate $y(\theta)=\ln\big(\tan(3\pi/8)\big)\approx 0.88$. Then we know that the equator is supposed to be $2\pi$ long, so we find that $y(\theta)$ should be roughly $0.88/(2\pi)\approx 14\%$ of however long the equator is drawn on our map, and that's how high on the map we shall draw Montreal.

**Lines tangent to the unit circle.** Let's practise the chain rule and our understanding of trigonometric inverses with one more geometric example. Suppose we are asked to find the lines tangent to the unit circle with a slope of $-1/\sqrt{3}$. Sketching a circle, we notice that for any given slope, there are exactly two points on the unit circle where the tangent line will have that slope, and these points will be opposite one another. So we first restrict ourselves to the upper semicircle; we do this because to calculate the slope, we need to find the derivative, and we only know how to find derivatives of well-defined functions at the moment. (The graph of the upper semicircle is a function, whereas the graph of the whole unit circle is not a function—it fails the vertical line test.) In other words, there is a function $f(x)$ such that whenever $(x, y)$ is a point on the upper half of the semicircle, we have $y = f(x)$. We will express $f$ in two ways and calculate its derivative $f'$ in two different ways as well.

We can express $f$ in terms of trigonometric functions as follows. Let $(x, y)$ be any point on the upper half of the unit circle. We want to determine what $y$ is as a function of $x$. Well, the line segment (of unit length) between the origin and $(x, y)$ creates an angle of $\arccos x$ from the $x$-axis. So since $y$ is equal to the sine of this angle, divided by the length of the hypotenuse (which is 1), we see that $y = \sin(\arccos x)$. So by the chain rule, we simply compute

$$y' = \sin'(\arccos x)\arccos' x = \cos(\arccos x) \cdot \frac{-1}{\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}.$$

There is another way to express the equation of $y$ as a function of $x$. We notice that the triangle whose vertices are $(0, 0)$, $(x, 0)$, and $(x, y)$ is a right triangle, so we have $x^2 + y^2 = 1$. From here, we just isolate $y = \sqrt{1-x^2}$, where we take the positive branch of the square root because we are considering the upper half of the circle. Hence we have

$$y' = \frac{d}{dx}(1-x^2)^{1/2} = \frac{1}{2}(1-x^2)^{-1/2} \cdot (-2x) = \frac{-x}{\sqrt{1-x^2}} \tag{31}$$

by the chain rule (and the power rule for rational powers, or the reciprocal rule combined with the root rule if you prefer). This is exactly the same formula for the derivative that we computed in the previous paragraph.

So much for calculating the derivative. How does this help us find the point? Well, we are looking for the point $(x, y)$ where the slope is $-1/\sqrt{3}$. So we have

$$\frac{-x}{\sqrt{1-x^2}} = \frac{-1}{\sqrt{3}},$$

and after some routine algebra, we find that $x = 1/2$ satisfies this equation. When $x = 1/2$, we have

$$y = \sqrt{1-x^2} = \sqrt{1-(1/2)^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}.$$

We are now ready to find the equation of the first line. In slope-intercept form, the line has an equation of the form $y = mx + b$, where $m = -1/\sqrt{3}$, and we know the point $\big(1/2, \sqrt{3}/2\big)$ lies on this line, so we have the equation

$$\frac{\sqrt{3}}{2} = -\frac{1}{\sqrt{3}} \cdot \frac{1}{2} + b,$$

and we find that $b = 2/\sqrt{3}$, after some more algebra. Hence the line we seek is given by the equation

$$y = -\frac{1}{\sqrt{3}}x + \frac{2}{\sqrt{3}}.$$

As we noted above, this is only one of the two lines with slope $1/\sqrt{3}$ that is tangent to the unit circle. There will be another line with the same slope that is tangent to the point opposite to $\big(1/2, \sqrt{3}/2\big)$, which is $\big(-1/2, -\sqrt{3}/2\big)$. We repeat the procedure to find the $y$-intercept $b$ of this new line. We have

$$\frac{-\sqrt{3}}{2} = \frac{-1}{\sqrt{3}} \cdot \frac{-1}{2} + b,$$

whence $b = -2/\sqrt{3}$, and we see that the second line is given by the equation

$$y = -\frac{1}{\sqrt{3}}x - \frac{2}{\sqrt{3}}.$$

Because we only know how to find derivatives of functions, to find the tangent lines to the circle from this example, we had to exploit the symmetry of the circle in order to sidestep the fact that the equation $x^2 + y^2 = 1$ defining a circle is not a well-defined function of $x$. For more complicated curves that are also well-defined functions of $x$, this kind of reasoning (using the fact that opposite points on a circle should have tangent lines with the same slope) may not work, and want to have a way to determine the slopes (and equations) of lines tangent to all such curves. In the next section, we introduce a technique for doing so.

## 9. Implicit differentiation

If $f$ and $g$ are two functions that are both differentiable and equal to each other on some interval $(a, b)$ (i.e., $f(x) = g(x)$ for every $a < x < b$), then we have $f'(x) = g'(x)$. This is obvious because if the graphs are the same on this interval, then the tangent lines must also be the same. This fairly innocuous statement forms the basis of a very powerful technique called *implicit differentiation*.

Consider again the equation defining the unit circle, namely $x^2 + y^2 = 1$. If we think of $x$ as a variable that changes, then $y$ must change whenever $x$ does in order to maintain the validity of the equation. Hence we can think of both sides as functions of $x$ whose derivatives must be equal; in other words,

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}1. \tag{32}$$

When calculating this derivative, we must be careful to think of $y$ as a function of $x$. This means that we must use the chain rule when calculating $(d/dx)y^2$. We have $(d/dy)y^2 = 2y$, so by the chain rule, $(d/dx)y^2 = 2y(dy/dx)$. Hence taking the derivatives of both sides in the equation (32) yields the equation

$$2x + 2y\frac{dy}{dx} = 0,$$

which means that

$$\frac{dy}{dx} = \frac{-x}{y}.$$

Does this tally with the equation (31) that we derived earlier? Well, recall that that equation was solely for the upper semicircle, where we have $y = \sqrt{1-x^2}$, so yes, indeed it does. But our new formula works equally well on the lower semicircle, where $y = -\sqrt{1-x^2}$. It tells us that on the lower semicircle we have $dy/dx = x/\sqrt{1-x^2}$. (Think about how the sign of the slope of the tangent line changes as you move around the unit circle, and convince yourself that this change in sign for the derivative of the lower semicircle makes sense.)

21.X **Another algebraic curve.** An *algebraic curve* is the set of all points $(x, y)$ that make an algebraic equation $f(x, y) = 0$ true. The unit circle is an algebraic curve; its defining algebraic equation is $x^2 + y^2 = 0$, so $f(x, y)$ here is $x^2 + y^2 - 1$. Let us find a tangent line on a curve whose defining equation is slightly more complicated than that of a circle.

**Proposition 54.** *The line tangent to the curve*

$$y^2 = x^3 - 2x + 5$$

*at the point* $(2, 3)$ *is*

$$y = \frac{5}{3}x - \frac{1}{3}.$$

*Apart from being tangent to the curve at* $(2, 3)$*, this tangent line intersects the curve at exactly one more point, namely,* $(-11/9, -64/27)$*.*

*Proof.* Let us first check that $(2, 3)$ is even on the curve. We have

$$2^3 - 2 \cdot 2 + 5 = 8 - 4 + 5 = 9,$$

and of course $3^2 = 9$, so the point is indeed on the curve.

Now we want to find the slope $dy/dx$ of the tangent line, so let us take the derivative with respect to $x$ of both sides of the curve's defining equation. We have

$$2y\frac{dy}{dx} = 3x^2 - 2,$$

so that

$$\frac{dy}{dx} = \frac{3x^2 - 2}{2y}.$$

To find the slope of the tangent line at the point $(2, 3)$, we simply plug in $x = 2$ and $y = 3$ into this formula for $dy/dx$, obtaining

$$\left.\frac{dy}{dx}\right|_{x=2,y=3} = \frac{3 \cdot 2^2 - 2}{2 \cdot 3} = \frac{10}{6} = \frac{5}{3}.$$

Hence the tangent line has equation $y = (5/3)x + b$ for some real number $b$. We can find the $y$-intercept $b$ by noting that $(2, 3)$ lies on this tangent line. We have $3 = (5/3) \cdot 2 + b$, so

$$b = 3 - \frac{10}{3} = \frac{9}{3} - \frac{10}{3} = \frac{-1}{3}.$$

This proves the first claim in the proposition.

* The rest of the proof from here onwards is marked as optional, as it is fairly long and relatively light on calculus concepts. The goal is to find the intersection points between a cubic equation and a line, which is high school material (though probably on the more complex end). It is good to skim through at least once to make sure you are familiar with these techniques.

We wish to find out where the curve $y^2 = x^3 - 2x + 5$ and the line $y = (5/3)x - 1/3$ intersect, besides at $(2, 3)$, where we already know the line is tangent to the curve. To do this, we substitute the tangent line equation into the curve equation and find all solutions. The substitution yields

$$\left(\frac{5}{3}x - \frac{1}{3}\right)^2 = x^3 - 2x + 5,$$

which means that

$$\frac{25}{9}x^2 - 2 \cdot \frac{5}{9}x + \frac{1}{9} = x^3 - 2x + 5;$$

moving all the terms to the right-hand side and then multiplying through by 9 lets us simplify this to

$$9x^3 - 25x^2 - 8x + 1 = 0.$$

This on the left-hand side here is a cubic polynomial, so finding all its solutions would be hard, except that we already know that $x = 2$ is a solution. So we can divide the cubic polynomial by the polynomial $x - 2$ to obtain $9x^2 - 7x - 22$.

From here, we can use the quadratic formula to find the roots of this quadratic polynomial. But here's a shortcut: $x = 2$ will be a root again, since it is a repeated root of $9x^3 - 25x^2 - 8x + 1$! One way to see this is to reason about the original cubic curve being tangent to the tangent line (naturally) at $x = 2$. This means if you substitute one equation into the other you will get a cubic curve that just touches the $x$-axis at $x = 2$, but doesn't cross it. The other way to see this is to calculate

$$\frac{d}{dx}(9x^3 - 25x^2 - 8x + 1) = 27x^2 - 25x - 8,$$

so the derivative of this polynomial is 0 at $x = 2$. Hence its graph is flat when $x = 2$, and we already know it equals 0 at $x = 2$. Hence it has a repeated root at $x = 2$. Either way, we now know that $x = 2$ is a root of the quadratic polynomial $9x^2 - 7x - 22$ we obtained in the previous paragraph, and dividing it by $x - 2$ once more results in the linear polynomial $9x + 11$. This has a zero at $x = 11/9$, meaning that original curve $y^2 = x^3 - 2x + 5$ and the tangent line to this curve at $(2, 3)$ intersect again when $x = 11/9$. It remains to find the $y$-value(s) when $x = -11/9$. To do this we simply use the equation

$$y = \frac{5}{3} \cdot \frac{-11}{9} - \frac{1}{3} = \frac{-55}{27} - \frac{9}{27} = \frac{-64}{27}$$

to find that $y = -64/27$, or $y = (-4/3)^3$ if you like. ∎

***Elliptic curves and cryptography.** Any curve of the form

$$y^2 = x^2 + ax + b$$

is called an *elliptic curve*, as long as $4a^3 + 27b^2 \neq 0$. You may check that the curve from Proposition 54 is an elliptic curve (there we have $a = -2$ and $b = 5$). Elliptic curves are *smooth*—their graphs don't have self-intersections or cusps. To see what cubic curves that fail the elliptic condition $4a^3 + 27b^2 \neq 0$ look like, use an online graphical calculator like Desmos to graph $y^2 = x^3$ and $y^2 = x^3 - 3x + 2$.

In general, the secant line between two points that lie on an elliptic curve will cross the elliptic curve exactly one more time, and lines tangent to an elliptic curve will cross the elliptic curve exactly once more, like we saw in Proposition 54. (There are some exceptions, like vertical tangent lines and tangent lines at inflection points, but we'll gloss over these exceptions as they won't come up for our cryptographic purposes.) This means that we can define a sort of "addition" for points on elliptic curves that will spit out another point on the elliptic curve. Here are the rules for this special addition.

i) If $P$ and $Q$ are two different points, then the secant line passing through these two points intersects the curve at one more point, call it $R$, and we define $P + Q = -R$. (Note the minus sign!)

ii) If $Q = P$, then we cannot take a secant line, but we can find the tangent line at $P$, and it will cross the elliptic curve at exactly one more point, call it $R$, and we defined $P + P = -R$. We can also write $P + P$ as $2P$, hence $2P = -R$.

If we want to find some larger multiple of $P$, we can combine methods (i) and (ii). Let's say we want to calculate $7P$. Well, we can calculate $2P$ by finding the tangent line to $P$, seeing where it crosses the curve again, and then taking the negative of that point. Now we can find $4P$ by noting that

$$4P = P + P + P + P = (P + P) + (P + P) = 2P + 2P,$$

and we know what $2P$ is, so we can find the tangent line to $2P$ and so on to find $4P$. Knowing what $P$, $2P$, and $4P$ all are, we may calculate

$$7P = (P + 2P) + 4P$$

using secant lines.

Before we get into how this helps with cryptography, let's briefly go over what cryptography even is. *Cryptography* is the study of keeping secret data safe from potential bad actors. Whenever you send your password to a website over a secure connection, you're not actually sending your password as a string to the website, since anyone intercepting the communication channel would then be able to steal your password and (since you use the same password for everything) hack your bank accounts, etc. The way to get around this is to *encrypt* it with a secret encryption key. In our simplified account of things, let's say that your password (after converting to binary) is $p = 10100111$. Suppose the secret encryption key, that both you and the website know but no one else does, is $k = 11011101$. Instead of sending your password, you first scramble it by performing *bit addition* $\oplus$ with the secret key. To do bit addition, line up the two binary numbers, and add them like you would ordinary numbers, but $1 + 1 = 0$ and you never have to carry anything:

$$\begin{array}{rr} & 10100111 \\ \oplus & 11011101 \\ \hline & 01111010 \end{array}$$

So instead of sending $p$, you send the *encrypted* string $s = 01111010$. How can the website recover your password $p$ from this? Very conveniently, you can get $p$ back from $s$ (i.e., you can *decrypt* the message) by just doing bit addition with the secret key $k$ once more! Behold:

$$\begin{array}{rr} & 01111010 \\ \oplus & 11011101 \\ \hline & 10100111 \end{array}$$

It seems like we have found a safe method for sending passwords. But this begs the question: *How do we communicate the secret key to each other?* The website can't possibly use the same secret key with all users, since then we could all steal each other's passwords. We also can't just send the secret key beforehand over the communication channel beforehand, since any malicious agents listening would then know the key, and could perform the decryption process for themselves.

This is where elliptic curves come in. It provides us a way for your computer and the website to collectively decide upon a secret key *without actually sending any sensitive data over the communication channel.* The key insight is that if you have a point $P$ on an elliptic curve, and you calculate $P + P$ using the tangent line method, then knowing $2 \cdot P = P + P$, it's kind of difficult to find out what

$P$ was in the first place. Here's something that's even harder. If you pick some largish integer $n$ and calculate $R = nP$, which is some other point on the elliptic curve, and then show me both $P$ and $R$, it is very difficult for me to find out what $n$ is.

So here's the scheme. You want to send a password to a website securely. The website picks some suitably large number $m$ randomly, and sends you both the point $P$ and the point $mP$ (after encoding this data in binary). You pick some suitably large number $n$ randomly, and send the website the binary string representing the point $nP$. Any evil person tapping the communication channel now knows $P$, $mP$, and $nP$, but as discussed earlier this does not allow them to find out what the numbers $m$ and $n$ are. Now you know $n$ and you know $mP$, so you can calculate $n(mP) = (nm)P$ using elliptic curve addition. The website knows $m$ and it knows $nP$, so it can calculate $m(nP) = (mn)P$. There's no difference between $(nm)P$ and $(mn)P$, because both just mean $P$ (elliptically) added to itself $mn = nm$ times. You and the website now have access to the exact same binary string, namely the binary representation of the point $(mn)P$! You can safely use this as the secret key, encrypt your password with it, and send that encrypted message over the communication channel. Any hackers listening in will see only the encrypted message but not know the secret key, so your password is safe from them, and the website can decrypt the message using the same secret key, in order to see your password.

**Rederiving old formulas implicitly.** With this technique of implicit differentiation, we could actually have worked out the derivatives of the natural logarithm, roots, and inverse trigonometric functions without appealing to Theorem 49. The only downside to this approach is that it does not actually tell us that the functions ln, $\sqrt[n]{x}$, arcsin, etc. are differentiable, unlike Theorem 49. We have to assume they are differentiable in order to use the technique. Putting that nitpick aside, rederiving some of these formulas is a good way to practise implicit differentiation, so let us do that now.

The natural logarithm ln is such that if

$$y = \ln x,$$

then

$$e^y = x.$$

By differentiating both sides (using the chain rule on the left-hand side), we see that

$$e^y \frac{dy}{dx} = 1,$$

whence

$$\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x},$$

as we found out in the previous section.

Let's try an inverse trigonometric function now. We'll find the derivative of arccsc, which was skipped over in the proof of Theorem 52 above. We have

$$\csc(\operatorname{arccsc} x) = x,$$

so by the chain rule,

$$-\csc(\operatorname{arccsc} x)\cot(\operatorname{arccsc} x) \cdot \frac{d}{dx}\operatorname{arccsc} x = 1$$

follows from differentiating both sides. By drawing a suitable triangle, we see that $\cot(\operatorname{arccsc} x) = \sqrt{x^2-1}$, and of course $\csc(\operatorname{arccsc} x) = x$, so we find that

$$\frac{d}{dx}\operatorname{arccsc} x = \frac{-1}{x\sqrt{x^2-1}},$$

which is exactly claim (iv) of Theorem 52.

It is possible to rederive Theorem 50 as well using implicit differentiation, but as we hinted at before, one can actually state and prove a general formula for the derivative of the function $x^r$, where $r$ is any real number (so this applies to all integer powers $r = n$ as well as all roots $r = 1/n$). The following theorem subsumes Theorems 40 and 50.

**Theorem 55** (*General power rule*). *Let $r$ be any real number. Then*

$$\frac{d}{dx}x^r = rx^{r-1}. \tag{33}$$

*(We define $0^0 = 1$ to make this formula work for $r = 1$ and $x = 0$.)*

*Proof.* We wish to find $dy/dx$, where

$$y = x^r.$$

The trick is to take the logarithm on both sides (we cannot take the logarithm of a negative number, but we get around this by taking the logarithm of the *absolute value* of both sides). We have

$$\ln|y| = \ln|x^r| = \ln\big(|x|^r\big) = r\ln|x|,$$

which, after differentiating both sides, yields

$$\frac{1}{y}\frac{dy}{dx} = \frac{r}{x}.$$

Isolating for $dy/dx$ gives us

$$\frac{dy}{dx} = \frac{ry}{x} = \frac{rx^r}{x} = rx^{r-1}.$$

The alert reader will notice that this does not allow us to find the derivative at $x = 0$, since in that case $y = x^r = 0$ and $\ln|y|$ is undefined. This special case has to be dealt with separately, directly from the definition of limits. If $r = 1$, then $x^r$ is just $x$, and the derivative is 1 everywhere, even at $x = 0$, which is what (33) prescribes. When $r \neq 0$, we have

$$\frac{d}{dx}x^r\Big|_{x=0} = \lim_{x\to 0}\frac{x^r - x^0}{x - 0} = \lim_{x\to 0} x^{r-1} = 0,$$

which means the claim is still true. (This is because $r \cdot 0^{r-1}$ is equal to 0 when $r > 1$ and blows up to infinity when $r < 1$, which the right-hand side of (33) does as well.) ∎

23.x **Logarithmic differentiation.** To prove the general power rule, we took the logarithm of both sides of the equation before differentiating. This allowed us to pull the exponent $r$ of $x$ down in front. Taking the logarithm before differentiating implicitly is useful in other cases as well. The laws

$$\ln(ab) = \ln a + \ln b \qquad \text{and} \qquad \ln\Big(\frac{a}{b}\Big) = \ln a - \ln b$$

tell us that if we take the logarithm before differentiating, then a complicated product or quotient can reduce to a sum or a difference instead, which is easier to take the derivative of. Of course this manoeuvre adds a bit of complexity because we have to invoke the chain rule with all the resulting logarithms, but anyone who has tried applying the product and quotient rules multiple times to a convoluted expression will be convinced that exchanging these two rules for the chain rule may be worth it.

In the following example that demonstrates logarithmic differentiation, we find the *normal line* to a graph at a point instead of the tangent line. This is the unique line intersecting the curve at that point *perpendicularly* to the tangent line. If the slope of the tangent line is $m$, then the slope of the normal line is $-1/m$. (If the tangent line is horizontal, then the normal line is vertical.)

**Proposition 56.** *The normal line to the graph*

$$f(x) = \frac{(x-3)^5 x^{3/2}}{\sqrt{3+x^2}}$$

*at the point* $(1, -16)$ *is given by the equation*

$$y + 16 = -\frac{1}{20}(x - 1)$$

*Proof.* Let $y = f(x)$; we want to find $dy/dx$, but first let us take the logarithm of (the absolute value of) both sides. This results in

$$\begin{aligned}\ln|y| &= \ln\left(\left|\frac{(x-3)^5 x^{3/2}}{\sqrt{3+x^2}}\right|\right)\\ &= \ln\big(|x-3|^5\big) + \ln\big(|x|^{3/2}\big) - \ln\big(|3+x^2|^{1/2}\big)\\ &= 5\ln|x-3| + \frac{3}{2}\ln|x| - \frac{1}{2}\ln|3+x^2|.\end{aligned}$$

(The identities

$$|ab| = |a| \cdot |b|, \qquad \left|\frac{a}{b}\right| = \frac{|a|}{|b|}, \qquad \text{and} \qquad |a^r| = |a|^r$$

are useful to keep in mind when carrying out logarithmic differentiation.) Now, taking the derivative of both sides gives us

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{5}{x-3} + \frac{3}{2x} - \frac{1}{2(3+x^2)} \cdot 2x,$$

from which we see that

$$\frac{dy}{dx} = y\left(\frac{5}{x-3} + \frac{3}{2x} - \frac{x}{3+x^2}\right).$$

Substituting the definition of $y = f(x)$ into this equation tells us that

$$f'(x) = \frac{(x-3)^5 x^{3/2}}{\sqrt{3+x^2}}\left(\frac{5}{x-3} + \frac{3}{2x} - \frac{x}{3+x^2}\right).$$

To find the slope of the normal line at $(1, -16)$, we first find the slope of the tangent line at that point, which is

$$\begin{aligned} f'(x) &= \frac{(1-3)^5 1^{3/2}}{\sqrt{3+1^2}}\left(\frac{5}{1-3} + \frac{3}{2} - \frac{1}{3+1^2}\right) \\ &= \frac{(-2)^5}{2}\left(-\frac{5}{2} + \frac{3}{2} - \frac{1}{4}\right) \\ &= -16\frac{-5}{4} \\ &= 20. \end{aligned}$$

Hence the slope of the normal line is the negative reciprocal $-1/20$ of 20, and of course $(1, -16)$ lies on this line, so the equation of the normal line, in slope-point form, is $y - (-16) = -(1/20)(x-1)$, as claimed. ∎

You could use the product and quotient rules to derive exactly the same result, but the intermediate expression would be slightly more complicated. That said, if you're just going to plug in $x = 1$, maybe it doesn't matter so much—ultimately it is a matter of taste which approach you choose to follow for problems like these.

Here is another derivative that is easy to find using logarithmic differentiation.

**Proposition 57.** *The function*

$$f(x) = x^x$$

*has as its derivative the function*

$$f'(x) = x^x(1 + \ln x).$$

*Proof.* Let $y = f(x)$ and observe that

$$\ln|y| = \ln\big(|x^x|\big) = x \ln|x|.$$

This means that

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + 1 \cdot \ln x,$$

whence

$$\frac{dy}{dx} = y(1 + \ln x).$$

But $y = f(x) = x^x$, so the proposition follows. ∎

This confirms our intuition that, for large $x$, the function $x^x$ grows much faster than $b^x$ for any constant $b$; whereas $(d/dx)b^x$ was $b^x$ times a constant (namely, $\ln b$), the derivative of $x^x$ is itself times something that still continues to grow with $x$. You could also try differentiating $x^x$ without resorting to logarithmic differentiation by writing $x^x = e^{x \ln x}$, and then using the chain rule.

**Functions characterised by differential equations.** When we differentiate an equation involving $y$ and $x$ implicitly with respect to $x$, we end up with an equation that relates $y$ with its derivative $y'$. For instance, the equation $x^2 + y^2 = 1$ defining a circle gave rise to the equation $y' = -x/y$. In general, an equation that contains a variable as well as its derivatives (including higher derivatives) is called a *differential equation*.

For instance, we saw that if $y = e^x$ then $y' = e^x$, so $y$ satisfies the very nice differential equation $y' = y$. We might then ask the question, *Is $e^x$ the only function $f(x)$ such that $f' = f$?* The answer is "not quite," since if $f(x) = 2e^x$, then we also have $f'(x) = 2e^x$, so again we have $f' = f$. Let's modify our question then: *Are the functions $f(x) = ce^x$, where $c$ is an arbitrary constant, the only functions $f$ satisfying the differential equation $f' = f$?* The answer is now yes.

**Theorem 58.** *If $f(x)$ is a function such that $f(x) = f'(x)$ for all real numbers $x$, then $f(x) = ce^x$ for some constant $c$.*

*Proof.* We take the derivative of the fraction $f(x)/e^x$ using the quotient rule. This gives us

$$\frac{d}{dx}\frac{f(x)}{e^x} = \frac{e^x f'(x) - f(x)e^x}{e^{2x}} = \frac{f'(x) - f(x)}{e^x},$$

but since $f(x)-f'(x)=0$ for all $x$, we see that $f(x)/e^x$ has derivative 0, meaning that it is a constant function. So there is some constant $c$ such that

$$\frac{f(x)}{e^x} = c$$

for all $x$, and hence $f(x)$ must equal $ce^x$ everywhere. ∎

As mentioned earlier, differential equations can also involve higher-order derivatives like $f''$ and $f'''$. This is often the result of differentiating implicitly multiple times. For instance, if we implicitly differentiate the equation

$$y^2 = x^3 + ax + b$$

of a general elliptic curve, we obtain

$$2yy' = 3x^2 + a.$$

Differentiating implicitly once more (using the product rule on the left-hand side), we obtain

$$2\big((y')^2 + yy''\big) = 6x,$$

whence one can isolate for

$$y'' = \frac{3x-(y')^2}{y} = \frac{3x-\big(3x^2+a/(2y)\big)^2}{y} = \frac{12xy^2-9x^4-6ax^2-a^2}{4y^3}.$$

If you want to solve for $y''$ at a particular $x$ on (say) the positive branch of $y$, then you can substitute $y=\sqrt{x^3+ax+b}$ to obtain a hideous formula for $y''$ solely in terms of $x$.

Let's consider the differential equation

$$y''+y=0. \tag{34}$$

Can you think of any functions $y=f(x)$ that satisfy it? Well, $\sin' = \cos$ and $\cos' = -\sin$, so we see that $y=\sin x$ and $y=\cos x$ *both* satisfy this differential equation. Are these the only two possible solutions to (34)? As with the equation $y'=y$, the constant function 0 is also a solution to $y''+y=0$, and in fact it is the only solution if we add the extra assumptions that $f(0)=0$ and $f'(0)=0$. This will help us solve the differential equation in general, so let us first prove this special case as a lemma.

**Lemma 59.** *If $g(x)$ satisfies $g''(x)-g(x)=0$ for all $x$, and furthermore we have $g(0)=0$ and $g'(0)=0$, then $g$ is identically 0 (i.e., $g(x)=0$ for all $x$).*

*Proof.* The trick is to consider the derivative

$$\frac{d}{dx}\big(g(x)^2+g'(x)^2\big).$$

By the chain rule, this derivative equals

$$2f(x)g'(x) + 2g'(x)g''(x) = 2g'(x)\big(g(x) + g''(x)\big).$$

But we know that $g(x) + g''(x) = 0$ for all $x$, so we see that the derivative of $g(x)^2 + g'(x)^2$ is 0. This means that it is constant, and the fact that

$$g(0)^2 + g'(0)^2 = 0 + 0 = 0$$

further informs us that this constant is 0. In other words,

$$g(x)^2 + g'(x)^2 = 0$$

for all $x$, and (since both terms on the left-hand side are nonnegative), $g(x) = 0$ for all $x$. ∎

26.x If you mess around a bit, you might notice that the function $y = \sin x + \cos x$ also satisfies $y + y'' = 0$. Then the experience points you gained from seeing that the general solutions to $y' = y$ were $ce^x$ for all constants $c$ would lead you to check whether, for example, $f(x) = 2\sin x + 3\cos x$ satisfies (34) as well, and you'll find that it does. So your final educated guess is that all of the solutions to $y'' + y = 0$ are of the form $y = f(x) = a\sin x + b\cos x$, where $a$ and $b$ are constants (which could both be 0, as Lemma 59 showed). This is indeed the answer.

**Theorem 60.** *If $y = f(x)$ satisfies $f''(x) + f(x) = 0$ for all $x$, then letting $a = f'(0)$ and $b = f(0)$, we have*

$$f(x) = a\sin x + b\cos x.$$

*Proof.* Let

$$g(x) = f(x) - a\sin x - b\cos x.$$

The goal is to show that $g$ is identically 0, and we have a handy lemma at our disposal for doing such a thing. So let's show that $g$ satisfies all three hypotheses of Lemma 59.

First, we want to show that $g'' + g = 0$. We calculate

$$g'(x) = f'(x) - a\cos x + b\sin x$$

and

$$g''(x) = f''(x) + a\sin x + b\cos x$$

. Hence

$$\begin{aligned} g''(x) + g(x) &= f''(x) + a\sin x + b\cos x + \big(f(x) - a\sin x - b\cos x\big) \\ &= f''(x) + f(x) \\ &= 0 \end{aligned}$$

for all $x$.

Now we have to check that $g(0) = 0$ and $g'(0) = 0$. Well,

$$g(0) = f(0) - a\sin 0 - b\cos 0 = f(0) - b = f(0) - f(0) = 0,$$

and

$$g'(0) = f'(0) - a\cos 0 + b\sin 0 = f'(0) - a = f'(0) - f'(0) = 0.$$

So indeed these identities both hold.

This means that $g$ satisfies the conditions of Lemma 59, and therefore $g(x) = 0$ for all $x$. In other words,

$$f(x) = a\sin x + b\cos x$$

is true for all $x$. ∎

In fact, we can *define* the function sin to be the unique function $y = f(x)$ satisfying

$$y + y'' = 0, \qquad f(0) = 0, \qquad \text{and} \qquad f'(0) = 1;$$

likewise we can define the function cos to be the unique function $y = f(x)$ with

$$y + y'' = 0, \qquad f(0) = 1, \qquad \text{and} \qquad f'(0) = 0; \tag{35}$$

Armed with only these two definitions, let $g = \sin'(x)$. By differentating the identity $\sin'' + \sin = 0$, we obtain

$$g'' + g' = \sin''' + \sin' = 0.$$

Then, since $g(0) = \sin'(0) = 1$ and

$$g'(0) = \sin''(0) = \sin(0) = 0,$$

we see that $g$ satisfies all the properties listed in (35), and we must therefore have $g(x) = \cos x$ for all $x$. This proves that $\sin' = \cos$. In a similar fashion, we can prove that $\cos' = -\sin$, and from there we'd be able to prove all the other properties of sin and cos that we know and love, having begun only with these alternative differential definitions of these two functions.

***A famous identity due to Euler.** We all know that there is no real number whose square is $-1$; in other words, the polynomial $x^2 + 1$ has no real roots. We can artificially define a number whose square is $-1$. It is customary to call this number $i$ (except in some engineering disciplines, where it is called $j$ instead). It is not a real number, but instead is used to construct the *complex numbers*. A full description of the rules governing $i$ and its complex brethren falls outside the scope of this course, but what we are able to do is prove a very famous identity of Euler.

Consider the function $y = e^{ix}$. Since $i$ is just a number, we can differentiate this function (applying the chain rule) to get $y' = ie^{ix}$ and $y'' = i^2 e^{ix}$. But we defined $i^2 = -1$, so in fact $y'' = -e^{ix}$, and we see that $y'' + y = 0$. Aha! So Theorem 60 applies to this function $f(x) = e^{ix}$. We have $f'(0) = ie^{i0} = i$ and $f(0) = e^{i0} = 1$, so the theorem gives

$$e^{ix} = i \sin x + \cos x$$

for all real numbers $x$. Some people call this *Euler's formula*, after Leonhard Euler. The famous special case known as *Euler identity* results from setting $x = \pi$ in this formula. We have

$$e^{i\pi} = i \sin \pi + \cos \pi = 0i + (-1) = -1.$$

Complex numbers have plenty of applications in the natural sciences and engineering. To learn more about them, you could take a class such as MATH 316 Complex Variables.

**The catenary.** We now know that any function $y = f(x)$ which is a solution to the differential equation $y' = y$ must be equal to $ce^x$ for some constant $c$, and any solution to $y'' + y = 0$ must be equal to $a \sin x + b \cos x$ for some constants $a$ and $b$. We now derive something similar for the differential equation $y'' = y$, which can also be written $y'' - y = 0$. In other words, we will find two functions $s(x)$ and $c(x)$ such that any function $y = f(x)$ satisfying $y'' = y$ must be equal to $as(x) = bc(x)$ for some constants $a$ and $b$.

It's easy to see that the constant function $y = 0$ satsifies $y'' = y$, and so does $y = e^x$. So whatever the mysterious functions $s(x)$ and $c(x)$ end up being, we must have $e^x = as(x) + bc(x)$ for some constants $a$ and $b$. We'll refrain from giving the formulas for $s(x)$ and $c(x)$ right away, but eagle-eyed readers will notice that already in these notes we have seen another function $y = f(x)$ that satisfies $y'' = y$. This is actually the function $s(x)$. You can try to find this function by scrolling back in the notes, or you can just strap yourself in for the ride.

To motivate the fact that $s(x)$ and $c(x)$ have some important applications, we'll introduce $c(x)$ as the solution to a real-life physics problem. Suppose that a wire is suspended between the tops of two tall poles (as on a telephone line). Let us take the origin to be the lowest point on the wire, so that as $x$ increases or decreases, $y$ increases. The question is, *What is an expression for $y$ in terms of $x$?* You might think that this shape is a parabola, and in fact the correct function describing $y$ is very close (in shape, not in formula) to a parabola, but ultimately it is not a parabola.

Let's make our lives easier by considering only half of this picture: the side where $x$ is positive. To determine what $y$ in terms of $x$, it would be prudent to analyse the angle at which the wire hangs, at any given point on the wire. This is because this angle controls how fast $y$ changes in terms of $x$ (if the angle is $\theta$, then the slope of the tangent line is $dy/dx = \tan \theta$).

Consider any point $P$ on the wire, and note that the point is stationary. Therefore, all of the forces acting upon $P$ are balanced. A moment's consideration convinces us that the downward force acting on that point must come solely from all the weight of the wire that is beneath that point. This depends on the weight wire used, but if the wire has a weight of $w$ per unit length, and the length of the wire below $P$ is $s$, then a downward force of $ws$ acts upon $P$ (and is counteracted by the tension of the wire above $P$, so that $P$ does not move along the $y$-direction).

The horizontal forces are also balanced. There is some horizontal force $h$ coming from the line being tied to the pole. The exact quantity $h$ depends on the type of wire used and how tightly it is strung at the top of the pole. What we do know is that since all the forces are balanced, we have $\tan\theta = ws/h$. (We reasoned earlier that $dy/dx = \tan\theta$, so we already have a formula for $dy/dx = ws/h$. But since we don't have a formula for $s$ in terms of $x$ yet, and since we might still not be able to find out a formula for $y$ even if we had $dy/dx$, this isn't so helpful at the moment.)

At the lowest point of the wire, $\theta = 0$, and as it increases $s$ increases as well. We have an exact formula for $s$ in terms of $\theta$, namely

$$s = \frac{h}{w}\tan\theta.$$

The parameters $h$ and $w$ depend only on the wire and how much slack it's given when it's tied to the pole, so let us suppose that we have an ideal wire under ideal conditions, making $h/w = 1$. Then we just have $s = \tan\theta$, which will make calculations a lot easier. From here, we can very easily calculate

$$\frac{ds}{d\theta} = \sec^2\theta.$$

But we want to find formulas for $x$ and $y$ (in the hopes of writing $y = f(x)$), so let's start by deriving formulas for $dx/d\theta$ and $dy/d\theta$. First we see that at any point on the wire where the angle is $\theta$, if we change $s$ by some small amount $ds$, then $x$ changes by $\cos\theta$ times that small amount, and $y$ changes by $\sin\theta$ times that small amount. In other words,

$$\frac{dx}{ds} = \cos\theta \qquad \text{and} \qquad \frac{dy}{ds} = \sin\theta.$$

This is very nice, except for the fact that the derivatives are with respect to $s$ and yet the formulas are stated in terms of $\theta$. So we use the chain rule to observe that

$$\frac{dx}{d\theta} = \frac{dx}{ds}\cdot\frac{ds}{d\theta} = \cos\theta\sec^2\theta = \sec\theta$$

and

$$\frac{dy}{d\theta} = \frac{dy}{ds}\cdot\frac{ds}{d\theta} = \sin\theta\sec^2\theta = \sec\theta\tan\theta.$$

We are now faced with the challenging task of determining what the functions $x(\theta)$ and $y(\theta)$ are. In other words, what is the function $x(\theta)$ whose derivative is $dx/d\theta = \sec\theta$, and what is the function $y(\theta)$ whose derivative is $dy/d\theta = \sec\theta\tan\theta$. If you paid attention during our discussion of the Mercator projection, you'll remember that

$$x(\theta) = \ln\left(\tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right)\right)$$

has derivative $dx/d\theta = \sec\theta$, and if you know the derivatives of the trigonometric functions well, you'll remember that

$$y(\theta) = \sec\theta$$

has derivative $dy/d\theta = \sec\theta\tan\theta$. (Technically we have to add an arbitrary constant to each of the formulas for $x(\theta)$ and $y(\theta)$, since arbitrary constants have derivative 0, but let's not worry about this too much. Rest assured that the extra constant doesn't matter here.) We now have formulas for $x$ and $y$! The final step is to try and find a nice formula for $y$ in terms of $x$.

Let's clean things up a bit. The formula for $x$ in terms of $\theta$ has a ln in it, so taking exponentials of both sides, we find that

$$e^x = \tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right) = \frac{\sin(\theta/2 + \pi/4)}{\cos(\theta/2 + \pi/4)}.$$

The angle $\theta/2 + \pi/4$ is a bit of a pest here; but one might notice that it is half of the "nice" angle $\theta + \pi/2$. We might try to use the double angle formula $\sin(2\alpha) = 2\sin\alpha\cos\alpha$ somehow, but we'd need sin times cos to appear somewhere. Well why don't we try flipping the fraction around? We have

$$e^{-x} = \frac{\cos(\theta/2 + \pi/4)}{\sin(\theta/2 + \pi/4)},$$

and now if we add $e^x$ to $e^{-x}$ and cross-multiply, everything simplifies very nicely:

$$\begin{aligned}
e^x + e^{-x} &= \frac{\sin(\theta/2 + \pi/4)}{\cos(\theta/2 + \pi/4)} + \frac{\cos(\theta/2 + \pi/4)}{\sin(\theta/2 + \pi/4)} \\
&= \frac{\sin^2(\theta/2 + \pi/4) + \cos^2(\theta/2 + \pi/4)}{\sin(\theta/2 + \pi/4)\cos(\theta/2 + \pi/4)} \\
&= \frac{1}{(1/2)\sin(\theta + \pi/2)} \\
&= \frac{2}{\cos\theta} \\
&= 2\sec\theta
\end{aligned}$$

But $\sec\theta = y$, so we finally obtain an equation $e^x + e^{-x} = 2y$ relating $x$ and $y$. Isolating for $y$ yields the equation

$$y = \frac{e^x + e^{-x}}{2}$$

of the *catenary*: an idealised model of a rope or wire hanging between two poles supporting only its own weight.

We now define the *hyperbolic cosine* function by

$$\cosh x = \frac{e^x + e^{-x}}{2};$$

so the catenary created by the very specific wire under ideal tension conditions is exactly described by $y = \cosh x$. For general weight-per-unit-length and tension parameters $w$ and $h$, respectively, the equation of the resulting catenary is given by

$$\cosh x = \frac{h}{w}\cosh\Big(\frac{w}{h}\Big).$$

By multiplying the numerator and denominator by $e^x$, we obtain the alternative formula

$$\cosh x = \frac{e^{2x} + 1}{2e^x}.$$

Maybe now you realise where you've seen this expression before—it is the derivative of the function

$$\frac{e^{2x} - 1}{2e^x}$$

from Proposition 44. Let us define the *hyperbolic sine*

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{e^{2x} - 1}{2e^x}$$

to be the function from that proposition; then we know that $\sinh' = \cosh$.

28.x Furthermore, the calculation

$$\frac{d}{dx}\frac{e^x + e^{-x}}{2} = \frac{e^x}{2} + \frac{-e^{-x}}{2} = \frac{e^x - e^{-x}}{2}$$

tells us that $\cosh' = \sinh$. This should remind you of the relationships between the derivatives of sin and cos, but note that this time there is no negative sign in the derivative of cosh. Further contemplation reveals that

$$\sinh'' = \sinh \qquad \text{and} \qquad \cosh'' = \cosh,$$

so $y = \sinh x$ and $y = \cosh x$ are both solutions to $y'' = y$. The following theorem is the hyperbolic analogue to Theorem 60.

**Theorem 61.** *If $y = f(x)$ satisfies $f''(x) = f(x)$ for all $x$, then letting $a = f'(0)$ and $b = f(0)$, we have*

$$f(x) = a\sinh x + b\cosh x.$$

**Proof.* The full proof is a little outside the scope of this course. The main idea is to mimic the proofs of Lemma 59 and Theorem 60. First we show that if $g''(x) = g(x)$ for all $x$ and $g(0) = 0 = g'(0)$, then $g$ must be identically zero. This mirrors Lemma 59 but the details are a bit harder. (See Problem 43 in Chapter 18 of Spivak's book for a roadmap.) From there the proof is extremely similar to that of Theorem 60. We let

$$g(x) = f(x) - a\sinh x - b\cosh b,$$

and show that this choice of $g$ has $g'' = g$ and $g(0) = 0 = g'(0)$. ∎

If $f(x) = a\sinh x + b\cosh x$, then since both sinh and cosh are defined in terms of $e^x$ and $e^{-x}$, then we can alternatively express any solution to $y'' + y = 0$ as a linear combination of $e^x$ and $e^{-x}$.

**Corollary 62.** *If $y = f(x)$ satisfies $f''(x) = f(x)$ for all $x$, then*

$$f(x) = Ae^x + Be^{-x},$$

*where*

$$A = \frac{f'(0) + f(0)}{2} \qquad \text{and} \qquad B = \frac{f(0) - f'(0)}{2}.$$

*Proof.* Since $f''(x) = f(x)$ for all $x$, by Theorem 61 we have

$$f(x) = a\sinh x + b\cosh x,$$

where $a = f'(0)$ and $b = f(0)$ Expanding the definitions of sinh and cosh, we find that

$$\begin{aligned} f(x) &= a \cdot \frac{e^x - e^{-x}}{2} + b \cdot \frac{e^x + e^{-x}}{2} \\ &= \frac{a+b}{2} \cdot e^x + \frac{b-a}{2} \cdot e^{-x}, \end{aligned}$$

which is exactly the claim. ∎

***A linear-algebraic interlude.** If you are taking (or have taken) MATH 133 or any other linear algebra class, you might feel that these statements have some kind of linear-algebraic flavour. Let us summarise the results of the preceding few pages in the language of linear algebra. The set of functions $y = f(x)$ that are solutions to $y' = y$ is a one-dimensional vector space. The function $e^x$ is a nonzero element of this vector space, and hence all elements in the vector space can be written as some multiple of $e^x$. Likewise, the set of functions solving $y'' + y = 0$ form a two-dimensional vector space, and a basis for this space is given by the two functions sin and cos. Lastly, the set of functions satisfying

$y'' = y$ is also two-dimensional, and there are two different but equally natural choices for bases we can choose from, namely $\{\sinh, \cosh\}$ and $\{e^x, e^{-x}\}$. The computation in the preceding corollary tells us that a vector written as $(a, b)$ under the first coordinate system can be converted to the second coordinate system by means of the matrix-vector equation

$$\begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}.$$

If you're wondering why the set of solutions to $y' = y$ is one-dimensional whereas the sets of solutions to $y'' + y = 0$ and $y'' = y$ are both two-dimensional, you'll have to take a higher-level course on differential equations, e.g., MATH 315. Suffice it to say, it isn't a coincidence that we saw an equation involving only the first derivative giving rise to a space of dimension 1, and the equations involving second derivatives creating spaces of dimension 2.

**Hyperbolic functions.** We now know the definitions of $\sinh x$ and $\cosh x$. They are analogous to sin and cos, but for the differential equation $y'' - y = 0$ instead of $y'' + y = 0$. The ordinary sine and cosine functions are used to define four more trigonometric functions, and we can do something analogous for hyperbolic functions. We define the *hyperbolic tangent* function tanh by

$$\tanh = \frac{\sinh}{\cosh};$$

since $\cosh x > 1$ for all $x$, $\tanh x$ is well defined for all real numbers $x$. As $x \to -\infty$ we have $\tanh x \to -1$, and as $x \to \infty$ we have $\tanh x \to 1$. This function is used for many statistical regression problems, including machine learning. (Often, tanh is shifted and scaled to something like $\tanh /2 + 1/2$ so that its range is $(0, 1)$ instead of $(-1, 1)$. Then it can be used to model *probabilities*. For example, the probability of passing a calculus exam graphed as a function of time spent learning the material looks a bit like this function.)

Just as we defined csc, sec, and cot as the inverses of sin, cos, and tan respectively, we can define

$$\operatorname{csch} = \frac{1}{\sinh}, \qquad \operatorname{sech} = \frac{1}{\cosh}, \qquad \text{and} \qquad \coth = \frac{1}{\tanh}.$$

Here are some important identities for hyperbolic functions. Note the similarities and differences with the analogous trigonometric identities.

**Proposition 63.** *We have*

*i)* $\sinh(-x) = -\sinh x$ *for all* $x$*;*

*ii)* $\cosh(-x) = \cosh x$ *for all* $x$*;*

*iii)* $\cosh^2 - \sinh^2 = 1$*;*

*iv)* $\tanh^2 + \operatorname{sech}^2 = 1$*;*

*v)* $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$ *for all* $x$ *and* $y$*; and*

*vi)* $\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$ *for all* $x$ *and* $y$*.*

*Proof.* Identities (i) and (ii) are easy to verify and left as an exercise. We prove (iii) in the same way that we "proved" (27) with calculus, only this time the argument isn't circular. Without using identity (iii), we know that $\sinh' = \cosh$ and $\cosh' = \sinh$, so by the chain rule we have

$$\frac{d}{dx}(\cosh^2 x - \sinh^2 x) = 2\cosh x \sinh - 2\sinh\cosh = 0.$$

This means that $\cosh^2 - \sinh^2$ is a constant function. Now we plug in, say, $x = 0$ to find that this constant is 1. Identity (iv) follows from dividing both sides of identity (iii) by $\cosh^2$.

To prove identity (v), it is perhaps best to expand the right-hand side and then simplify:

$$\begin{aligned}
\sinh x \cosh y + \cosh x \sinh y &= \frac{e^x - e^{-x}}{2} \cdot \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \cdot \frac{e^y - e^{-y}}{2} \\
&= \frac{e^{x+y} - e^{-x+y} + e^{x-y} - e^{-x-y}}{4} \\
&\qquad + \frac{e^{x+y} + e^{-x+y} - e^{x-y} - e^{-x-y}}{4} \\
&= \frac{2e^{x+y} - 2e^{-x-y}}{4} \\
&= \sinh(x+y)
\end{aligned}$$

Identity (vi) is proved similarly; this is left as an exercise. ∎

The following proposition records the derivatives of all the hyperbolic functions.

**Theorem 64.** *We have*

*i)* $\sinh' = \cosh$*;*

*ii)* $\cosh' = \sinh$*;*

*iii)* $\tanh' = \operatorname{sech}^2$*;*

*iv)* $\operatorname{csch}' = -\coth \cdot \operatorname{csch}$*;*

*v)* $\operatorname{sech}' = -\operatorname{sech} \cdot \tanh$*; and*

*vi)* $\coth' = -\operatorname{csch}^2$*.*

*Proof.* We already proved (i) and (ii) in the preceding discussion. To prove (iii) we use the quotient rule; it gives

$$\tanh' = \left(\frac{\sinh}{\cosh}\right)' = \frac{\cosh^2 - \sinh^2}{\cosh^2} = \operatorname{sech}^2,$$

where in the last equality we have invoked identity (iii) from Proposition 63.

The derivatives (iv), (v), and (vi) are proved using the reciprocal law. We prove (v) here and leave the other two as an exercise. We have

$$\operatorname{sech}' = \left(\frac{1}{\cosh}\right)' = \frac{-\cosh'}{\cosh^2} = \frac{-\sinh}{\cosh^2} = -\operatorname{sech}\tanh. \quad \blacksquare$$

We finish our discussion of hyperbolic functions by briefly mentioning the inverse hyperbolic functions. These are often denoted by $\sinh^{-1}$, $\cosh^{-1}$, etc., but we will use the notation arsinh, arcosh, etc. instead. The reason the prefix is "ar-" and not "arc-" is pedantry. If $x = \cos\theta$, then $\theta$ is the length of the arc of the unit circle from $(1,0)$ to $\big(x, \sin(\arccos x)\big)$, which is where the term $\arccos x$ comes from. By analogy, people started using terms like "arcsinh" and "arccosh" for hyperbolic inverses as well. But technically none of these functions represent the arc of anything, so eventually this was corrected to arsinh and so on, where "ar-" is short for "argument." If $y = \sinh x$, then $x$ is the argument of the sinh function that yields $y$, so $x = \operatorname{arsinh} y$.

Since the hyperbolic functions admit explicit formulas in terms of $x$, their inverse functions can be solved explicitly.

**Proposition 65.** *The explicit formulas for the inverse hyperbolic functions are*

*i)* $\operatorname{arsinh} x = \ln\big(x + \sqrt{x^2+1}\big)$;

*ii)* $\operatorname{arcosh} x = \ln\big(x + \sqrt{x^2-1}\big)$;

*iii)* $\operatorname{artanh} x = (1/2)\ln\big((1+x)/(1-x)\big)$;

*iv)* $\operatorname{arcsch} x = \ln\big(1/x + \sqrt{1/x^2+1}\big)$;

*v)* $\operatorname{arsech} x = \ln\big(1/x + \sqrt{1/x^2-1}\big)$; *and*

*vi)* $\operatorname{arcoth} x = (1/2)\ln\big((x+1)/(x-1)\big)$.

*Proof.* All of these formulas are proved the same way, but the method is fairly tedious and involves no calculus, so we only prove (i) and leave the rest as an exercise. The idea is to substitute a variable in for $e^y$, and then use the quadratic formula.

Let $y = \operatorname{arsinh} x$. We want a formula for $y$. By the definition of inverses, we have

$$x = \sinh y = \frac{e^y - e^{-y}}{2} = \frac{e^{2y}-1}{2e^y}.$$

To make things notationally easier, write $u = e^y$. Then we have

$$x = \frac{u^2-1}{2u},$$

or equivalently

$$2xu = u^2 - 1.$$

Now we can use the quadratic formula to solve the equation $u^2 - 2xu - 1 = 0$. We get

$$u = \frac{2x \pm \sqrt{(-2x)^2 - 4(-1)}}{2} = x \pm \frac{\sqrt{4x^2+4}}{2} = x \pm \sqrt{x^2+1}.$$

To check which branch of the $\pm$ we take, note that $u = e^y > 0$, but $x - \sqrt{x^2+1} < 0$, so only the positive branch makes sense. We find that

$$e^y = u = x + \sqrt{x^2+1},$$

so

$$y = \ln\bigl(x + \sqrt{x^2+1}\bigr),$$

proving (i). ∎

From these formulas it is clear that we can also find the derivatives of all the inverse hyperbolic functions using the chain rule. We could also use implicit differentiation or the formula for the derivatives of inverses.

**Proposition 66.** *The derivatives of the inverse hyperbolic functions are*

*i)* $\operatorname{arsinh}' x = 1/\sqrt{1+x^2}$;
*ii)* $\operatorname{arcosh}' x = 1/\sqrt{x^2-1}$;
*iii)* $\operatorname{artanh}' x = 1/(1-x^2)$;
*iv)* $\operatorname{arcsch}' x = -1/\bigl(|x|\sqrt{x^2+1}\bigr)$;
*v)* $\operatorname{arsech}' x = -1/\bigl(x\sqrt{1-x^2}\bigr)$; *and*
*vi)* $\operatorname{arcoth}' x = 1/(1-x^2)$.

*Proof.* We'll sketch three different ways to prove (i). The first way is to simply use the explicit formula

$$\operatorname{arsinh} x = \ln\bigl(x + \sqrt{x^2+1}\bigr).$$

Differentiating the right-hand side using the chain rule gives the desired formula, after a rumble and a tumble.

The second way is to use implicit differentiation. If $y = \operatorname{arsinh} x$, then $x = \sinh y$. Differentiating both sides of this, we have

$$1 = \cosh y \frac{dy}{dx},$$

so

$$\frac{dy}{dx} = \frac{1}{\cosh y}. \tag{36}$$

But $\cosh y = \sqrt{1+\sinh^2 y} = \sqrt{1+x^2}$, so we see that $dy/dx = 1/\sqrt{1+x^2}$.

The third way is to simply use the formula for the derivatives of inverses. We have

$$\operatorname{arsinh}' x = (\sinh^{-1})'(x) = \frac{1}{\sinh'(\operatorname{arsinh} x)} = \frac{1}{\cosh(\operatorname{arsinh} x)},$$

which is exactly the same thing as (36). From there we could use the identity $\cosh^2 y - \sinh^2 y = 1$ to get the correct answer as in the previous paragraph. There is a more geometric interpretation of this argument. Since $\cosh^2 - \sinh^2 = 1$, you can draw a right triangle with hypotenuse $\cosh(\operatorname{arsinh} x)$ and the other sides set to the lengths $\sinh(\operatorname{arsinh} x) = x$ and 1. This means that

$$\cosh(\operatorname{arsinh} x) = \sqrt{1 + x^2}.$$

Proving (ii) through (vi)—using whatever method you wish—is left as an exercise. If you use the triangle method, you may have to draw a triangle using the identity $\tanh^2 + \operatorname{sech}^2 = 1$ instead, but the ideas remain the same. ∎

Note that the derivatives of artanh and arcoth have exactly the same formula, but they have disjoint domains: artanh is defined only on $(-1, 1)$ and arcoth is defined only *outside* this interval.

## 10. Related rates

30.x In the previous section we got used to looking at equations involving two variables and differentiating both sides with respect to one of the variables in order to obtain some relation between the other variable and its derivative. This section continues this practice of differentiating both sides of an equation, but this time we are concerned with taking derivatives with respect to another variable not present in the equation—often this variable will represent *time*. The resulting equation will relate the derivatives of the different variables, which can be useful to determine some rates of change that are difficult to calculate on their own. This method is best explained by means of short examples, so that's what this section will mostly consist of.

**Traffic radar systems.** Radar can give us two pieces of information about a nearby object: how far away this object is from the radar transmitter and how fast the object is moving towards or away from the transmitter. Suppose that a radar system is placed 5 metres away from a straight road and picks up a car moving away from it. At the instant that the car is 13 metres away from the transmitter, the system measures the speed of the car at 36 metres per second. The question is, How fast is the car actually moving on the road?

Call the car's position on the straight road $x$, where $x = 0$ indicates the point on the road closest to the photo radar. Call the car's distance to the radar $h$. Then since the radar is 5 metres from the road, by the Pythagorean theorem $x$ and $h$ satisfy the equation

$$x^2 + 5^2 = h^2.$$

By differentiating this entire equation with respect to a variable $t$ representing time, we have

$$2x\frac{dx}{dt} = 2h\frac{dh}{dt}.$$

The radar can only measure $dh/dt$; when $h = 13$ metres, it senses that $dh/dt$ is 36 metres per second. We want to know $dx/dt$; fortunately we already have all the information we need to calculate this figure. When $h = 13$, we have

$$x^2 = 13^2 - 5^2 = 169 - 25 = 144,$$

so $x = 12$ and

$$\frac{dx}{dt} = \frac{h(dh/dt)}{x} = \frac{13 \cdot 36}{12} = 39$$

metres per second. (This is roughly 140 kilometres per hour.)

**A conical water tower.** Suppose that a cone-shaped water tower dispenses water at a constant rate of 10 cubic metres per minute. The cone has an angle of $\pi/4$, so that the radius of the top of the water is the same as the total depth of the water (distance from the top of the water to the deepest point of the cone). Suppose we want to find out how fast the radius of the top of the water is changing when the depth of the water is 15 metres.

We need to know the formula for the volume of a cone, which is

$$V = \frac{1}{3}Ah,$$

where $A$ is the area of the cone's base (the top of the water in this example) and $h$ is the height (the water's depth). But for this specific cone, we have $r = h$, so $A = \pi r^2 = \pi h^2$. This means that $h = \sqrt{A/\pi}$, and plugging this in above yields

$$V = \frac{1}{3\sqrt{\pi}}A^{3/2}.$$

Differentiating both sides with respect to time gives us

$$\frac{dV}{dt} = \frac{1}{3\sqrt{\pi}} \cdot \frac{3}{2}A^{1/2}\frac{dA}{dt} = \frac{\sqrt{A}}{2\sqrt{\pi}} \cdot \frac{dA}{dt}.$$

We want to know what $dA/dt$ is when $h = 15$ metres, at which time $A = \pi h^2 = 225\pi$ square metres. We also know that $dV/dt = -10$ cubic metres per minute, so we see that

$$-10 = \frac{\sqrt{225\pi}}{2\sqrt{\pi}} \cdot \frac{dA}{dt} = \frac{15}{2} \cdot \frac{dA}{dt},$$

and we conclude that the area changes by $dA/dt = -20/15 = -4/3$ square metres per minute.

**James Bond and Jaws.** In this example, we will use the fact that the position of a freefalling object $t$ seconds after being dropped is $s = 4.9t^2$ metres. James Bond and Jaws are in a plane that is cruising at 100 metres per second. Suppose Jaws pushes James Bond out of a plane, then jumps out himself one second later. At what rate is the distance between Jaws and James changing one more second after that? (Ignore the effects of air drag.)

Because Jaws exits the plane exactly one second after James, the horizontal distance between the two men is 100 metres, and remains that way throughout their fall. Let $t$ be the number of seconds that has elapsed since Jaws pushed James out of the plane. Let $y_1$ be the distance that James has fallen, and let $y_2$ be the distance that Jaws has fallen. For all $t \geq 0$, we have

$$y_1 = 4.9t^2,$$

and for all $t \geq 1$, we have

$$y_2 = 4.9(t-1)^2 = 4.9t^2 - 9.8t + 4.9.$$

So the vertical distance, call it $h$, between the two men at any time $t \geq 1$, is

$$h = |y_2 - y_1| = |-9.8t + 4.9| = 9.8t - 4.9.$$

We can easily calculate $\frac{dh}{dt} = 9.8$, but this is just the rate of change of the vertical distance between the two men, not the total distance.

By the Pythagorean theorem, the actual distance $r$ between the two men is equal to

$$r^2 = 100^2 + h^2 = 10000 + h^2.$$

Differentiating both sides with respect to $t$ gives

$$2r\frac{dr}{dt} = 2h\frac{dh}{dt},$$

whence

$$\frac{dr}{dt} = \frac{h}{r} \cdot \frac{dh}{dt} = \frac{9.8h}{r}.$$

It remains to plug in $h$ and $r$ at time $t = 2$. We have

$$h = 9.8 \cdot 2 - 4.9 = 14.7$$

metres and

$$r = \sqrt{10000 + (14.7)^2} \approx 101.1$$

metres, so

$$\frac{dr}{dt} = \frac{9.8 \cdot h}{r} \approx 1.43$$

metres per second.

02.XI **A pedestrian under a streetlamp.** A 2-metre-tall man walks away from a 6-metre-tall streetlamp at a constant rate of 3/2 metres per second. Let's calculate how fast the tip of the man's shadow is moving.

Call the distance between the tip of the shadow $s$ and the distance between the person and the streetlamp $x$. We know that $dx/dt = 3/2$ and want to find $ds/dt$. The triangle formed by the top of the streetlamp, the base of the streetlamp, and the shadow's tip is similar to the triangle formed by the person's foot, his head, and the tip of the shadow. Since the streetlamp is 6 metres tall and the person is 2 metres tall, the first triangle is exactly three times as large as the second. In other words, $s = 3(x - s)$, which means that $s = 3x/2$. Differentiating both sides of this equation with respect to $t$ gives us

$$\frac{ds}{dt} = \frac{3}{2} \cdot \frac{dx}{dt} = \frac{3}{2} \cdot \frac{3}{2} = \frac{9}{4}$$

metres per second. (Note that in this problem it did not matter how far the man was from the streetlamp, since $x$ disappeared from the equation after differentiating.)

**The distance between hands on a watch.** Suppose that you have a watch whose second hand and minute hand both have the same length of 1 centimetre. At exactly ten minutes past the hour, at what rate is the distance between the tip of the second hand and the tip of the minute hand changing?

Because both hands have the same length, drawing an imaginary line between the tips of the two hands completes an isosceles triangle. Let $h$ be the length of this imaginary line. If the angle between the two hands is in the interval $(0, \pi)$—which it is at exactly ten past the hour—then $h$ satisfies the formula

$$h = 2\sin\Big(\frac{\theta}{2}\Big).$$

(Draw a picture to see this!) We want to know $dh/dt$, and differentiating both sides of this formula tells us that

$$\frac{dh}{dt} = 2\cos\Big(\frac{\theta}{2}\Big)\frac{1}{2} \cdot \frac{d\theta}{dt} = \cos\Big(\frac{\theta}{2}\Big)\frac{d\theta}{dt}. \tag{37}$$

So we try and reason about the rate of change of $\theta$ with respect to time.

The second hand moves around the watch once per minute, or at a speed of $2\pi$ radians a minute. Meanwhile the second hand only moves 1/60 of the way around the watch every minute, or $\pi/30$ radians per minute. Hence the angle between the two hands, measured clockwise from the second hand to the minute hand, changes at a rate of $\pi/30 - 2\pi = -59\pi/30$ radians per minute. To use (37), we'll also need to know $\cos\theta$ at ten minutes past the hour. This is a sixth of the way around the clock; that is, $\theta = \pi/3$. Hence we see that

$$\frac{dh}{dt} = \cos\Big(\frac{\pi}{3} \cdot \frac{1}{2}\Big)\frac{-59\pi}{30} = \frac{-59\pi\sqrt{3}}{60}$$

centimetres per minute, or

$$\frac{-59\pi}{60 \cdot 60} = \frac{-59\pi\sqrt{3}}{3600} \approx -0.089$$

centimetres per second.

# III. APPLICATIONS

*Überhaupt verhält es sich*
*mit allen solchen neuen Calculs so,*
*dass man durch sie nichts leisten kann,*
*was nicht auch ohne sie zu leisten wäre;*
*der Vortheil ist aber der, dass* [...]
*jeder, der sich ihn ganz angeeignet hat,*
*auch ohne die gleichsam unbewussten*
*Inspirationen des Genies,*
*die niemand erzwingen kann,*
*die dahin gehörigen Aufgaben lösen,*
*ja selbst in so verwickelten Fällen*
*gleichsam mechanisch lösen kann,*
*wo ohne eine solche Hülfe*
*auch das Genie ohnmächtig wird.*

— CARL FRIEDRICH GAUSS (1843)

## 11. Extreme values and the mean value theorem

At this point of the course, we are no stranger to the concept of modelling some quantity or phenomenon by means of a function. For many such applications, it may be desirable to find a point at which the function is *maximised* or *minimised*. In other words, we may want to find the *extreme values* of a function and where they are attained.

First we must precisely define what we mean by a maximum or a minimum value. Let $f$ be any function, with some domain $D$. For $c$ in this domain $D$, we say that $f(c)$ is the *absolute maximum* of $f$ on $D$ if $f(c) \geq f(x)$ for all $x$ in $D$. Likewise we say that $f(c)$ is the *absolute minimum* of $f$ on $D$ if $f(c) \leq f(x)$ for all $x \in D$. Note that a function may not have an absolute or minimum value (an example is the function $f(x) = 1/x$). Even if a function attains an absolute maximum or minimum value, it may be the case that these values are attained by multiple choices of $c$ in the domain. For example, the absolute maximum of the function $f(x) = \cos x$ is 1, but we could pick $c$ to be 0, $2\pi$, $4\pi$, and so on in order to attain this extreme value.

In many applications, it is fitting to restrict the domain of the function. (For instance, $x$ might be the amount of some resource, so $x < 0$ doesn't make sense, and $x$ might have some limit from above representing how much we can afford.) Once the domain is restricted, it may be the case that some function that does not have an absolute maximum or minimum over all of $\mathbf{R} = (-\infty, \infty)$ does have an absolute maximum or minimum over this smaller domain.

04.XI For certain functions $f$, a number $c$ may be such that $f(c)$ is not an absolute maximum or minimum, and yet $f(c)$ is an extreme value if we restrict $f$ to a small interval around $c$. Concretely, we say that $f(c)$ is a *local maximum* of $f$ if there exists some $\delta > 0$ such that $f(c) \geq f(x)$ for all $x$ with $|x - c| < \delta$. And we say that $f(c)$ is a *local minimum* of $f$ if there is some $\delta > 0$ such that $f(c) \geq f(x)$ for all $x$ with $|x - c| < \delta$.

As an example, consider the polynomial $f(x) = x^3 - x^2 - x + 1$, whose domain is $\mathbf{R}$. By graphing the function we see that $f(1) = 0$, and moreover in some small interval around 1, say $(0, 2)$, $f$ is nonnegative. (Here we have picked $\delta = 1$, since $1 - 1 = 0$ and $1 + 1 = 2$.) That means that every $x$ in the interval $(0, 2)$ satisfies $f(x) \geq 0 = f(1)$, and 1 is a local minimum of $f$. But 1 is not an *absolute minimum* of $f$, since $f(x) \to -\infty$ as $x \to -\infty$. If you want a concrete counterexample, take $x = -2$; one has $f(-2) = -8 - 4 - (-2) + 1 = -9$, which is less than $0 = f(1)$.

If we restrict $f(x) = x^3 - x^2 - x + 1$ to the range $(-1, 2)$, then the absolute minimum of $f$ is 0 (attained at $x = 1$). On the other hand, $f$ still does not have an absolute maximum on this interval. As $x$ increases towards 2, the values $f(x)$ increase towards 3. But since the number 2 itself is not in the range $(1, 2)$, we see that $f(x)$ never actually equals 3 on this domain.

The issue in the preceding discussion lies in the fact that the endpoint 2 is not included in the interval. The absolute maximum of $f$ on $(-1, 2]$ is 3, attained at $x = 2$. So you might suspect that if we restrict the domain of a continuous

function to a *closed* interval (endpoints included on *both* sides) then the following theorem guarantees the existence of both an absolute maximum and minimum.

**Theorem 67** (*Extreme value theorem*). *Let $f$ be continuous on an interval $[a, b]$. Then there are numbers $c$ and $d$ in $[a, b]$ such that $f(c)$ is the absolute maximum of $f$ on $[a, b]$ and $f(d)$ is the absolute minimum of $f$ on $[a, b]$.*

*Proof.* Take MATH 242 Analysis 1 or MATH 254 Honours Analysis 1 to see this proof. ∎

Just like the intermediate value theorem, the extreme value theorem is *nonconstructive*—it asserts the existence of a maximum $f(c)$ and a minimum $f(d)$ without telling you what $c$ and $d$ are. (In fact, the extreme value theorem is closely related to the intermediate value theorem; their proofs both use something called the *least upper bound property* of the real numbers.)

Unlike with the intermediate value theorem however, the methods of calculus can help us find out where these extreme values are attained, so long as the function $f$ is differentiable. This is due to the following theorem of Pierre de Fermat, which is found in some treatises dating to between 1636 and 1642. (Because derivatives hadn't formally been invented at that point, the actual wording Fermat used differs quite a lot from the proof given below, but nevertheless the germ of these ideas originate with him.)

**Theorem 68** (*Fermat's interior extremum theorem*). *If $f$ has a local maximum or a local minimum at $c$ and $f'(c)$ exists, then $f'(c) = 0$.*

*Proof.* First we do the proof when $f$ has a local maximum at $c$. By definition, this means that there is some $\delta > 0$ such that $f(c) \geq f(x)$ for all $x$ with $|x - c| < \delta$. Since $f(x) \leq f(c)$ for all $c - \delta < x < c$, the numerator in the left-hand limit

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}$$

must be negative, and since the denominator $x - c$ is negative as well, we have

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \geq 0.$$

Similar logic tells us that the numerator in the right-hand limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

is also negative, but this time the denominator $x - c$ is positive, so

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} \leq 0.$$

But $f'(c)$ exists and equals both of these one-sided limits, so both $f'(c) \geq 0$ and $f'(c) \leq 0$ hold at the same time. This forces $f'(c) = 0$.

We have proved the theorem in the case that $f(c)$ is a local maximum. If instead $f(c)$ is a local *minimum*, then one could repeat the argument above with slight modifications to show that $f'(c) = 0$ in this case as well. More simply, one can consider the function $-f$, which now has a local maximum at $c$, and use the above paragraph to see that $-f(c) = 0$. This implies that $f'(c) = 0$ as well. ∎

Now we are sure that $f(x) = x^3 - x^2 - x + 1$ attains both an absolute maximum and minimum on any closed interval—let's consider $[0, 2]$ this time and use Fermat's theorem to help us find the absolute extreme values on this interval. The word "interior" in the name of Theorem 68 suggests its use: if some input $c$ in the *interior* $(0, 2)$ of the interval $[0, 2]$ attains an extreme value and $f'(c)$ exists, then we must have $f'(c) = 0$. The derivative $f'$ of $f$ is given by $f'(x) = 3x^2 - 2x - 1$, which factors as $(3x+1)(x-1)$, so the only $c$ with $f'(c) = 0$ that lies in the range $(0, 2)$ is $c = 1$. We check that $f(1) = 0$. Because no other $x$ in $(0, 2)$ has $f'(x) = 0$ (but all of these derivatives exist), we deduce that no other point in $(0, 2)$ can be an extreme value. To finish off the procedure, we still need to check $x = 0$ and $x = 2$, since these points are not covered by Fermat's theorem. We have $f(0) = 1$ and $f(2) = 3$, so we conclude that the maximum value of $f$ on $[0, 2]$ is 3, attained at $x = 2$, and the minimum value of $f$ on $[0, 2]$ is 0, attained at $x = 1$.

If we had considered the wider interval $[-1, 2]$, then the input $c = -1/3$ would have also satisfied $f'(c) = 0$, so we would have had to check $f(-1/3) = -1/27 - 1/9 + 1/3 + 1 = 32/27$ as well, but this is not larger than $f(2) = 3$ or smaller than $f(1) = 0$, so this is not an extreme value. We'd also have had to check the left endpoint $f(-1) = 0$ to see that the minimum of $f$ on $[-1, 2]$ is still 0, but now it's attained at both $x = -1$ and $x = 1$.

**Two words of warning.** Suppose $f$ has a local maximum or minimum at $f(c)$. Fermat's theorem only tells us that $f'(c) = 0$ under the condition that $f'(c)$ exists. For example, the minimum of $f(x) = |x|$ is 0, attained at $x = 0$, but $f'(0)$ is not 0, because it does not exist. This example suggests that we should also add points $c$ where $f'(c)$ does not exist to our checklist when looking for possible candidates for the absolute maximum or minimum on an interval.

The other thing to be wary of is that Fermat's theorem only says that if $f$ is a local extremum, then $f'(c) = 0$. The converse is not true! It is possible for $f'(c)$ to equal 0 even when $f(c)$ is not a local extreme value. A concrete example is given by the function $f(x) = x^3$. Then $f'(0) = 0$, but $f$ is increasing on both sides of 0.

**The closed interval method.** We have seen that a continuous function must attain a maximum value and a minimum value on a closed interval, and that this must happen either at the endpoints, or at some $c$ in the interior with $f'(c) = 0$ or $f'(c)$ nonexistent. So we can sum up what we know about extreme values in the the following procedure.

**Algorithm E** (*Find extreme values in a closed interval*). Given an interval $[a, b]$ and a function $f$ that is continous on $[a, b]$ and differentiable on $(a, b)$, this

procedure finds the absolute minimum and maximum values of $f$ on $[a, b]$.

**E1.** [Take derivative.] Find the derivative $f'$ of $f$.

**E2.** [Set derivative to 0.] Add all points $c$ in $(a, b)$ with $f'(c) = 0$ to the list of candidates.

**E3.** [Check where derivative does not exist.] Add all points $c \in (a, b)$ where $f'(c)$ does not exist to the list of candidates.

**E4.** [Check endpoints.] Add the points $a$ and $b$ to the list of candidates.

**E5.** [Evaluate $f$ on candidates.] Calculate $f(c)$ for all candidates $c$ in the list.

**E6.** [Output.] The smallest value calculated in step E5 is the absolute minimum of $f$ on $[a, b]$, and the largest is the absolute maximum. ∎

06.XI An input $c$ in the domain of $f$ such that $f'(c) = 0$ or $f'(c)$ does not exist is called a *critical number*. (The value $f(c)$ is sometimes called a *critical value*.) The closed interval method can be summarised thus: *Evaluate $f$ on the two endpoints on the interval and on all the critical numbers in the interval, then output the maximum and minimum critical values.*

Before moving on to bigger and better things, let's apply the closed interval method to a couple of concrete examples. Suppose we want to find the maximum and minimum values of the function

$$f(x) = \begin{cases} x \ln x, & \text{if } x > 0; \\ 0, & \text{if } x = 0 \end{cases}$$

on the closed interval $[0, e]$. This function is continuous, since $x \ln x \to 0$ as $x \to 0$ from the right. (The easiest way to prove this is using L'Hospital's Rule, which we'll see in a few classes' time.) So by the extreme value theorem there is both a maximum and minimum on this interval. We take the derivative $f'(x) = x(1/x) + 1 \cdot \ln x = 1 + \ln x$, and observe that the equation $\ln x + 1 = 0$ is equivalent to $\ln x = -1$. Taking the exponential of both sides, $f'(x) = 0$ only if $x = e^{\ln x} = e^{-1} = 1/e$. This critical number lies in the range $[0, e]$, so we evaluate

$$f\Big(\frac{1}{e}\Big) = \frac{1}{e} \ln \frac{1}{e} = -1/e.$$

We also evaluate $f$ on the endpoints:

$$f(0) = 0 \qquad \text{and} \qquad f(e) = e \ln e = e.$$

The largest number in the set $\{0, -1/e, e\}$ is $e$ and the smallest is $-1/e$, so those are the extreme values of $f$ on $[0, e]$.

Here's a harder one. Let's find the maximum and minimum values of the function $f(x) = x - \sqrt[3]{x}$ on the interval $[-1, 4]$. The derivative of $f$ is

$$f'(x) = 1 - \frac{1}{3} x^{-2/3} = 1 - \frac{1}{3t^{2/3}}.$$

Solving for $f'(x) = 0$ leads us to the equation

$$x^{2/3} = \frac{1}{3},$$

whence we find that two of the critical numbers are $c = \pm 1/\sqrt{27}$. There is a third critical number, namely $c = 0$, since $f'(0) = -\infty$. So in the closed interval method, the set of points on which we evaluate $f$ is $\{-1, -1/\sqrt{27}, 0, 1/\sqrt{27}, 4\}$. We calculate

$$f(-1) = (-1) - \sqrt[3]{-1} = -1 - (-1) = 0 \qquad \text{and} \qquad f(0) = 0 - \sqrt[3]{0} = 0,$$

as well as

$$f\Big(\frac{-1}{\sqrt{27}}\Big) = \frac{-1}{\sqrt{27}} - \sqrt[3]{\frac{-1}{\sqrt{27}}} = \frac{-1}{3^{3/2}} - \frac{-1}{\sqrt{3}} = \frac{-1-(-3)}{\sqrt{27}} = \frac{2}{\sqrt{27}}.$$

Then since both $x$ and $\sqrt[3]{x}$ are odd functions, we easily deduce that

$$f\Big(\frac{1}{\sqrt{27}}\Big) = -\frac{2}{\sqrt{27}},$$

and it's also easy to plug in

$$f(4) = 4 - \sqrt[3]{4}.$$

The only negative number here was $-2/\sqrt{27}$, so this is the minimum of $f$ on $[-1, 4]$. The maximum is $4 - \sqrt[3]{4}$, since this number is greater than $4 - \sqrt{4} = 2$ whereas $2/\sqrt{27} = 2/\big(3\sqrt{3}\big)$ is less than 1.

**The mean value theorem.** Imagine you toss a ball into the air and catch it. The ball starts in your hand, ends in your hand, and moves continuously over time, so by the extreme value theorem, it attains a maximum height at some point. Because the ball rises to this height and then starts falling after it reaches it, a local maximum is reached at this point, and (assuming the ball's position is a differentiable function) Fermat's theorem says that the derivative of the ball's vertical position must be zero at that point. In other words, there is some point in time at which the ball's velocity is zero along the vertical axis.

This exact line of reasoning can be generalised to give us the following theorem.

**Theorem 69** (*Rolle's theorem*). *Let $a$ and $b$ be numbers with $a < b$. Suppose that*

*i) $f$ is continuous on $[a, b]$;*
*ii) $f$ is differentiable on $(a, b)$; and*
*iii) $f(a) = f(b)$.*

*Then there is a number $c$ in $(a, b)$ such that $f'(c) = 0$.*

*Proof.* The proof is morally the same argument as the ball-tossing example given above, but to be completely precise we have to consider three cases. Either $f$ is completely flat, or it increases at some point (as in the example above), or it decreases at some point. The first case is easy. If $f$ is completely flat on $[a, b]$, then $f'$ is equal to some constant on the whole interval, and any $x$ in $(a, b)$ satisfies $f'(x) = 0$.

If $f$ is not constant on $[a, b]$, then there is some $x$ in $(a, b)$ such that $f(x) \neq f(a)$. Either $f(x) > f(a)$ or $f(x) < f(a)$. If $f(x) > f(a)$, then $f(a) = f(b)$ is not the maximum value of $f$ in $[a, b]$. By the extreme value theorem, $f$ must attain a maximum value in $[a, b]$, and we know it doesn't do so at the endpoints, so it must attain a maximum $f(c)$ at some input $c$ in the interior $(a, b)$. We assumed that $f$ is differentiable on $(a, b)$, so $f'(c)$ exists, and by Fermat's interior extremum theorem, we must have $f'(c) = 0$.

The case where $f(x) < f(a)$ is extremely similar. It tells us that $f(a) = f(b)$ is not the *minimum value* of $f$ in $[a, b]$, but $f$ must attain a minimum value by the extreme value theorem, so this happens at some $c \in (a, b)$. The derivative $f'(c)$ exists by hypothesis (ii), so $f'(c) = 0$ by Fermat's theorem. ∎

09.XI Rolle's theorem is a special case of a very important theorem called the mean value theorem. In the ball-tossing scenario above, the net displacement of the ball was zero, and there was some point at which its velocity was zero as well—its instantaneous velocity matched its average velocity.

It is an intuitive fact that this should generalise to situations where the net displacement isn't zero. Let's revisit the scenario from our very first class back in August, in which you and your friend were driving from Montreal to Quebec City. Suppose your car was photographed by a traffic camera in Montreal at 9:00 P.M. and then by another traffic camera in Quebec City at 10:40 P.M. You weren't speeding on either of these occasions, but they recorded your location and the time you were there. A week later, you receive a speeding ticket in the mail. Do the traffic cops have a case?

Of course in real life this would not happen because the cops seemingly have other important things to do like roam around McGill campus. But let's follow the logic here. The time between the two photos is 100 minutes, and during that time, your car's displacement was roughly 230 kilometres (the distance as the crow flies between Montreal and Quebec). So your average velocity throughout your journey was 2.3 kilometres per minute, or 138 kilometres per hour. Your average *speed* is at least this, and in fact it should be even higher, because the highway between the two cities isn't perfectly straight so you surely drove more than 230 kilometres.

Traffic tickets are issued not for high average (or *mean*) speed, but instead for high instantaneous speed. But is it possible for you to have driven at an average speed of 138 kilometres per hour without *at some point* having gone *exactly* 138 kilometres per hour? The mean value theorem, which is a generalisation of Rolle's theorem, says that this is not possible. The full mean value theorem is

not very difficult to prove now that we have access to Rolle's theorem.

**Theorem 70** (*Mean value theorem*). *Let $a$ and $b$ be numbers with $a < b$. Suppose that*

*i) $f$ is continuous on $[a, b]$;*

*ii) $f$ is differentiable on $(a, b)$; and*

*Then there is a number $c$ in $(a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Proof.* The trick is to try and reduce the statement to Rolle's theorem. In that setting, the line segment connecting $f(a)$ and $f(b)$ was horizontal, since $f(a) = f(b)$, and we wanted to find a horizontal tangent line somewhere on the graph of $f$. This time the line segment connecting $f(a)$ and $f(b)$ has slope $\big(f(b) - f(a)\big)/(b - a)$, and we want to find some tangent line to the graph of $f$ with this slope.

The right thing to do is to consider the line passing through $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$ as a function $g(x)$, and then subtract this function from $f$. Since the slope of the line is

$$m = \frac{f(b) - f(a)}{b - a}$$

and $\big(a, f(a)\big)$ is a point on this line, we have

$$g(x) - f(a) = \frac{f(b) - f(a)}{b - a}(x - a),$$

and we have

$$g(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

We then define

$$h(x) = f(x) - g(x)$$

for all $x$ in $[a, b]$; the claim is that $h(x)$ satisfies the hypotheses of Rolle's theorem.

Since $g(x)$ is the equation of a line, it is both continuous and differentiable on $[a, b]$. Hence $h = f - g$ is continuous on $[a, b]$ and differentiable on $(a, b)$ by our hypotheses on $f$ and the fact that continuity and differentiability are preserved by the subtraction operation. Furthermore, the graph of $g$ passes through the two points $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$, so $g(a) = f(a)$ and $g(b) = f(b)$. This means that

$$h(a) = f(a) - g(a) = 0 \qquad \text{and} \qquad h(b) = f(b) - g(b) = 0.$$

So $h$ does satisfy the conditions of Rolle's theorem, and we deduce that there must be some $c \in (a, b)$ with $h'(c) = 0$.

What does it mean for $h'(c)$ to be zero? We have $h' = f' - g'$, so $h'(c) = 0$ implies that $f'(c) = g'(c)$. But $g$ is the equation of a line, with slope $m = \big(f(b) - f(a)\big)/(b - a)$ everywhere, so we have

$$f'(c) = g'(c) = \frac{f(b) - f(a)}{b - a},$$

which is what we wanted. ∎

Throughout much of this course so far, the statements we've been establishing about derivatives mostly told us that if $f$ had some property, then $f'$ must have some (other) property. Rolle's theorem and the mean value theorem are no different in this regard. The miracle of the mean value theorem is that now the information we gain about $f'$ is so strong that it can give us information *in the other direction*. In other words, what we know about the derivative $f'$ can now tell us something about the function $f$ itself. Here's an example.

To prove Theorem 58, Lemma 59, and Proposition 63, we used the seemingly obvious fact that if a function's derivative is identically zero, then the function must be constant. This is something we didn't actually prove. (Theorem 38 asserts that the derivative of a constant function is identically zero, but that is the converse statement.) Now that we have the mean value theorem, we finally can. You can check that none of the results listed above are invoked in the following proof, so there is no circularity here.

**Theorem 71.** *Let $f$ be a function such that $f'(c) = 0$ for all $c \in (a, b)$. Then $f$ is constant on $[a, b]$; that is, $f(x) = f(a)$ for all $x \in [a, b]$.*

*Proof.* Let $x$ in $[a, b]$ be given. If $x = a$, then clearly $f(x) = f(a)$, so we may assume for the rest of the proof that $x > a$. By the mean value theorem, there is some $c$ in $(a, x)$ such that

$$f'(c) = \frac{f(x) - f(a)}{x - a}.$$

But by hypothesis, $f'(c) = 0$, so the right-hand side in the equation above must also be zero. This tells us that $f(x) - f(a) = 0$. Hence $f(x) = f(a)$. ∎

11.IX As a corollary, two functions with the same derivative must have the exact same shape; if they are not the same function then they must differ by exactly a constant.

**Corollary 72.** *Let $f$ and $g$ be continuous on $[a, b]$ and differentiable on $(a, b)$. If $f' = g'$, then there is a constant $c$ such that*

$$f(x) = g(x) + c$$

*for all $x$ in $[a, b]$.*

*Proof.* Let $h = f - g$. Since $f' = g'$, we have $h' = f' - g' = 0$, so by Theorem 71, $h$ is a constant function; there is some constant $c$ such that $h(x) = c$ for all $x \in [a, b]$. Then we have

$$f(x) = g(x) + h(x) = g(x) + c$$

for all $x$ in $[a, b]$. ∎

A similar type of statement is encapsulated by the following scenario. Suppose two cars participate in a drag race. At time $t = 0$, the cars are at the same horizontal position. If, for all $t \geq 0$, Car A's velocity at time $t$ is always greater than Car B's velocity at the same time, then Car A will always have travelled a greater horizontal distance than Car B. The precise mathematical formulation of this principle is the following theorem. Its proof is another consequence of the mean value theorem.

**Theorem 73.** *Let $f$ and $g$ be continuous of $[a, b]$ and differentiable on $(a, b)$. If $f(a) = g(a)$ and $f'(x) > g'(x)$ for all $x$ in $(a, b)$, then*

$$f(x) > g(x)$$

*for all $x$ in $(a, b)$.*

*Proof.* Let $h = f - g$ as in the previous proof. Let $x$ in $(a, b)$ be arbitrary. By the mean value theorem applied to the function $h$ on the interval $[a, x]$, there is some $c$ in $(a, x)$ such that

$$h'(c) = \frac{h(x) - h(a)}{x - a}.$$

Since

$$h'(c) = f'(c) - g'(c) > 0$$

and $x - a > 0$, we see that $h(x) - h(a) > 0$ as well. But

$$h(a) = f(a) - g(a) = 0,$$

so we conclude that

$$f(x) - g(x) = h(x) > 0.$$

This means that $f(x) > g(x)$. ∎

**Increasing and decreasing functions.** A function $f$ is said to be *increasing* on an interval $[a, b]$ if for all numbers $s$ and $t$ in $[a, b]$ with $s < t$, we have $f(s) < f(t)$. Likewise, $f$ is *decreasing* on $[a, b]$ if for all $s$ and $t$ with $a \leq s < t \leq b$ one has $f(s) > f(t)$. A differentiable function $f$ that is increasing on some interval $[a, b]$ has positive derivative everywhere on the open interval $(a, b)$. (This can be proved from the limit definition of the derivative—for $x$ in $(a, b)$, expand $f'(x)$ as a limit and reason about the signs of the numerator and denominator.) Similarly, a decreasing function has negative derivative. In the spirit of Theorem 71, we prove converses to these two statements.

**Theorem 74.** *Let $f$ be a function that is continuous on $[a, b]$.*

*i If $f'(c) > 0$ for all $c \in (a, b)$, then $f$ is increasing on $[a, b]$.*

*ii) If $f'(c) < 0$ for all $c \in (a, b)$ then $f$ is decreasing on $[a, b]$.*

*Proof.* The first part of the proof is common to both (i) and (ii). Let $s$ and $t$ be any two numbers in $[a, b]$ with $s < t$. Since $f$ is continuous on $[s, t]$ and differentiable on $(s, t)$ by the mean value theorem there is some $c$ in $(s, t)$ with

$$f'(c) = \frac{f(t) - f(s)}{t - s}.$$

Since $s < t$¡ the denominator on the right-hand side is positive.

First we prove part (i). If $f'(c) > 0$, then $f(t) - f(s)$ must be positive, so $f(s) < f(t)$ and since $s$ and $t$ were arbitrary, we see that $f$ is increasing. For part (ii), we assumed that $f'(c) < 0$, so $f(t) - f(s)$ is negative this time and hence $f$ is decreasing. ∎

This method of determining when a function is increasing or decreasing can help us determine whether a critical number $c$ gives rise to a local maximum or a local minimum.

**Corollary 75** (*First derivative test for local extrema*). *Let $c$ be a critical number of a continuous function $f$.*

*i) If $f'(x)$ changes sign from positive to negative at $x = c$, then $f$ attains a local maximum at $c$.*

*i) If $f'(x)$ changes sign from negative to positive at $x = c$, then $f$ attains a local minimum at $c$.*

*iii) If $f'$ has the same sign on both sides of $c$, then $f$ attains neither a local maximum nor a local minimum at $c$.*

*Proof.* In case (i), Theorem 74 tells us that $f$ is an increasing function immediately to the left of $c$ and a decreasing function on the interval to the right of $c$, so $f(c)$ is a local maximum. Case (ii) is similar but with the roles of "increasing" and "decreasing" swapped, so $f(c)$ is a local minimum. In case (iii), $f$ is either increasing on both sides of $c$ or decreasing on both sides of $c$. In either of these two situations, $f(c)$ is not a local maximum or minimum. ∎

Let's see an example. Suppose we want to find out where the function

$$f(x) = \frac{1}{2}x^4 + x^3 - x^2 + 3,$$

which is defined on the whole real line $\mathbf{R}$ has local maxima and minima. To do this we might like to chart out where it is increasing and where it is decreasing. We compute its derivative

$$f'(x) = 2x^3 + 3x^2 - 2x,$$

which admits the factorisation

$$f'(x) = x(2x-1)(x+2).$$

The critical numbers of $f$ are the roots of the derivative, which are 0, $1/2$, and $-2$.

To the left of $-2$, the factor $x$ is negative, $2x-1$ is negative, and $x+2$ is negative, so $f'(x)$ is negative. This means that $f$ is decreasing on $(-\infty, -2)$. On the interval $(-2, 0)$, the factors $x$ and $2x-1$ are still negative but now the factor $x+2$ is positive, so $f'(x)$ is positive and $f$ is increasing. On the interval $(0, 1/2)$, $x$ and $x+2$ are both positive, but $2x-1$ is negative, so $f'(x)$ is negative and $f$ is decreasing. And lastly, to the right of $1/2$, all three factors are positive, so $f'$ is positive and $f$ is increasing. The results of this paragraph are summarised in the following table:

| interval | $x+2$ | $x$ | $2x-1$ | behaviour of $f$ |
|---|---|---|---|---|
| $(-\infty, -2)$ | $-$ | $-$ | $-$ | decreasing |
| $(-2, 0)$ | $+$ | $-$ | $-$ | increasing |
| $(0, 1/2)$ | $+$ | $+$ | $-$ | decreasing |
| $(1/2, \infty)$ | $+$ | $+$ | $+$ | increasing |

By the first derivative test, we see that $f(x)$ attains local minima when $x = -2$ and $x = 1/2$, as well as a local maximum when $x = 0$. Because $f$ goes to $\infty$ as $x \to \pm\infty$, there is no absolute maximum of $f$. But there is an absolute minimum, which must be one of the local minima $f(-2)$ and $f(1/2)$. We compute

$$f(-2) = -1 \qquad \text{and} \qquad f\Big(\frac{1}{2}\Big) = \frac{93}{32}$$

and conclude that $f(-2) = -1$ is the absolute minimum of the function $f$ over the domain $\mathbf{R}$.

**Concavity.** The derivative of a differentiable function $f$ contains enough information for us to find its local extreme values as well as the intervals on which it is increasing or decreasing. This lets one make a rudimentary sketch of the graph of $f$. However, there is a subtler aspect to the shape of a function's graph that can only be ascertained by means of the second derivative.

We say that a function $f$ is *concave upward* or *convex* on an interval $I$ if for all $x$ in $I$, the graph of $f$ lies *above* the tangent line to $f$ at the point $\big(x, f(x)\big)$. Similarly, we say that a function $f$ is *concave downward* or simply *concave* on an interval $I$ if $f$ lies *below* all of its tangents in $I$.

The following theorem, which ties the second derivative to concavity, is another consequence of the mean value theorem.

**Theorem 76** (*Concavity test*). *Let $f$ be a differentiable function such that $f'$ is also differentiable on some interval $I$.*

*i) If $f''(x) > 0$ for all $x$ in $I$, then the graph of $f$ is concave upward on $I$.*

*ii) If $f''(x) < 0$ for all $x$ in $I$, then the graph of $f$ is concave downward on $I$.*

*Proof.* We shall spend the bulk of this proof proving (i); the proof of (ii) will follow easily after that.

Let $a$ be any number in $I$ and let $L$ denote the tangent line to $f$ at $a$. This line has slope $f'(a)$ and contains the point $\big(a, f(a)\big)$, so $L$ admits the expression

$$L(x) = f(a) + f'(a)(x - a)$$

as a function of $x$. We wish to show that on the domain $I$, the graph of $f$ lies entirely above the tangent line $L$, except at $x = a$ where it touches the graph exactly once. In other words, we want to prove that

$$f(x) > L(x) = f(a) + f'(a)(x - a) \tag{38}$$

for all $x$ in $I$ with $x \neq a$.

First we consider numbers $x$ with $x > a$. By applying the mean value theorem to the interval $[a, x]$, we find that there is some $c$ in the interval $(a, x)$ for which

$$f'(c) = \frac{f(x) - f(a)}{x - a}.$$

Since the derivative $f''$ of $f'$ is positive on all of $I$, the function $f'$ must be increasing on $I$. So the inequality $a < c$ implies that $f'(a) < f'(c)$, and we have

$$f'(a) < f'(c) = \frac{f(x) - f(a)}{x - a}.$$

Multiplying both sides by $x - a$ (which is a positive quantity), we obtain

$$f'(a)(x - a) < f(x) - f(a),$$

which rearranges to (38).

The case where $x < a$ is not so different. The mean value theorem, this time applied to the interval $[x, a]$, furnishes some $c$ in $(x, a)$ with

$$f'(c) = \frac{f(x) - f(a)}{x - a}.$$

Again, the function $f'$ is increasing on all of $I$, so the inequality $a > c$ implies that $f'(a) > f'(c)$. Hence we have

$$f'(a) > f'(c) = \frac{f(x) - f(a)}{x - a}.$$

This time, we multiply both sides by the negative number $x - a$, which necessitates flipping the inequality $>$ to $<$. Once again, we obtain

$$f'(a)(x-a) < f(x) - f(a),$$

which is equivalent to (38) and settles the proof of (i).

To prove (ii), we assume that $f''$ is negative on $I$. Since

$$(-f)'' = (-f')' = -f'',$$

we see that the second derivative of $-f$ is *positive* on $I$. By (i), the graph of $-f$ must be concave upward on $I$; the graph of $-f$ lies above all of its tangents. Hence the graph of $f$ lies below all of its tangents, and $f$ must be concave downward. ∎

Let us apply Theorem 76 to analyse the concavity of the function $f(x) = x^4/2 + x^3 - x^2 + 3$ from the previous example. We had already calculated its derivative $f'(x) = 2x^3 + 3x^2 - 2x$, and we take the derivative one more time to obtain

$$f''(x) = 6x^2 + 6x - 2 = 2(3x^2 + 3x - 1).$$

The factor 2 obviously has no roots, so the roots of $f''$ occur at the roots of $3x^2 - 3x - 1$. By the quadratic formula, these are

$$x = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 6(-1)}}{2 \cdot 3} = -\frac{1}{2} \pm \frac{\sqrt{21}}{6},$$

and the sign of $f''$ can only change at these two points. When $x < -1/2 - \sqrt{21}/6$, we have $f''(x) > 0$ (you can choose a very large negative number to test this fact). Since $\sqrt{21}/6 > \sqrt{9}/6 = 1/2$, the number 0 lies in the interval

$$\left(-\frac{1}{2} - \frac{\sqrt{21}}{6},\ -\frac{1}{2} + \frac{\sqrt{21}}{6}\right), \tag{39}$$

so we can test $f''(0) = -2$ to see that $f''$ is negative on this whole interval. Lastly, it's not hard to check that $f''(x) > 0$ for $x > -1/2 + \sqrt{21}/6$. Here's a chart of our findings

| interval | $f''(x)$ | behaviour of $f$ |
|---|---|---|
| $(-\infty,\ -1/2 - \sqrt{21}/6)$ | $+$ | concave upward |
| $(-1/2 - \sqrt{21}/6,\ -1/2 + \sqrt{21}/6)$ | $-$ | concave downward |
| $(-1/2 + \sqrt{21}/6,\ \infty)$ | $+$ | concave upward |

By Theorem 76, we conclude that $f$ is concave downward on the interval (39), and concave upward elsewhere. The numbers $x = -1/2 \pm \sqrt{21}/6$ are points at which the concavity of $f$ changes. Such points are called *inflection points*.

Recall that the critical numbers of $f$ were $-2$, 0, and $1/2$. We discovered earlier that $f(-2)$ and $f(1/2)$ were local minima, and $f(0)$ was a local maximum. Keen-eyed readers might look at the concavity chart above for $f$ and notice that $-2$ and $1/2$ both belong to intervals where $f$ is concave upward, and 0 belongs to an interval where $f$ is concave downward. This observation generalises to other functions as well, and provides an alternative test for determining whether a critical number $c$ with $f'(c) = 0$ gives rise to a local maximum or a local minimum.

**Theorem 77** (*Second derivative test for local extrema*). *Suppose that $f$ is a twice differentiable function with $f'(c) = 0$.*

*i) If $f''(c) > 0$, then $f(c)$ is a local minimum of $f$.*

*ii) If $f''(c) < 0$, then $f(c)$ is a local maximum of $f$.*

*Proof.* This follows fairly easily from Theorem 76 and the definitions of concavity. By hypothesis, we know that $f'(c) = 0$, which means that the tangent line to $f$ at $c$ has slope 0; it must therefore be the horizontal line $y = f(c)$. If $f''(c) > 0$, then since $f$ is continuous there must be some interval $I$ containing $c$ where $f''$ is positive. By Theorem 76, $f$ is concave upward on $I$, meaning that the graph of $f$ lies above all of its tangents on this interval. In particular, it lies above the line $y = f(c)$, and we see that $f(x) \geq f(c)$ for all $x$ in $I$. Hence $f(c)$ is a local minimum, proving part (i).

If on the other hand we had $f''(c) < 0$, then

$$(-f)''(c) = (-f')'(c) = -f''(c) > 0,$$

so $-f(c)$ is a local minimum of $-f$ by (i). This means that $f(c)$ is a local maximum of $f$, proving part (ii). ∎